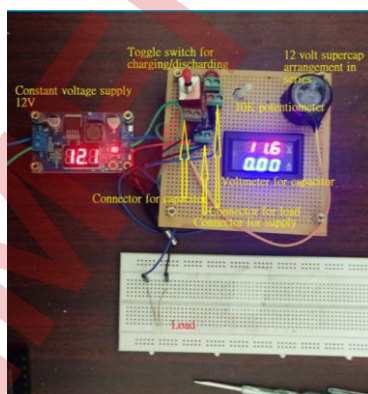




DEPARTMENT OF METALLURGICAL AND
MATERIALS ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY
MADRAS
CHENNAI – 600 036

Battery-supercapacitor hybrid energy storage systems (BSHESS): Technology assessment and market analysis-based business plan



A Thesis

Submitted by

Sandeep Kumar

For the award of the degree

Of

**MASTER OF SCIENCE (ENTREPRENEURSHIP) PRENEURSHIP) by
Research**

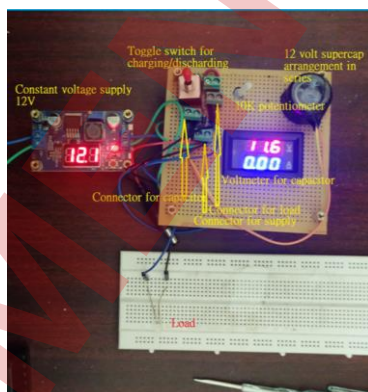
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DEPARTMENT OF ENGINEERING DESIGN
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True meanings of life are so distributed and different to many, that's why so much literature is written and appreciated worldwide!

Dr. Tiju Thomas

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I dedicate this thesis to my supervisor Dr. Tiju Thomas for all his inspiration to me !!

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THESIS CERTIFICATE

This is to certify that the thesis titled **Battery-supercapacitor hybrid energy storage systems (BSHESS): Technology assessment and market analysis-based business plan** submitted by me to the Indian Institute of Technology, Madras for the award of the degree of **MASTER OF SCIENCE (ENTREPRENEURSHIP)**, is a bona fide record of research work carried out by me under the supervision of **Professor Tiju Thomas**. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Chennai 600 036

Sandeep Kumar

Date: Monday, 20 September 2021

Dr. Tiju Thomas

Associate Professor

Materials Processing Section

Department of Metallurgical and Materials
Engineering, Indian Institute of
Technology Madras Chennai – 600 036

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LIST OF PAPERS BASED ON THIS THESIS

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This program has given me a sense of professional and emotional empowerment, and I am now completely hopeful about the future.

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ABSTRACT

KEYWORDS: POTENTIAL CUSTOMER, BSHESS, PLAN, BUSINESS, HYBRID, ENERGY

A market-analysis-based business plan has been developed for Battery-Supercapacitor Hybrid Energy Storage Systems (BSHESS) to meet the requirements of end-users and potential customers in the off-grid and portable power segments. Based on the needs and value analysis done through a market survey, few technology solutions have been proposed. To test these technology solutions, a test bed has been designed and developed. BSHESS delivery allows 10 times fast charging (compared to a pure battery source) for pulse power demand. The calculated peak shaving percentages range from 20% to over 50%. This offers obvious cost-cutting possibilities. This is due to the capability of handling a minimum of 3 times more current for a specified time without implementing voltage-boost electronics.

The technology has been experimented with using a load-based switching method for BSHESS on the testbed developed. Using BSHESS to store energy during nominal loads and discharging it during abnormal loads, businesses can save a minimum of ₹3.33 per KWh right away, whenever they exceed the base load tariff. From a business sustainability standpoint and to facilitate completely Indigenous production to lower the cost of the technology solution, lab-scale initial supercapacitors using activated carbon are being made and tested. These show a total capacitance of >200mF at 1 volt working voltages.

In light of the above, a business plan is presented for launching our technology product, which includes the company's vision, mission, and goals. The business plan will follow a 24-step lean startup development modality. Financial projections are given for the proposed solutions' economic feasibility.

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GLOSSARY

The following are some of the commonly used terms in the thesis:

Engineering, Procurement and Construction (EPC)	Engineering, Procurement and Construction (EPC) firms deliver a complete package of resources to complete infrastructure projects. EPC services typically provide a single responsible source for executing a project, thus alleviating risk for the owner
CAGR	The yearly rate of return that an investment must generate to raise the account from its beginning balance to its ending balance, over a specific time frame
Phytosynthesis	Biological synthesis processes which employ different kinds of secondary metabolites from different parts of the plant (whole plant, leaves, fruit peel, root, bark, seed, and stem) for the non-hazardous, eco-friendly reduction of nanoparticles.
Equivalent series resistance (ESR)	Key parameter of supercapacitors that measures the internal resistance within the device
pH	It is logarithmic scale which is used to express the acidity or basicity of an aqueous solution.
Ammonothermal	The ammonothermal method is a synthesis method that involves use of ammonia particularly in its supercritical fluid state to grow high quality crystals. This processes is similar to hydrothermal synthesis but use ammonia instead of water capable of synthesizing nitride materials especially GaN under high pressure and high temperature.

Soft-Urea	Soft urea method is one of the newly developed methods for the synthesis of materials using the unique property of urea at mild conditions. These uses include synthesizing biodegradable polymers and composites, as well as making it an important technique as it is environmentally friendly in materials science especially in fields of materials that require environmentally friendly synthesis processes.
-------------------------	---

ABBREVIATIONS

AC	Activated Carbon
AC	Alternate Current
AEMWE	Anion Exchange Membrane Water Electrolysis
ARPU	Average Revenue Per Unit
B2B	Business-To-Business
B2C	Business To Customer
BESS	Battery Energy Storage Systems
BET	Brunauer-Emmett-Teller Method
BGP	Bitter Gourd Pericarp
BHM	Beachhead Market
BSHESS	Battery-Supercapacitor Hybrid Energy Storage System
CAC	Customer Acquisition Cost
CAGR	Compound Annual Growth Rate
CB	Carbon Black
CEA	Central Electricity Authority
CLV	Customer Lifetime Value
C-MET	Centre For Materials For Electronics Technology
CNT	Carbon Nanotube
CP	Chronopotentiometry
CPI	Consumer Price Index
CSAT	Customer Satisfaction Score
CV	Cyclic Voltammetry
CVD	Chemical Vapour Deposition
DC	Direct Current
DMU	Decision Making Unit
DRDO	Defence Research And Development Organisation
EA	Electricity Act

EDLCs	Electric Double-Layered Capacitors
EIS	Electrochemical Impedance Spectroscopy
EPC	Engineering Procurement And Construction
ESCOs	Energy Efficiency Service Companies
ESR	Equivalent Series Resistance
ESS	Energy Storage Systems
EV	Electric Vehicle
eVTOL	Electric Vertical Take-Off And Landing
FACTS	Flexible Alternating Current Transmission System
FOR	Forum Of Regulators
FTIR	Fourier-Transform Infrared Spectroscopy
GCD	Galvanostatic Charge-Discharge
GDP	Gross Domestic Product
GNF	Graphitic Nanofiber
GOI	Government Of India
GPS	Global Positioning System
GST	Goods And Services Tax
GW	Giga Watt
HCL	Hydrochloric Acid
HEV	Hybrid Electric Vehicle
IESS	Independent Energy Storage Systems
IITM	Indian Institute Of Technology Madras
IoT	Internet Of Things
ISA	International Solar Alliance
ISRO	Indian Space Research Organisation
KOH	Potassium Hydroxide
KPI	Key Performance Indicators
LOI	Letters Of Intent/Interest
LTV	Lifetime Value
MCU	Microcontroller Unit

MEA	Membrane Electrode Assembly
MEH	Micro-Energy Harvesting
MSME	Micro, Small & Medium Enterprises
MSP	Meter Service Providers
MVBP	Minimum Viable Business Product
NMP	N-Methyl Pyrolidine
OEMs	Original Equipment Manufacturers
PCB	Printed Circuit Board
PEDOT	Poly(3,4-Ethylene-Dioxythiophene)
pH	Potential Of Hydrogen
PHEV	Plug-In Hybrid Electric Vehicles
PPY	Polypyrrole
PSA	Principal Scientific Adviser
PSP	Pumped Storage Plants
PVDF	Polyvinylidene Fluoride
R&D	Research And Development
R2R	Roll To Roll
RE	Renewable Energy
ROI	Return On Investment
SC-ESS	Supercapacitor Energy Storage Systems
SE	Secondary Electrons
SEM	Scanning Electron Microscopy
SMART	Specific, Measurable, Achievable, Relevant, Time-Bound
SOC	State Of Charge
SS	Solid-State
SS	Stainless Steel
SWOT	Strengths, Weaknesses, Opportunities, And Threats
TAM	Total Addressable Market
TMO	Transition Metal Oxide
TOC	Table Of Contents

TRL		Technology Readiness Level
Wh, kWh		Watt-Hour, Kilo-Watt-Hour
XRD		X-Ray Diffraction

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NOTATION

English Symbols		
m		market size
m		meter
m		Mass in kilograms
p		Innovation parameter
q		Imitation parameter
W or W		Watt
h		hour
k_g		Kilogram
g		gram
F		Farad
G		Giga or 10^9
V		Volt or Volts
M		Mole
d		Distance between atomic-layers
R		Resistance
X		Reactance
i		Imaginary constant
Z		Complex impedance
L		Lakhs or 10^5
nm		Nanometer or 10^{-9}
I_d		Signal intensity in the defect-related region
I_g		Signal intensity in the non-defective region
L_d		distance in nm between the said defects
s		Second or Seconds
n_d		Density of Faults
c_p		Specific Capacitance
R_{ct}		Charge Transfer Resistance

P_g		Power Density
R_s		Series Resistance
W		Warburg Resistance
P		Power in Watts
E		Energy in Joules
J		Joule or Joules
Hz		Hertz

Symbols		
$> \text{ or } +$		More than
$\$$		Doller sign
$\%$		Percentage sign
₹		Indian Rupees
μ		Micro or 10^{-6}
$\text{\AA}$		Angstrom or 0.1 nanometer
$^{\circ}\text{C}$		Degree Celsius
θ		Theta (angle between incident-ray and the scattering-planes)
λ		Wavelength in nanometers
Δt		Discharge Time
Δv		Voltage Window
Ω		ohms

CHAPTER 1.

INTRODUCTION

In later a long time, the worldwide energy space has experienced gigantic alteration, moving toward more feasible and efficient energy sources. This is spurred by the need to fulfil the rising demand, lower carbon emissions, and extend the energy storage capacity. In this regard, battery-supercapacitor hybrid energy storage systems (BSHESS) and supercapacitors have become appealing innovations (Habib & Sou, 2018). These could completely transform power supply and energy storage systems. The business prospects offered by energy solutions based on applications of supercapacitors are examined in this thesis. The market dynamics, technological developments, and strategic approaches that can propel the acceptance and commercialization of these across a range of industries are examined in this book. The financial possibility, competitive environment, and advancement potential of supercapacitor innovations are inspected in this paper. It looks to shed light on modern commerce procedures and prospects in vitality arrangements based on these solutions. Viable client association and a thinking approach to advertising passage are fundamental for the fruitful commercialization of this innovation. As a result, a strong business strategy can help young companies to expand.

For strategically developing a business model, the objectives of this work include, (a) developing a hypothesis for a beachhead market, (b) ensuring value proposition, (c) validating this hypothesis by field studies/surveys/interviews, (d) finding the targeted customer and beachhead market, (e) finding the desired performance parameters for

targeted value-proposition, (f) developing an experimental setup to test these performance parameters, and (g) matching the value proposition with the experimental outcomes. Finally, these outcomes will establish a business model for the supercapacitors and BSHESS market. The scope of the thesis, its objectives, and supporting literature review are done to gain strong technical and business insights into existing technologies and existing markets (Dhanraj Nehate et al., 2019). This chapter describes the need for this thesis and how it will benefit society by employing BSHESS and supercapacitors to provide practical technological and commercial solutions. More specifically, this chapter explains (a) the background of the research, (b) the motivation of the research, (c) the scope of this thesis, (d) the objective of the thesis, and (e) the organization of the thesis in detail.

1.1 BACKGROUND

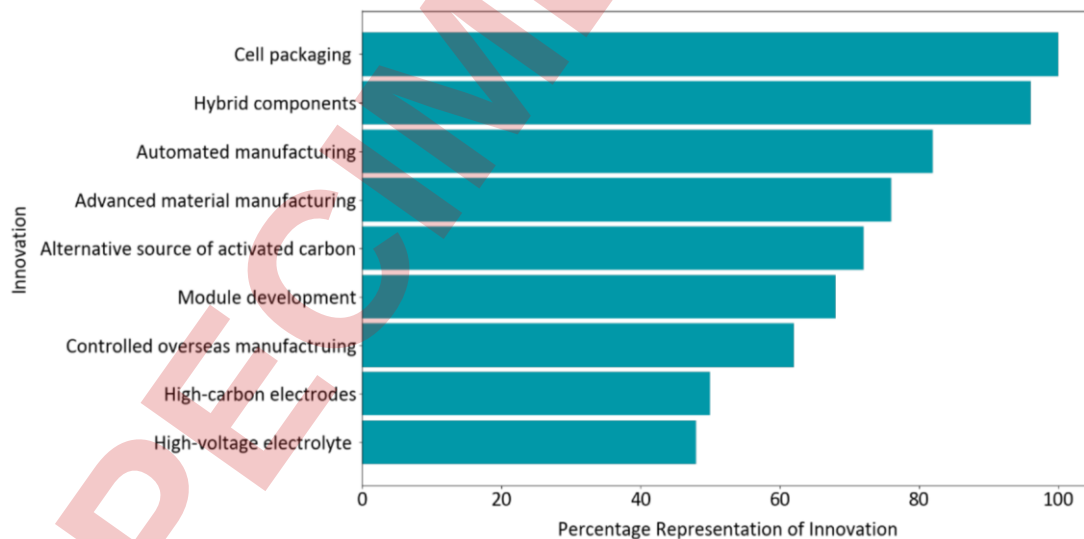


Figure 1.0.1 Representation of top 10% innovations in supercapacitor-enabled applications. This is to understand the efforts for cost reduction for adoption in applications (Durvasulu et al., 2023)

The distinctive features of supercapacitors as an appropriate alternative are highlighted in this background research, which explores the technological and financial elements of contemporary energy storage technologies. The increasing demand for energy storage technologies, particularly in the integration of renewable energy, highlights the value of supercapacitors and BSHESS (S. Wang et al., 2008). It moreover looks at the necessity for productive energy storage systems and how industry developments are influencing the selection of these innovations. The major discussion points for the background study are explained in the next subsections.

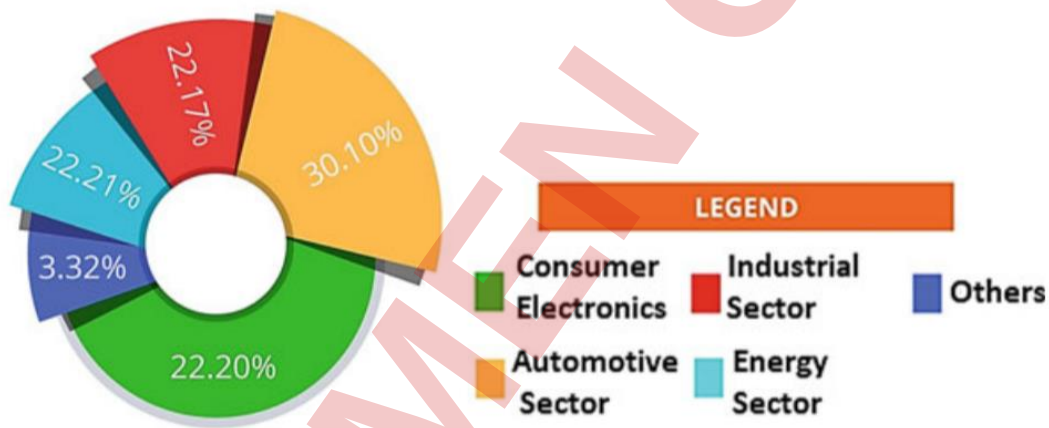


Figure 1.0.2 The market share of supercapacitors in different domains (Lasrado Dylan and Ahankari, 2021)

1.1.1 Financial background

In 2021, the global supercapacitor market was estimated to be worth \$1.5 billion. It is projected to increase at an average yearly growth rate of 30% between 2021 and 2030, reaching a \$15 billion market. The primary driver for this growth is the automotive, portable energy units in the renewable energy industry and consumer electronics sectors, which currently make up 32% and 30% market share, respectively, followed

by the energy sector at 21% market share. The Indian supercapacitor market is anticipated to grow at a robust CAGR due to increasing sales of electronic products and automobiles (Durvasulu et al., 2023). In contrast, the Indian supercapacitor sector is still in its infancy but is showing promise, with only a few companies like as Godi India and Keltron (Aman Mohd, 2024). However, the limited use of supercapacitors as long-term energy storage solutions poses a challenge for the Indian market. Key drivers for the Indian market include the use of supercapacitors in renewable energy systems, electric vehicles and the electronics industry (Office of the PSA to the GOI, 2014).

1.1.2 Looks alike analysis

Hybrid EV is a look-alike so far as it is related to decarbonisation efforts. It is not so, in terms of the fact that it's a specific mobility product, while this work offers an energy storage product. The plot shows that imitation is more than the innovation coefficient and sales will increase in the future. The potential market size (m) is estimated to be 11,555,000 units. Innovation parameter p is found as 0.008. This implies that the product will gain its peak sales in the near future if our product is connected to the mobility segment and immediate adopters will accept the product. Imitation parameter q is found as 0.143. This implies the product has penetration into the market via word-of-mouth. $p > q$ implies a successful product and scope of further advertisement for more sales.

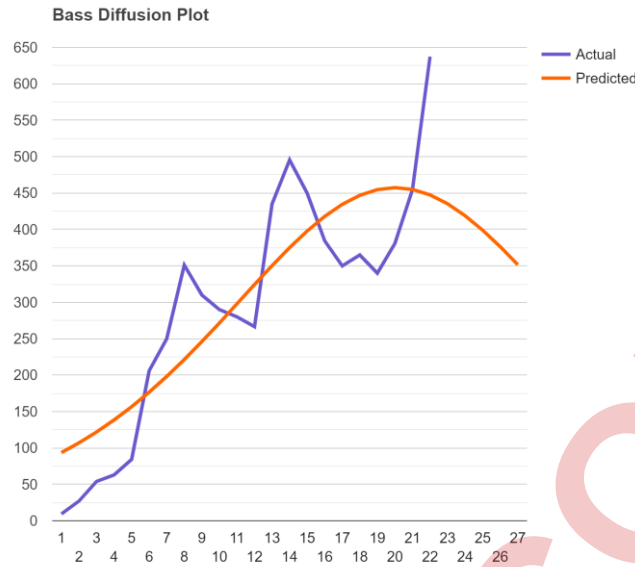


Figure 1.0.3 Calculated innovation coefficient for a near-matching product based on the data collected from (Davis & Boundy, n.d.)

1.1.3 Technical background

Because of supercapacitors' unique qualities, they are well-suited for applications such as power burst requirements, portable electronics, renewable energy systems, and electric cars. This is also due to their unique advantages over traditional chemical batteries. These benefits include high power density, fast charging/discharging, and long cycle life compared to batteries (Dhanraj Nehate et al., 2019). When combined in a hybrid system, the supercapacitor can handle the high power demands while the battery provides the high energy density. Some key benefits of using BSHESS include (a) improved power and energy density compared to using just batteries or supercapacitors alone, (b) ability to handle rapid charge/discharge cycles and handling high power pulses when needed, (c) enhanced reliability and performance for applications like electric vehicles, renewable energy integration, and micro-grids. Also, the hybrid system requires careful control and energy management strategies to

optimize the complementary characteristics of the battery and supercapacitor (Jayawickrama & Rajakaruna, 2004).

Hybridizing supercapacitors with battery systems enhances thermodynamic efficiency and rate capabilities, addressing irregular energy consumption from batteries (Jayawickrama & Rajakaruna, 2004). This technology also can benefit applications like consumer electronics, microgrids, renewable energy, and portable electric appliances (Latha et al., 2007). For such purposes, researchers are exploring the design of native BSHESS systems. They focus on novel electrode materials and coating methodologies to enhance supercapacitor commercialization (Kathi & Kadali, 2022). For example, researchers at the Indian Institute of Science developed a novel ultra-micro supercapacitor using field-effect transistors as charge collectors. Medical equipment, electric vehicles, and consumer electronics can all benefit from this small energy storage solution (Panwar et al., 2023). The wide range and efficient operation of hybrid supercapacitors are demonstrated by their possible use in a variety of energy storage systems.

1.2 LITERATURE REVIEW

This literature review examines many affordable materials, such as carbon-based compounds, transition metal oxides, and conductive polymers to improve performance and lower production costs. It also looks at the Industry 4.0-influenced fabrication processes. These include the synthesis of electrode materials, fabrication techniques, preparation of electrolytes, and device assembly. This research also looks at the legislative frameworks and market conditions that are currently affecting energy

storage systems. It also looks at how current business paradigms are influencing the future in a quickly changing technological landscape. The subsequent extensive investigation of the topic is intended to bring attention to novel research achievements and future development prospects as well as existing obstacles in the context of the sector to create an understanding of energy integration solutions.

1.2.1 Technology opportunities and gaps for supercapacitors and BSHESS

Supercapacitors visibility has been effectively demonstrated in sectors such as consumer electronics, automotive, renewable energy, and the IoT. Businesses can create more reliable, sustainable, and functional goods by looking into creative designs and increasing the utility of supercapacitor-based solutions (N. Kumar et al., 2022). It can further attract investments and create new commercial opportunities. Additionally, the increasing need for sustainable solutions supports this kind of innovation. These technological innovations must satisfy the changing demands of companies and consumers and follow global trends toward environmental responsibility and clean energy (Durvasulu et al., 2023). Going deeper into the world of current opportunities, the following part identifies the primary demand drivers and gaps for the currently available opportunities.

1.2.1.1 Demand drivers for opportunities:

A. Demand of high power output and rapid charge/discharge:

High power output and fast charge/discharge capabilities are becoming increasingly crucial, particularly in the variety of applications. The interest in them has increased

due to performance difficulties with traditional lithium-ion batteries. Because of this, supercapacitors are perfect for usage in the renewable energy and high-efficiency hybrid electric vehicle sectors. The lack of adequate charging infrastructure, especially in many Asian regions, further highlights this need (Office of the PSA to the GOI, 2014).

B. Demand of nature-friendly materials:

The shift to environmentally friendly solutions also promotes research into sustainable materials and supercapacitor production methods that match global sustainability goals (Durvasulu et al., 2023; Kausar et al., 2023). This is a current global need (N. Kumar et al., 2022). These efforts have led to the development of modified electrode materials created through photosynthesis. This kind of process utilizes plant extracts as reducing agents and catalysts to produce nano-energy materials with desirable properties (Meena et al., 2024). For supercapacitor applications, phytosynthesized nanomaterials and composites can show higher capacitance and energy density, among other features. The environmental impact of supercapacitors is further reduced by the fact that their main constituents, metal oxides and activated carbon, are easily recyclable or reusable.

C. Demand for innovation for business opportunities:

Furthermore, the market for supercapacitors will provide opportunities for innovation as businesses look into new materials and hybrid systems that can improve effectiveness while lowering the drawbacks of conventional energy storage techniques (Office of the PSA to the GOI, 2014). A dynamic market for energy storage technologies across

several industries is defined by these demand factors. Some businesses have been known to employ the use of supercapacitors in combination with batteries and other forms of energy storage (Kathi & Kadali, 2022). To that end, hybrid solutions are developed to cover the technical gaps and adapt to needs specific to different industries.

D. Import substitution for materials and technology

India has developed supercapacitors using locally available materials, reducing the country's reliance on imported technologies (Panwar et al., 2023; Vijayakumar et al., 2018). For this reason, enhancing indigenous supercapacitor technology is a critical part of India's Atma Nirbhar Bharat (self-reliant India) mission to reduce imports in strategic fields. The supercapacitor industry in India has tremendous growth potential, even if it is still in its infancy. Researchers and manufacturers in India have been improving the performance and cost-effectiveness of domestically produced supercapacitors to make them competitive with imported alternatives (Office of the PSA to the GOI, 2014). These gadgets can be integrated into consumer electronics, electric vehicles and different renewable energy systems (Durvasulu et al., 2023).

1.2.1.2 Gaps

A. Tackling the issue of lower energy density and high production cost:

Supercapacitor's energy storage capacity, often lower than that of traditional batteries, limits their use in applications requiring sustained energy delivery over longer periods. The major goals of research for supercapacitors are to store more energy density yet

maintain all the beneficial characteristics, making it a vital development area in energy storage technology (N. Kumar et al., 2022). To overcome this, the solutions are being considered as high-surface-area electrodes and improved electrolytes and fabrication process (Pathak et al., 2024).

High surface area, carbon-rich materials are used to create efficient supercapacitor electrodes, which enable quick cycles of charge and discharge and effective charge storage (Yang et al., 2023). Metal-oxides provide a higher energy density due to their electrochemically active activity, and the use of aqueous electrolytes enhances safety because they are non-flammable and non-toxic (An et al., 2019). Another tactic to increase energy density while having low production costs and security is the fabrication of metal oxide-based asymmetric supercapacitor devices.

New resources have been explored, such as saltwater and waste materials from the cotton industry. These are cost-effective and abundant. The total cost of production can be significantly decreased by recycling waste materials and making use of an abundance of natural resources (Challagulla et al., 2020; Vijayakumar et al., 2018). Low-density aerogels derived from carbon-rich materials have also shown potential in enhancing the capacitance and performance of supercapacitor electrodes. This approach of hybrid technology integration improves sustainability while lowering production costs (Zuo et al., 2015). Further, additional operational challenges, such as the control algorithm, can help reduce costs and be solved using proper electronic control techniques (P. Kumar et al., 2021a).

B. Need of control algorithms, state of charge management and fault detection in hybrid integration:

Intelligent control algorithms restrict overcharging, over-discharging, and heating problems as a measure that may negatively affect the system performance. They balance the energy flow between the two storage units and improve the effectiveness of the system. Typically, this is achieved through active regulation of the fastening and unfastening of the battery and the supercap. By changing the values of the charge and discharge rates during operating, commensurate with the energy needs during operation, this situation prolongs their life cycle (Xiong et al., 2017). One popular example is droop control scheme (Papageorgiou et al., 2023).

Advanced control algorithms govern the SOC of each element in hybrid energy storage so that it does not allow any deterioration, enables efficient use of energy, and provides acceptable levels of performance throughout the system's life cycle. They also improve the system's life span by ensuring optimal energy distribution between the two storage devices (Chen & Zhang, 2021). It is equally important to detect and neutralize battery or supercapacitor component faults to avert system collapse and improve functional safety (Torregrossa et al., 2015). Adeyinka, (Adeyinka et al., 2024) explains that further significant aspect of successful integration and scaling of any hybrid energy storage systems is connected with supercapacitors' realistic electrical parameter models.

C. Need for scalability, cost-effectiveness and modular design of hybrid systems

As the system grows, space, complexity and costs also increases. Moreover, the power electronics needed to control energy flow between parts can become complicated and pricey. It is needed to create energy storage solutions that can scale by using modular designs, which allow the addition of different energy storage units (Díaz-González et al., 2012). It is also required to design advanced control systems to handle the energy flow between parts, which helps keep everything running efficiently and reliably (Chen & Zhang, 2021).

Costly materials like ruthenium oxide (RuO_2) has a high theoretical capacitance and good electrochemical performance. However, due to its rarity and high cost, it is not a good choice for large-scale, affordable supercapacitor production (Díaz-González et al., 2012; Majumdar, 2018). Researchers are developing more affordable, more scalable alternatives to this kind of exotic material for use as scalable supercapacitor electrodes. For instance, activated carbon is more cost-effective and scalable than other exotic materials such as RuO_2 . (Challagulla et al., 2020; Vijayakumar et al., 2018). It can be synthesized in tones for little cost from carbonaceous precursors, from petroleum polymers to metal organic frameworks or mxenes. (Yang et al., 2023), (Majumdar, 2018).

D. Temperature and safety-related issues with batteries

Supercapacitors can function in a larger temperature range—between -40°C and 85°C —without experiencing appreciable performance loss. Conversely, batteries – such as lithium-ion batteries – typically work only in the narrow temperature range of -20°C to 40°C ; extreme temperatures can drastically affect performance. In addition,

supercapacitors lower the chance of explosions and flames because they are non-flammable. A typical problem with lithium-ion batteries is thermal runaway. Additionally, overcharging can harm batteries and raise safety issues, which is why supercapacitors are chosen in hybrid-mode. Supercapacitors are also designed to prevent short circuits, which are a frequent source of fires and other safety risks. They don't have the potential to leak, which could cause problems for the environment and public safety. (Du Pasquier et al., 2003; S. Wang et al., 2008).

E. Issues with batteries in consumer electronics (flexible and wearable applications)

Existing battery technologies like lithium-ion have some of the highest energy densities (100–265 Whkg⁻¹ but they still cannot meet the requirements of the wearable market. As wearables require miniaturization and are lightweight, the limited space for batteries is a major constraint. Also, conventional batteries have rigid prismatic or cylindrical shapes with rigid encapsulations, making them difficult to use in flexible and stretchable wearable electronics. The rigid battery components can peel away from each other during deformation, leading to performance degradation or failure or safety hazard (L. Wang et al., 2023). Supercapacitors do not pose a risk of thermal runaway or fire hazards, they are intrinsically safer than lithium-ion batteries, enabling user safety in wearable applications (Du Pasquier et al., 2003). Supercapacitors can be made to fit into a variety of shapes and sizes by integrating them onto flexible substrates. They are therefore perfect for wearable technology, such as fitness trackers and smart clothes. Their longevity reduces maintenance and replacement costs significantly (Teymoory et al., 2023).

F. Issues with standardization and testing

Super-capacitors performance evaluation is still empirical and this has necessitated formulation of testing protocols to enable reproducible results to be generated. Super capacitors capacitance distributions and ESR are critical to performance estimation which has not yet been modeled accurately. The preparation of guidelines and standards for performance assessment of supercapacitor is vital in order to obtain consistent and replicable results. Higher level modeling techniques, numerous simulations and affecting variables like temperature, voltage, and aging can all be used to achieve this (Zhang & Pan, 2015). Developing automated testing systems with precise control over test conditions and data acquisition can help minimize human errors and improve the repeatability of test results. Supercapacitors can be made more dependable and widely used by establishing industry standards and uniform testing. Hence there is a necessity to develop testing hardware as well. Various popular tests for BSHESS, such as charge and discharge profiles, load profile capture etc., are being developed to create a set of testing procedures (Gómez et al., 2023).

1.2.2 Materials

1.2.2.1 Carbon as a cost-effective materials:

A. Design variations for carbon based electrodes

Activated carbon may be economically manufactured from a variety of precursors, including mxenes, metal-organic frameworks, and polymers. It has been demonstrated that using low-density aerogels made from carbon-rich materials can improve the

capacitance and functionality of supercapacitor electrodes. Researchers are also investigating the use of natural resources like saltwater and industrial waste materials like cotton-based electrodes as affordable substitutes in the creation of supercapacitors (Challagulla et al., 2020; Vijayakumar et al., 2018).

These materials are generally converted into porous carbon by such common techniques as activation, hydrothermal treatment, and pyrolysis. A sustainable and effective method for producing activated carbon from different biomasses results in a variety of surface textures and morphologies that improve supercapacitor performance (Aparna et al., 2022; Forouzandeh et al., 2020a). The addition of heteroatoms like oxygen, nitrogen, and sulfur in activated carbon improves a supercapacitor's performance by promoting ion transport, conductivity, and wettability. By creating ion bilayers at the electrode-electrolyte interfaces, activated carbon's hierarchical pore structure and morphology improve capacitance and energy storage properties (Senthilkumar et al., n.d.). With this much flexibility for material-design, they are perfect for industrial applications, portable electronics, and emergency energy systems (Yang et al., 2023).

B. Alternatives to carbon and their comparison

Graphene, a single sheet of carbon atoms organized in a two-dimensional honeycomb lattice, has remarkable surface area, mechanical strength, and electrical conductivity. Carbon nanotube (CNT) and graphitic nanofiber (GNF) nanocomposites are being investigated as supercapacitor electrode materials because they provide great cycling stability and particular energy values. Because of their special electrical and structural characteristics, carbon nanotubes (CNTs) are attractive options for supercapacitor

electrodes. High conductivity, mechanical flexibility, and superior electrochemical stability are characteristics of CNT-based electrodes that improve energy storage performance (Aparna et al., 2022; Forouzandeh et al., 2020a).

Research indicates that the performance and stability of supercapacitors are much improved when activated carbon is combined with graphene or carbon nanotubes. When compared to other materials like graphene, activated carbon is more affordable, which makes it a better choice for large-scale commercial applications. Because of its strict quality criteria and the specialized equipment required for its synthesis, graphene is more expensive to create than activated carbon. It is challenging for graphene-based supercapacitors to achieve economies of scale because of this complexity. According to the sources, activated carbon is therefore a more economical choice for supercapacitor technologies than graphene (Forouzandeh et al., 2020a). Transition metal oxides (TMOs) are another well-liked option because of their high specific capacitance characteristics, which are covered in the following paragraph.

1.2.2.2 Transition metal oxides (TMOs):

RuO_2 , CoO_4 , and MnO_2 are examples of transition metal oxides (TMOs) that are being researched as supercapacitor electrode materials. ZnO , AMoO_4 ($A = \text{Co, Mn, Ni, Zn}$), and XCo_2O_4 ($X = \text{Mn, Cu, Ni}$) are further TMOs. These TMOs are appropriate for energy storage applications because of their large theoretical capacitance, quick Faraday redox reactions, strong electrical conductivity, and chemical stability. Other TMOs such as Co_3O_4 , MnO_2 , and ZnO are more affordable and plentiful than RuO_2 , and they can be used as alternatives to more costly materials. Despite having a high

theoretical capacitance and quick redox reactions, TMOs are not widely used in supercapacitors due to their high cost and environmental toxicity. (Forouzandeh et al., 2020a; Liang et al., 2021). Research focuses on methods like nano-structuring materials, modifying quantum dots, and forming composites for improvements in electrochemical performance, stability, and cost-effectiveness (Liang et al., 2021). But still, AC continues to be the best choice for a scalable, economical, and time-efficient approach.

1.2.2.3 Conductive polymers:

Conductive polymers can also be used as supercapacitor electrode materials, according to the literature (Zuo et al., 2015). While the specific cost-effectiveness of conductive polymers is not discussed, they can potentially be a more affordable alternative to some traditional electrode materials. These polymers have several benefits, including chemical stability, quick faradaic redox reactions, high specific capacitance, high charge storage capacity, flexibility, and superior electrical conductivity. Applications for conductive polymers in supercapacitors include asymmetric arrangements with n-type and p-type doped electrodes. Higher output power and charge storage capacity result from this method's expansion of the voltage window between the two electrodes. The primary problem with conductive polymers, though, is their instability, particularly when they are utilized as negative electrodes or when conductive polymers are used for both the positive and negative electrodes. The breakdown of polythiophene derivatives is the cause of this instability. Their instability is still a problem, though, and research is being done to improve their functionality and deal with this constraint (Forouzandeh et al., 2020a; Raza et al., 2018).

1.2.2.4 Comparison:

The cost of producing conductive polymers and transition metal oxides (TMOs) is usually higher than that of activated carbon. In terms of stability, AC is most stable with millions of cycles and poses exceptional electrochemical stability. Other TMOs have the largest theoretical specific capacitance, but because of their weak electronic conductivity, they have a lower real capacitance (Aparna et al., 2022; Raza et al., 2018). Conductive polymers can provide high specific capacitance due to their pseudocapacitive behavior but their stability may be a concern. TMOs provide high specific capacitance but suffer from poor stability and higher costs (Liang et al., 2021).

By adjusting the pore structure, activated carbon can be designed to have extraordinarily high specific surface areas (up to 3000 m²/g). Optimizing the surface chemistry and pore structure can improve AC's performance (Forouzandeh et al., 2020a; Raza et al., 2018). TMOs and conductive polymers generally have lower specific surface areas than activated carbon. Activated carbon offers the best overall performance, balancing cost-effectiveness, stability, and high surface area. Furthermore, the production of high-performance, economically feasible energy storage supercapacitors is made possible by the simplicity of process of fabrication (Forouzandeh et al., 2020a).

1.2.3 Supercapacitor fabrication and Industry 4.0 manufacturing

1.2.3.1 Electrode material synthesis:

Supercapacitor electrode materials have been synthesized using a variety of methods. Chemical vapour deposition (CVD), hydrothermal synthesis, and electrospinning are common techniques for material synthesis. Additional approaches include ammonothermal, electrodeposition, thermal reduction, solid-state (SS) synthesis, and soft-urea procedures. Every approach has a unique set of difficulties. These include assuring the necessary qualities of the electrode material, controlling impurities, managing reaction parameters precisely, and scalability. For example CVD requires good control of pH, temperature and reaction time while in hydrothermal synthesis temperature, pressure and gas flow rate must be well controlled. Whereas, for thermal reduction the conditions such as temperature and reduction atmosphere needs to be maintained and altered in the case of electrospinning the properties of the precursor solution needs to be modified. Electrodeposition requires changes to be made to the composition of the electrolyte and the deposition environment, while solid-state synthesis requires the maintenance of the continuity of mixing and contact between the precursors. Ammonothermal and soft-urea methods require exact control over the reaction's temperature, pressure, and duration. These challenges must be overcome in order to develop high-performance supercapacitor electrodes using these synthesis methods (Akhtar et al., 2023; Yaseen et al., 2021).

Among these synthesis methods, CVD has been used in combination with other methods like hydrothermal synthesis to prepare electrode materials. For example,

MnO₂ interlinked nanowires were synthesized on carbon fabric using a hydrothermal electrodeposition method incorporated with CVD. The combination of CVD with hydrothermal synthesis has also been used to prepare 3D graphene/MnO₂ foam with enhanced supercapacitor performance. One of the primary synthesis techniques for supercapacitor electrode material is presented in this study: hydrothermal synthesis. By altering the product's crystallization during hydrothermal synthesis, the cooling rate can be controlled to improve the electrode material's electrochemical performance (Akhtar et al., 2023; R. Wang et al., 2017). Electrode materials with a specific capacitance of 400 F/g have been created via thermal reduction and oxidation with metal-organic frameworks (MOFs) (J. Liu et al., 2022). Additionally, MnO₂ electrode materials with a specific capacitance of 448 F/g have been produced using solid-state (SS) synthesis (Ren et al., 2022). MnO₂ electrodes with a specific capacitance of 556 F/g have been prepared by chemical bath deposition and electrodeposition (Y. Liu, Zeng, et al., 2017). Numerous electrode materials have been successfully synthesized using these synthetic procedures. They have high specific capacitance values that make them appropriate for use in supercapacitors.

The desired structure and material properties are among the factors that must be considered while selecting the best synthesis procedure for supercapacitor applications. The surface area and porosity of activated carbon, which are critical for supercapacitor electrodes, are commonly increased by chemical activation. In order to manage morphology and crystallinity for improved electrochemical performance, transition metal oxides (TMOs), such as manganese oxide (MnO₂), are frequently manufactured via hydrothermal techniques. To customize film thickness and shape, electrochemical polymerization is commonly used to create conductive polymers like polypyrrole

(PPY) and poly (3,4-ethylenedioxythiophene) (PEDOT). In this case, the researchers have the ability to enhance these attributes of supercapacitors through a synthesis method fit for the material of the device. This then translates to improved performance in energy storage applications as the manufacturing of the electrodes is enhanced (R. Wang et al., 2017; Yaseen et al., 2021).

1.2.3.2 Electrode fabrication methods:

The choice of coating method for supercapacitor electrode fabrication depends on factors like scalability, uniformity, and material compatibility. Spray-coating allows for uniform and conformal coatings on large-area substrates, while electro-spinning produces highly porous and interconnected nanofiber networks with high surface area. Both of which are ideal for supercapacitor electrodes. Drop-casting is a simple and cost-effective method suitable for customised fabrication and material screening. The doctor-blade method enables the deposition of thick, uniform coatings on large-area substrates commonly used for commercial supercapacitors. Brush coating is suitable for customised and irregular-shaped substrates, though it requires manual skill and may result in non-uniform coatings (Forouzandeh et al., 2020b; Vandeginste, 2022). The selection of the appropriate coating technique depends on the desired electrode structure, electrolyte preparation, material properties, and production scale requirements for optimizing supercapacitor performance.

1.2.3.3 Electrolyte preparation:

Solvent casting, ion gel synthesis, and additional techniques like electrochemical polymerization, thermal polymerization, and radiation-induced polymerization are used in the production of electrolytes for supercapacitors. To create a solid polymer electrolyte film, a polymer and salt are dissolved in a solvent and then cast onto a substrate. Ion-gel synthesis blends the mechanical stability of polymers with the strong ionic conductivity of ionic liquids. With chemical or physical cross-linking, it produces solid-state and flexible electrolytes. Other methods, such as electrochemical, thermal, and radiation-induced polymerization, allow for the in-situ formation of conductive polymer electrolytes or cross-linked gel polymer electrolytes. The choice of synthesis method depends on the desired electrolyte properties. It also considers compatibility with electrode materials and the target application for the supercapacitor. In this way, development of the electrolyte solutions can now be done in accordance to a particular sample (González-Gil et al., 2022; Hong et al., 2020).

There is a high dependency on the compatibility of both the electrode materials and the electrolyte to develop customized solutions of electrolytes. The energy storage device's stability and performance are maximized by this compatibility. The physical and chemical interactions between the electrode materials and the electrolyte must be considered during the fabrication process. For instance, compatibility with the electrolyte may influence the selection of binders and conductive additives. Performance can occasionally be enhanced by pre-treating the electrolyte before building the electrode-electrolyte membrane. The ohmic resistance can be considerably decreased in alkaline anion exchange membrane water electrolysis (AEMWE) by pre-

feeding the electrolyte to the membrane electrode assembly (MEA). By reducing resistance by up to 94%, this technique can increase conductivity (Xing et al., 2022).

Ionic conductivity and electrochemical performance are significantly impacted by the electrode's wettability by the electrolyte. The wettability may be affected by the electrode fabrication procedure, including calendaring. Under specific processing circumstances, certain electrolytes may break down or react with the electrode materials. Effective ion transport depends on the electrolyte's capacity to penetrate the porous electrode structure. The pore structure and electrolyte infiltration may be affected by the electrode fabrication procedure, including the mixing and casting stages (Abdollahifar et al., 2023). During the electrode manufacturing process and assembling of the device, the chemical and electrochemical stability of the electrolyte must be taken into account.

1.2.3.4 Device assembly:

Usually, a separator is positioned between two electrodes composed of bio-derived carbon materials to create supercapacitors. The assembly is immersed in an electrolyte solution with electrodes that have been coated with a current collector. Laser patterning and roll-to-roll (R2R) manufacturing are two essential methods for the flexible and scalable production of supercapacitor devices. R2R manufacturing makes it possible to produce flexible, large-area supercapacitors continuously. It deposits electrode materials onto flexible substrates to achieve cheap costs and high throughput. Among the methods are inkjet printing, gravure printing, and slot-die coating. Supercapacitors for uses including flexible displays, wearable electronics, and energy harvesting

devices can be produced using this scalable method (Lokhande et al., 2019; Vandeginste, 2022).

By enabling complex designs on electrodes, laser patterning improves R2R manufacturing. By producing interdigitated electrode architectures, this technique raises supercapacitors' power and energy density. Electrode materials can be altered or removed selectively using this method. It works well with conductive polymers, metal oxides, and carbon-based materials. It also makes direct writing on pliable substrates possible. Fabricating wearable and on-chip supercapacitors benefits from this capability (Kim et al., 2024). Other techniques for fabricating devices include direct growth, vacuum filtering, inkjet printing, and screen printing. These techniques advance the creation of high-performing supercapacitor devices. Each method has special benefits in terms of accuracy, material compatibility, and cost-effectiveness. The manufacture of scalable supercapacitors is thus made possible (Lokhande et al., 2019; Vandeginste, 2022). Advanced manufacturing methods combined with various electrode material synthesis techniques ensure the development of supercapacitors. This supports Battery-Supercapacitor Hybrid Energy Storage Systems (BSHESS) that meet growing energy storage demands across diverse applications.

1.2.3.5 Fabrication of BSHESS

The advantages of both technologies are combined in batteries and supercapacitors hybrid energy storage systems (BSHESS). They are suggested for a number of uses, such as smart grids, electric cars, and the integration of renewable energy. BSHESS can be configured in a number of ways to optimize performance, each having unique

advantages and disadvantages. As can be seen in the figure, the battery and supercapacitor are connected in series with the loading system. During acceleration, high power pulses are derived from the supercapacitor as mentioned earlier in this manuscript. The primary energy source is battery. Furthermore, this arrangement reduces pressure on batteries and enhances the production processes' performance. Whereas the battery and the supercapacitor are connected in parallel to the load to, thereby both supply the energy required. A work of supercapacitor deals with high power pulse, meanwhile battery supplies most of the energy. This simpler configuration requires more complex control strategies to distribute power between the two storage devices. A semi-active BSHESS connects the battery or supercapacitor to the load via a DC-DC converter. Independent control and ideal power management are made possible by this configuration. However, it is less efficient and more complicated. The battery and supercapacitor are connected to the load via a fully operational BSHESS. Separate DC-DC converters are used to accomplish this. Maximum power management flexibility is provided by this configuration. It is the most expensive and difficult, though. Cost, complexity, efficiency, and particular application needs are some of the elements that influence the configuration decision. This enables customized BSHESS integration to satisfy a range of energy storage requirements during regular and peak load periods (Kouchachvili et al., 2018).

Energy from renewable sources, such solar panels, or during off-peak hours can be stored via BSHESS. When demand is high or renewable energy is not available, it then releases this energy to heat water (S. Wang et al., 2008). Supercapacitors' high power density and quick reaction can also be used to manage abrupt changes in load and offer auxiliary services to the grid (Xiong et al., 2017). Optimizing BSHESS performance

and efficiency requires effective control mechanisms. In DC microgrids, droop-control techniques have been put forth to reduce bus voltage variations and accomplish battery state-of-charge (SOC) balancing (Papageorgiou et al., 2023). To improve BSHESS performance, Wirtinger-based integral inequality-based optimal control algorithms have also been created. Integrating of these with renewable energy sources like solar or wind can help smooth power fluctuations and improve overall system efficiency (Zhong et al., 2021). For example, Duke Energy collaborated with Aquion Energy and Maxwell Technologies. They built a Battery-Supercapacitor Hybrid Energy Storage System (BSHESS). This system uses supercapacitors for real-time solar smoothing. Batteries are utilized for load shifting (Kathi & Kadali, 2022). These companies are giants of the supercapacitors market following the recent Industry 4.0 revolution.

1.2.4 Market landscape and business practices and industry 4.0 trends:

1.2.4.1 Current market trend and sectors

The current market trends as explained in the section 1.1.1 Financial background. Moreover, It is in the Asia Pacific region, especially in China and India, where the growth and expansion of the market is likely going to be much boosted. These include increase in electronics sector and the growth of automotive industry in many countries as a cause of this increase. These applications include smart power grids, renewable energy systems, hybrid electrical cars, laptop computers, and GPS. However, there may be some obstacles to industry expansion due to the high initial costs of materials and a lack of understanding about supercapacitors. The market for supercapacitors is divided into three segments: type, end-use, and geography.

Leading manufacturers include Maxwell Technologies, CAP-XX, KEMET Corporation, Eaton, Murata Manufacturing Co. Ltd., AVX Corporation, Nippon Chemi-On, and Panasonic Corporation. These companies operate in a dynamic and rapidly evolving market landscape. The automotive segment will be one of the biggest potential applications for supercapacitors in HEVs, PHEVs, and electric buses. Supercapacitors are utilized in the electric vehicle (EV) industry primarily for regenerative braking, start-stop systems, and power assist features, which enhance vehicle performance and energy economy. In contrast, other rapidly expanding industries for supercapacitors and BSHESS are consumer electronics and renewable energy (Lasrado Dylan and Ahankari, 2021). The application where supercapacitors are most commonly used today is the renewable energy industry. They are widely used in the reserve systems of wind and solar electric facilities. Owing to relatively high power density and short charging time, supercapacitors find their application in portable consumer electronics such as smart wearables (Office of the PSA to the GOI, 2014).

Some notable case studies of use of applications:

In order to capitalize on the high power density, rapid charge/discharge capability, and energy recovery from braking system applications of hybrid electric vehicles, such as the Lamborghini Sián and the Toyota Yaris-R, which were developed in partnership with the Massachusetts Institute of Technology, hybrid gasoline supercapacitor powertrains have been used in these vehicles. Supercapacitors have been used for ten years in the transportation industry in Europe, particularly in Germany, for their low-floor trams. In addition to reducing the requirement for overhead lines, these trams' quick recharging at each stop and lack of overhead lines increase efficiency by 10% to

25% (CAPTOP S.R.L., n.d.). The addition of a combination of flywheels and a supercapacitor module to the lead-acid battery storage installed in a microgrid on the Scottish Isle of Eigg has improved the life and reduced maintenance of the lead-acid battery storage system. This energy storage system helped with frequency control for smooth grid operation and helped Eigg (Durvasulu et al., 2023). A few more cases are also explained in section 6.3.

1.2.4.2 Industry 4.0 practices

The integration of technology across numerous industries is the focus of Industry 4.0, often known as the fourth industrial revolution. Such a revolution is significantly applicable in the manufacturing of energy modules and supercapacitors, where advanced technology brings better products, high-yield, and cost reduction. All of this is being done using sophisticated techniques that make use of BIG data, data mining, and machine learning for design considerations, optimization, affordability, and domestication (Mouchou et al., 2021). The creation of completely automated production lines is simplifying the manufacturing of supercapacitors. Another example is Skeleton Technologies, which is constructing the first fully automated supercapacitor production line in collaboration with Siemens Digital Industries.

This cutting-edge production line aims to increase production capacity by 40 times. It significantly reduces production costs and increases the competitiveness of supercapacitors. Skeleton Technologies plans to implement fully automated ultracapacitor production technologies in its facilities, which would significantly reduce manufacturing costs by approximately 90% over five years (Skeleton, 2021). Adoption

of advanced technologies usually requires a significant upfront financial investment, which may be unaffordable for smaller firms. Market expansion may be hampered by potential consumers' general lack of knowledge about supercapacitors and their benefits. The problem of the high cost of advanced materials used in supercapacitors still requires research and development (Lasrado Dylan and Ahankari, 2021; Office of the PSA to the GOI, 2014)

Interest in the growth and development of the supercapacitor industry and its applications could be therefore catalyzed by governments through the introduction of several regulations and or initiatives. One of the measures includes offering of non-cash incentives such as grants and tax credits to reduce the cost of the R & D as well as the production process. The supercapacitor technology lifespan can be increased and the cost of building them be decreased through funding for the chosen research project. Besides, it helps in the determination of the acceptance standards and more to do with capacity demonstration. For example, in Kerala the Keltron factory has emerged as one of the first centers for manufacturing of supercapacitors in India. The project has been developed with technical support from the Indian Space Research Organisation (ISRO). Keltron has also been collaborating with organisations such as C-MET and the Defense Research and Development Organisation (DRDO) for several years (Onu et al., 2023; thehindubusinessline, 2024). Industry 4.0 practices also helps in waste and time reduction and strengthen the regulatory frameworks for agile development.

1.2.4.3 Regulatory frameworks:

By section 166(2) of the Electricity Act 2003 (EA 2003), the Forum of Regulators (FOR) was established in India by Notification dated February 16, 2005. Also, As indicated in Figure 1.0.4, state governments have their separate support system for incentivize the energy landscape in India (Aayog, 2020). To improve the legal and operational environment for energy storage systems (ESS) in India, the working Group on ESS has put out several important recommendations. They contend that the Electricity Act 2003's current provisions are adequate to cover ESS and enable it to discharge energy into the grid in a manner akin to that of a producing plant. According to the recommendations, producing stations and Independent Energy Storage Systems (IESS) ought to be treated equally so that IESS can enter into storage service contracts and make use of different business models.

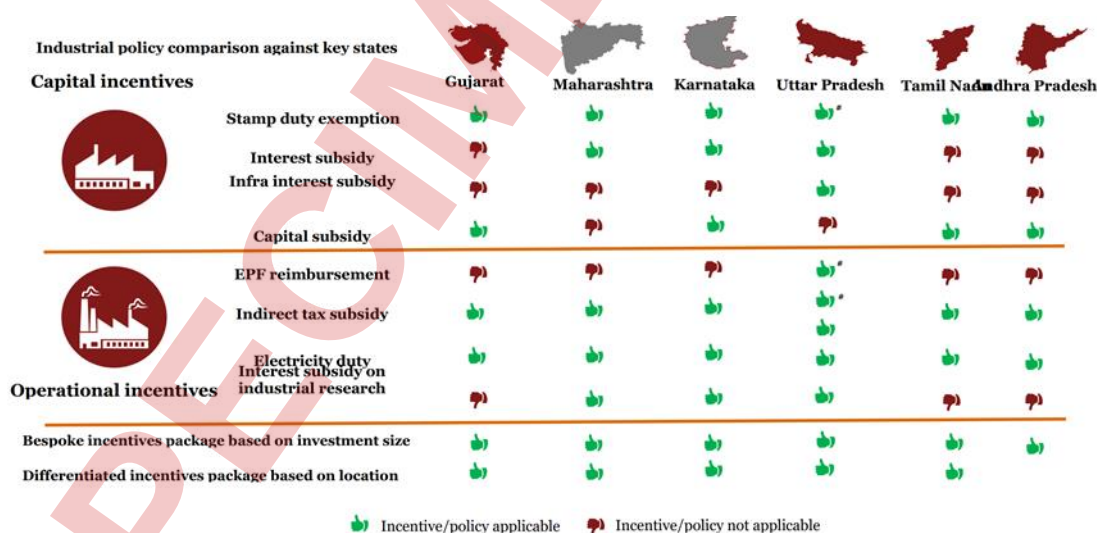


Figure 1.0.4: Comparison of industrial policies across states (Aayog, 2020)

These require either the purchase of energy from renewable resources or the signing of tolling agreements with regional distribution companies as part of integrating

renewable electricity into India's power system. The paper also explains how ESSs are categorized as both transmission and distribution assets, emphasizing their role in traffic control and meeting renewable procurement standards. Moreover, Using BSHESS to store energy during nominal loads and discharging it during abnormal loads, businesses can save a minimum of ₹3.33 per KWh right away, whenever they exceed the base load tariff (Lemian & Bode, 2022; Singh et al., 2005). Along with highlighting their function in traffic management and fulfilling renewable procurement requirements, the report also describes how ESS are classified as both transmission and distribution assets. It highlights the necessity of open access and connectivity by guaranteeing that ESS are regarded in the same manner as producing stations under regulatory frameworks. ESS operations should be scheduled in line with those of traditional generators to optimize their use during periods of peak demand. The recommendations also call for competitive tariff determination processes and tax policies that encourage investment in ESS, such as exemptions from customs tariffs and GST reductions, to foster a robust market for energy storage solutions in India (Forum of Regulators, 2022).

The driving force behind India's energy transition is currently efficient and sustainable energy management for less greenhouse gas emissions and the best use of energy resources. By 2030, the nation wants to reduce its GDP emission intensity by 45% from 2005 levels and attain 50% of its installed capacity from non-fossil fuel sources. This approach includes investments in electric vehicles and programs such as the National Solar Mission. However, because renewable energy (RE) sources like wind and solar are intermittent, including a sizable portion of RE presents grid stability issues.

BSHESS are seen as a crucial means of storing energy generated by renewable energy sources in order to enhance grid management and reduce reliance on fossil fuels.

Unfortunately, a national framework for ESS must be created to promote a friendly setting in which everyone can obtain cheap, effective, and sustainable electric power. The ESS framework involves use of legislative measures and financial incentives in a bid to achieve better grid stability, lower total energy costs and ensure dispatchable renewable energy round the clock. Energy storage demand is expected to reach 73.93 GW by 2031–2022, according to the Central Electricity Authority, requiring large investments. The framework also emphasizes the importance of the fund research and development in energy storing technologies, safety enhancements and domestic technologies. By so doing, India might ensure a just distribution of energy storage technology for all classes of individuals, and pave way for a transition to a cleaner energy regime (Ministry of Power, 2023).

1.2.5 Literature review on existing business models and techniques

At the moment, major companies with a sizable market share include Eaton Corporation PLC, Maxwell Technologies Inc. (Tesla Inc.), and others. To boost their market share and profitability, these well-established businesses take use of smart joint efforts. Power systems have relied heavily on asset owner/operator/service business models, which apply to both regulated utilities and unregulated industrial sectors. Energy aggregators, merchant transmission line operators, and Meter Service Providers (MSP) are a few examples in the regulated sector. However, there are Energy Efficiency service companies (ESCOs) & Independent power producers, which evidently participate in the unregulated markets. Such responsibilities were formerly the preserve of vertically

integrated utilities, and would enable the prevention of wasteful asset duplication. Nonetheless, the change started as specific companies appeared in the late 1990s and offered customers even greater value as services were unbundled. There is clear evidence of this phenomena since the year 2000 when, regulatory agencies have responded through new regulations that allow various parties to play an active role (Kathi & Kadali, 2022; NITI Aayog, 2019).

These economic models become more complicated with the introduction of Energy Storage Systems (ESS), which need permits to operate from both generators and consumers. Profitability may be delayed since ESS must deal with a two-part tariff system and usually recover investments over 20–25 years as opposed to the more common 8–10 years. Still, there are threats of independent market places targeting the industrial, commercial and residential segments with robust growth opportunities. These markets can deliver localized ESS solutions providing short tariff returns by lowering demand charges as well as incorporating dynamic pricing techniques. To harness these opportunities firms may be forced to adopt the asset leasing models. These models reduce the risks associated with large capital expenditures and operating costs while guaranteeing dependable service through long-standing financial and maintenance agreements (NITI Aayog, 2019).

1.3 SCOPE

The extent of this thesis involves systematically identifying business opportunities to position supercapacitors and its application on BSHESS based on targets that can be achieved. The research will focus on:

1. Developing a hypothesis for a beachhead market within the supercapacitor domain:

The research will focus on a specific geographical region, industry, and application, and will be bounded by the availability and quality of market data and customer insights. The performance parameters, competitive landscape, technology scope, and regulatory and policy considerations will also be defined to ensure a clear and manageable research scope. The research will stay focused if these restrictions and limitations are acknowledged. This will assist in formulating a logical theory for a beachhead market in the field of supercapacitors.

2. Making sure the chosen market has a strong value proposition: This study will focus on a particular market sector that offers the greatest potential for the business model, rather than the supercapacitor market as a whole. The customer's viewpoint is the main emphasis of this effort, which also highlights reciprocal value, finds distinctive differentiators, tests, and improves the value proposition. To make sure it appropriately reflects their demands and opinions of the business model, the propositions will be validated with customers.

3. Verifying the theory by doing field research to determine its viability in the market:

Consider the viewpoint of the customer when creating a compelling value proposition. Identify the things that make the idea unique. Emphasize how the business plan and clients benefit each other. Incorporating primary data and feedback into a customer-centric approach will be the primary emphasis of the validation process. The data collected from clients may differ from that of general secondary research, depending on demographics and other broad tendencies.

4. Determining the beachhead market, target client and business plan within the limitations of study: The following are some restrictions and boundaries on the research scope:

- It focuses on specific regions.
- Due to operational reasons certain areas are excluded.
- The study area is on sectors such as consumer electronics, transportation and renewable energy.
- Such markets where the availability of products is already high, the analysis does not consider them.
- The business operation activities and sustainable development objectives help define the customer categorization.
- The analysis is likely to exclude those markets that exist in complete or substantial saturation.
- The scale of the business and sustainability objectives determine customer segmentation.
- Components that don't fit the value proposition are not included.
- The availability of the large size of the market and the potential for growth in these markets receive especial attention.
- Growth rates which are slower than the average are ignored.
- Those main competitive environments that are not very much crowded are the key concern here.
- Markets with high rivalry are also excluded looking at the SWOT analysis on the strategic position of the business.
- Distribution and legal necessity remain important to the priority given to markets available to physically challenged individuals.
- Overly challenging or risky markets based on organizational capabilities are excluded.

These defined boundaries ensure a focused and coherent approach to identifying the most promising target customers and beachhead market for the supercapacitor and BSHESS business model, enhancing the strategy's viability and success.

5. Establishing the performance standards needed to provide the desired value proposition: Certain essential performance characteristics will be the main focus of the

study. These consist of cycle life, safety, power density, and energy density. It won't include every potential trait. The limitations of existing BSHESS and supercapacitor technologies will be taken into consideration when defining the performance requirements. This will take into account limits and inherent trade-offs. These include cost against performance and energy density versus power density. The prototype or experimental setup will be made to test the specified performance parameters. The amount of time, resources, and equipment that are available will determine its limitations. When setting performance parameters, the study will take into account the associated presumptions and uncertainties. This is especially important in a rapidly evolving technical and business context. It will state them clearly to make it easier to explain uncertainty in later improvements.

6. Establishing a prototype or experimental setup to verify and test the performance parameters: Critical performance metrics for the intended value proposition and beachhead market will be the main focus of this study. Acknowledging the inability to fully reproduce market intricacies, it will build the setup to mimic real-world settings as nearly as possible. While taking into consideration inherent constraints, accurate and trustworthy measurement methodologies will be employed. Although the difficulties associated with large-scale production may not be fully recognized, scalability and manufacturability will be taken into account. The project will focus on the most important elements of business model validation while working within time and resource restrictions. Lastly, it will admit that results might be unique to the market conditions, consumer preferences, and performance metrics that were studied.

7. Testing the value proposition through the experimental setup/prototype to assess its viability:

Using a prototype or conducting experiments to assess the viability of the value proposition to identify value-keyed features that consumers are willing to pay has some restrictions and limitations. One of these is that there may have been variability in the experimental conditions, such that it did not match real circumstances fully. Also, there is the difficulty and limitations of scale with regard to data collection. Testing restrictions: source scarcity and belief about the nature of systems or markets that may affect the outcome add other restrictions to the testing process. A thorough business plan for supercapacitors and the BSHESS market will be the end result of this thesis. It will incorporate knowledge from laboratory experiments and field research. Market analysis will also direct market entrance plans and strategic decision-making. This inquiry will probably be carried out in a certain amount of time. In terms of long-term market trends and customer validation in particular, this might restrict the breadth and depth. Additionally, the accessibility and availability of consumer data, market data, and other resources needed for field research and market analysis may limit this study. The next part outlines the goals of this work while stating the aforementioned limits.

1.4 OBJECTIVE:

The objectives are (1.) To develop a hypothesis for a beachhead market in the supercapacitor domain; (2.) to analyze the target market's pain points validate the hypothesis through field studies, needs, and preferences; (3.) creating a unique selling proposition that would explain the needs and explaining how it would be different

among similar products; (4.) establishing performance targets and specifications matching with value proposition; and (5) establishing a strong business idea. In this way, the thesis will guarantee that it will meet the market needs and offer significant added value as a basis for the future commercialization of and a business model of supercapacitors and BSHESS.

1.5 ORGANISATION OF THESIS

Chapter 1: A brief idea of the research, thesis background, its motivation, scope, the objective of the research, and literature review are discussed.

Chapter 2: This chapter discusses customer discovery and market segmentation. It covers selecting the beachhead market and building an end-user profile. Additionally, it includes the calculation of TAM for the beachhead market and developing a persona profile for the BHM. Finally, it identifies the next 10 customers.

Chapter 3: The insight of the revenue and business models is introduced in this chapter. It describes issues like customer value obtained by the firm, and the overall cost of gaining a customer. Furthermore, it encapsulates features about data acquisition, field research and the validation phase. Last followed by the cost estimates of the system.

Chapter 4: This chapter explains the synthesis and testing process. The development process of BSHESS prototype is also explained.

Chapter 5: In addition to technical goals, this chapter presents data and findings from the targeted market study.

Chapter 6: Summarizes and concludes the discussion. The important results and future scope of the study are discussed.

SPECIMEN COPY

CHAPTER 2.

MARKET OPPORTUNITIES AND CUSTOMER DISCOVERY

Customer discovery for a product is typically done by first defining a hypothesis, and then testing and validating that hypothesis through customer research and feedback. The key steps are (a) defining a hypothesis about the problem to solve and its proposed solution, (b) testing the hypothesis by asking relevant questions to determine the problem, (c) once the hypothesis is validated, testing the product concept with customers to get their feedback and refine the idea, (d) after incorporating customer feedback, validate the refined product concept to ensure it meets their needs and has the market potential (Hart, 2012). In this chapter, this work defines the hypothesis. The factors associated with the gaps will be primarily analysed focusing on the problem statement. Defining a problem statement or hypothesis involves several steps considering the risks and potential discussion about the market segments to identify immediate target customers.

2.1 DEVELOPING THE PROBLEM STATEMENT/HYPOTHESIS FOR CUSTOMER DISCOVERY:

To craft a compelling hypothesis or problem statement for potential customer discovery, it is essential to follow a structured approach. This begins with market segmentation, identification of the correct market segments as niche market and sub-segmentation is done further. We call this subsegment as beachhead market as the sample population will be tested out for specific pain points. After selecting a

beachhead market that serves as a testing ground for the existing pain point and proposed solution, an end-user profile is established. This profile will make it easier to determine the wants and frustrations of the specified group of clients. Total Addressable Market known by the acronym TAM is used to compute various aspects of potential revenue. It also helps in identifying the customer segments in the beachhead market, which the business idea elaborates on here. Lastly, the identification of the role of the subsequent 10 customers serves to fine-tune and define this approach for implementing and delivering goal-pertaining solutions to meet customers' requirements (Aulet, 2024). By following these steps, this thesis will achieve a well-founded hypothesis or problem statement part. The hypothesis is as follows: **"Customised energy solution providers must enhance their services with wider market reach. They must be able to integrate renewable sources and meet high power demands overcoming compatibility issues. This will help clients navigate modern energy markets, achieve energy capacity expansion, and attain more customers for easy portability needs."**

2.2 SELECTION OF MARKET SEGMENTATION: SUPERCAPACITOR ENABLED ENERGY STORAGE

This work is motivated by the supercapacitor technology-based market. The supercapacitor market can be segmented further into various categories which are explained in Table 2.1. The business potential of supercapacitors extends to various stakeholders in the energy sector. This includes energy integrators, utility companies, and grid operators. This hypothesis does not include market research about microgrid developers and electric vehicle infrastructure providers. Industrial and commercial

facilities, which can benefit from supercapacitor technology are focused for this research.

Table 2.1: Supercapacitor classification for market segment

Parameters	Description
Type	1. Electric Double-layered Capacitors (EDLCs): Fast charge/discharge, excellent cycle life (up to 1M cycle life), easy to fabricate, lower cost, Low energy density (less than 8 Wh.kg⁻¹), higher self-discharge rate (N. Kumar et al., 2022). 2. Pseudo-capacitors: Higher energy density (over 10 Wh kg⁻¹) and capacitance, slightly lower cycle life (up to 200k cycles), higher cost compared with EDLCs (J. Wang et al., 2017). 3. Hybrid Capacitors: Higher power and up to 10 times the energy density of EDLC, lower discharge rate, Lower cycle life compared with other types (up to 100k cycle life); complex design, leading to a higher cost (Yaseen et al., 2021)
Voltage	1. Less than 25V: Consumer electronics (e.g., cameras, portable devices), energy harvesting systems, backup power for low-power devices 2. 25V-100V: Industrial applications (e.g., uninterruptible power supplies, power smoothing), electric vehicles and hybrid systems, Renewable energy systems (e.g., solar and wind energy storage) 3. >100V: Grid energy storage solutions, high-power applications (e.g., rail transport, heavy machinery), large-scale renewable energy integration (Holtkamp & Alonso, 2020)
Electrode Material	1. Carbon-based supercapacitors: Chosen for best cost reduction scenario. 2. Metal-oxide-based 3. Conducting polymer-based. 4. Composite-based. (Dhanraj Nehate et al., 2019; Forouzandeh et al., 2020b)
Application and existing instances	1. Consumer electronics: cameras, portable applications 2. Aerospace: Electric vertical take-off and landing (eVTOL) aircraft and CubeSats. 3. Industrial: Black-start support, variable-frequency drives 4. Automotive: acceleration and regeneration. 5. Medical: Defibrillators, medical implants (e.g., pacemakers), patient monitoring equipment, and other assorted equipment. 6. Sensors, wireless technologies and Internet of things devices, and portable point-of-sale devices to reduce battery cycling 7. Renewable energy: Photovoltaic system, hydropower (black-start support), microgrids, bulk power systems and backup power (Dhanraj Nehate et al., 2019)

This suggests that energy integrators and contractors can use supercapacitors to improve the energy storage systems' dependability and performance. They can also be used by grid operators and utility corporations to increase the toughness and power of the system. Developers of micro grids can incorporate super capacitors to increase the dependability and effectiveness of these systems. They could be applied by organizations involved in the offering of infrastructure for electric vehicle charging in

order to enhance enhanced charging rates and thereby increasing the efficiency of the networks. In the commercial or industrial building, supercapacitors are incorporated to improve energy control, store energy and integrate with renewable energy systems.

For the final hypothesis, it is considered that, among these, energy integrators/contractors can play a crucial role. They can provide efficient solutions using supercapacitors for various end-user segments. In the automotive & transportation sectors, they can create hybrid energy storage systems that combine supercapacitors with batteries to improve energy efficiency and power delivery. In the industrial sector, they can integrate supercapacitors into energy storage and management systems to provide backup power and improve energy efficiency. In the energy & power sector, they can incorporate supercapacitors into grid-scale energy storage solutions to provide frequency regulation and voltage stabilization. In the electronics sector, they can design supercapacitor-based energy storage solutions for electronic devices and systems to provide backup power and improve power quality. In the military & defense sector, they can develop supercapacitor-based energy storage systems for critical systems and high-power pulsed loads. In the aerospace & aviation sector, they can integrate supercapacitors into aircraft and spacecraft energy storage systems to provide backup power and support high-power demands (Kathi & Kadali, 2022). Due to the wide potential of opportunity, energy integrators/contractors are considered as target customer who have a wide reach in supercapacitors enabled various markets (Durvasulu et al., 2023; Habib & Sou, 2018; *Revolutionizing Energy Storage with Supercapacitors - CE+T Power*, 2024). The selection of the target customers for market segment of interest is also supported by the various specific requirements and technical characteristics.

These specific requirements and technical characteristics of supercapacitors include some key features. These are rapid charging, high power density, and long cycle life. They also support grid stabilization, regenerative braking in electric vehicles, and peak shaving in microgrids. Therefore, the selection of market segments is directly linked to these properties. For this EDLCs are chosen. The characteristics outlined in Table 2.1 are responsible for this. This strategy of selection is in enhanced agreement with the increasing need for energy storage technologies (Durvasulu et al., 2023; S. Wang et al., 2008). Supercapacitors are thus marketed in this context as a promising special market segment that exploits these characteristics. To solve certain energy storage issues, energy integrators, contractors, and suppliers develop specialized devices like BSHESS. Such specific problems that need such highly customised products are selected by analysing various demand drivers, market trends and challenges associated (Bhardwaj et al., 2018; NITI Aayog, 2019).

2.2.1 Demand drivers and market trend

The demand for supercapacitor market is experiencing robust growth driven by several key demand drivers across various applications. These include consumer electronics, industrial applications, and energy-harvesting, telecommunications, and grid applications. Renewable energy and consumer electronics are among the second most demanding fields for the supercapacitor business (Habib & Sou, 2018; S. Wang et al., 2008). Supercapacitors' primary characteristics have put them in a position to flourish in the industry over the next several years. For these industries to stand out from rivals in the EV market, application-specific tailoring is necessary. This choice will generate multiple sources of income. It will take advantage of advantageous government

assistance and investigate novel uses for supercapacitors. Also, it will exploit the affordability feature of the supercapacitors in several client applications with scaling (NITI Aayog, 2019; Office of the PSA to the GOI, 2014). Hence, as an evaluation of the problems of the present and the future in the development of specialized solutions in these fields is required. This will lead to constant demand in customized BSHESS systems to be installed for the company to have a strong market base for sustainability and long-run business growth.

2.2.2 Analysis of challenges and potentials in renewable energy and consumer electronics segments for BSHESS

The market for supercapacitors is expanding, but there are still a number of obstacles facing the sector. Limited energy storage capacity, high cost in comparison to conventional batteries, and fierce competition are some of the main obstacles. With BSHESS's combination of features, restricted energy storage capacity can be restored without sacrificing current qualities (Huang et al., 2019). Ensuring seamless interoperability and integration across different hybrid energy storage technologies remains a challenge. The development of complex control schemes can overcome the integration issues of supercapacitors with other components. Hybrid energy storage solutions are therefore more reliable and effective (Katsanevakis et al., 2017). The market may not be sufficiently satisfying the need for specialized and customized supercapacitor solutions at the moment. Manufacturers ought to concentrate on creating supercapacitor designs that are versatile and flexible. These designs can be modified to satisfy the unique requirements of different sectors and uses, such as consumer electronics and renewable energy systems (Pathak et al., 2024).

Additionally, tailored and portable storage devices face further challenges. These include fabrication challenges, electrode material development, electrolyte issues, cell balancing, market awareness, higher cost, ion dynamics, and scalability (Pathak et al., 2024). These difficulties show how much more research and development is required to get beyond these barriers and get supercapacitors widely used. Moreover, other obstacles can be surmounted and awareness raised through marketing and branding, strategic alliances, certification and compliance, and consumer interactions (Okonkwo, 2024). Credibility among end users can be raised by obtaining pertinent certifications and adhering to industry norms (Paiss et al., n.d.). While manufacturers may institute growth strategies that produce new products, this can increase their product ranges and so the expense strategies may allow them (Jarbratt et al., 2023). Solution providers of energy like the energy-integrators and contractors enhance end users loyalty and trust when dealing with them. The highest level of support, training, and communication contribute to market development as well as business growth (IRENA, 2020; Tantau et al., 2021).

Supercapacitors have enormous commercial potential for energy integrators and contractors. Their dependability serves a wide range of industries and applications (IRENA, 2020; Lasrado Dylanand Ahankari, 2021). Energy storage systems operate more effectively and efficiently when supercapacitors are incorporated into solutions. This results in increased customer satisfaction and better overall system reliability. Energy solutions using supercapacitors can benefit contractors in their endeavors by providing customers with better enhanced environmentally friendly energy storage systems. Suppliers can satisfy the demand of fast charge and high power density through using supercapacitors in energy project. In a variety of sectors, such as

consumer electronics, automotive, aerospace, and renewable energy, these technologies also provide a long cycle life. Energy storage solution customization enables suppliers to modify their products to satisfy particular customer needs. Customization and specialization can help vendors of portable power solutions stand out in the market. This approach draws a wider network of clients searching for superior energy accumulation technologies (International Solar Alliance (ISA), 2023; Lasrado Dylanand Ahankari, 2021). The key benefits of using supercapacitors in energy storage systems include (a) rapid charging and high power density; (b) improved system reliability and stability; (c) enhanced energy management and control; and (d) customization and specialization for specific needs. Supercapacitors can be used in customized and specialized solutions for various applications and industries. This enables solution providers to create customized energy storage solutions for the particular requirements of their clients. In the end, this increases customer value and opens up new business prospects in the expanding market for portable power solutions (Bhardwaj et al., 2018). Vendors of customized energy systems can increase their market competitiveness by implementing these advantages. In the targeted beachhead market, this strategy gives far more value than in other industries such as; consumer electronics and renewable energy.

2.3 SELECTING BEACHHEAD MARKET: RENEWABLE ENERGY AND CONSUMER ELECTRONICS

The term "beachhead market" describes a specific commercial niche. Businesses can focus their resources here in order to grow quickly. This aids in building credibility prior to entering larger markets. There are several dangers associated with targeting a beachhead market. These include restricted market potential and imprecise market

assessment. A special emphasis is placed on initial customer's dependence. Market saturation and failure to scale could be issues of concern. Market saturation is a universal concept that might not need any explanation, but for the benefit of those who are new to the business, scaling is the ability to increase the size of the market one is serving by finding new customers to meet the demand that existing customers are not meeting. Other challenges are problems with resource management and the absence of product diversification. These risks might even threaten the very existence of a company and its long-term development if it heavily leaned on the first identified market niche. To mitigate these risks, businesses must carefully assess market conditions, customer needs, and the suitability of their product or service for the identified beachhead market. Further, they also need to offer more homogenized product portfolios and look for new niche markets for growth, sustainability and innovation (Aulet, 2024). By understanding the above-sized risks, businesses can easily develop approaches to expand beyond the beachhead market therefore attaining a larger market opportunity.

As a result of the growing popularity of electric and hybrid vehicles, the automotive sector is the one with the quickest rate of growth in the supercapacitor industry. After the electric vehicle segment, the sectors for the most growth are renewable energy systems, consumer electronics, and industrial applications (Lasrado Dylanand Ahankari, 2021). This development has been stimulated by increasing depletion of fossil fuels and rising consumer preference for renewable sources of energy, as well as political efforts to reduce carbon emission (Office of the PSA to the GOI, 2014). When selecting the best market sub-segment for supercapacitors' enabled solutions, key factors must be carefully evaluated. These primary parameters for evaluation of the

beachhead market include pain points, performance characteristics, market demand and growth potential, competitive landscape, and regulatory and policy landscape.

2.3.1 Parameters for evaluation of the beachhead market:

2.3.1.1 Application requirements and pain points:

Application requirements involve identifying specific applications that demand rapid response times, high power density, frequent charge/discharge cycles, and peak shaving to reduce peak power tariffs. The solution providers generally use a customized BSHESS system to fulfil this need based on the customised design. For instance, changing the control approach in a DC micro-grid system with a BSHESS enhances the battery SOC balance and minimizes the bus voltage fluctuations (Tian et al., 2022). BSHESS are increasingly adopted in residential, commercial, and industrial applications. They offer users flexibility in storing and utilizing electricity efficiently. Also, customized BSHESS focuses on energy sustainability and backup power, and the potential for rapid adoption in smaller communities with quicker decision-making processes (Dong et al., 2022).

Communities looking for portable power solutions have a greater demand for decentralized power generation and the integration of renewable energy sources. They also prioritize energy sustainability and backup power, and they have the possibility for successful technological validation and rapid acceptance. These communities often face challenges in accessing the main grid and require reliable power solutions, making them a suitable market for customized hybrid energy storage systems (Sanjel & Baral, 2021). Portable power users needing such systems can be considered end users in the

beachhead market. Opportunities for new entrants to provide customized solutions are presented by the growing demand for efficient, dependable, and sustainable power sources, especially in areas with frequent power outages (Li et al., 2021).

2.3.1.2 Performance characteristics and market opportunities:

A. Market growth for customized hybrid energy storage systems:

The customized BSHESS market is poised for significant growth. The growth is therefore enabled by enhanced technologies and the adoption of integrated renewable energy solutions. Additionally, industry trends favor high-performance and sustainable energy solutions. By 2030, the market is expected to grow to a size of \$20.12 billion. This is predicated on a compound annual growth rate (CAGR) of 19.68% from 2024 to 2030. The international marketplace for energy storage is anticipated to expand quickly, increasing installed storage capacity worldwide from a projected 56% to 270GW+ by 2026. Recent advances in hybrid energy storage systems integrated with renewable power generation are driving innovation and expanding the capabilities of customised energy storage solutions. There is also a need for efficient energy storage solutions to support renewable power generation (Dong et al., 2022; NITI Aayog, 2019). A CAGR of about 7.5% of the HESS market is expected for 2024 to 2030, and an estimated market size of around USD 43.78 billion in 2023 that is expected to rise to \$72.64 billion by 2030 (Lasrado Dylanand Ahankari, 2021). Having a positive outlook for the sector, these market indicators substantiate the fact that the hybrid energy storage system market represents a significant value, and the growth of this market.

B. Key trends for customised hybrid energy storage systems”

- Decentralized power generation is becoming more and more popular (Li et al., 2021).
- Use of renewables is gradually emerging as crucial in power system integration (NITI Aayog, 2019).
- The market is expanding due to an emphasis on energy sustainability (Sanjel & Baral, 2021).
- The commercial, industrial, and residential sectors are seeing an increase in adoption (Dong et al., 2022).
- The capabilities of BSHESS are being improved by technological developments (Gómez et al., 2023).
- Government incentives and policies support market development (Ministry of Power, 2023).
- There is an emphasis on grid resilience and backup power solutions (NITI Aayog, 2019).
- Cost optimization and reduced carbon footprints are key objectives (Office of the PSA to the GOI, 2014).

C. Competitive and regulatory landscape for customised hybrid energy storage systems:

The focus here is to differentiate BSHESS from other existing technologies. The current market for EVs is still occupied by long-existing companies. But in the energy sector, new entrants can also advance by providing tailored products and services. The competitive landscape in the BSHESS market is less intense than in the EV market. These solutions can target renewable power, backup power, and consumer electronics segments. The new business can successfully compete if it has the correct plan. New businesses can set themselves apart from more established rivals by emphasizing sustainability, expertise, and personalization. The use of renewable energy is being encouraged by government regulations and incentives. The development of specialized hybrid energy storage systems is being impacted by this. As a result, investment in

sustainable energy solutions is being encouraged, and it is driving this as a potential beachhead market for new businesses (Ministry of Power, 2023).

2.3.2 Decision on beachhead market

Customized solutions for renewable power backup, consumer electronics backup power, and specialized applications could serve as promising beachhead markets for a business in energy integration. The growing demand for reliable, portable and sustainable power solutions, driven by environmental consciousness is growing rapidly. BSHESS can be customized and specialized for specific application requirements. This flexibility allows businesses to differentiate themselves. It helps cater to the unique needs of various segments. This is also evident from the customer discovery and market research for this work. The details are concluded from interviews in this context which are available in the appendix.

2.4 PROFILE FOR THE END-USER

2.4.1 End user persona: Portable power users

Table 2.2 End user persona

Market segment/segment	Supercapacitor market/Renewable energy and consumer electronics markets.
(BHM)/Proposed product	Portable-power-solution market/customized BSHESS systems.
Customers	Portable power users.
Geographic	Located in rural, island, or remote areas with limited access to the main power grid. Ex: Residential communities, remote small-industrial sites, emergency response teams, and outdoor recreation enthusiasts across Africa, Asia, Latin America, and the Pacific Islands.
Demographics	Income levels: Low to middle, with some high-income individuals. Age: Wide range of ages, from young families to elderly residents. Genders: Mix of genders but more male Education levels: From primary to university level. Occupations: Farming, fishing, mining, construction, emergency services, etc., MSME

	Racial and ethnic backgrounds: Diverse
Psychographics	Social presence: Limited. Concerns: Environmental impact, carbon emissions, cost, energy independence and self-sufficiency
Pain Points	Reliance on costly, polluting, and unreliable diesel generators for power. Frequent power outages and blackouts due to grid instability or natural disasters. High energy costs and lack of access to affordable electricity. Environmental concerns about carbon emissions and noise pollution from generators.
Goals	Obtain reliable, sustainable, and affordable electricity to power homes, businesses, and operations. Reduce dependence on diesel generators and grid-supplied power. Lower energy costs and carbon footprint. Access to backup power during emergencies and outages. Ability to power remote locations and outdoor activities.
Needs	Decentralized power generation solutions that can integrate with renewable energy sources. Systems for energy storage that have a high power density, a long cycle life, and the ability to charge quickly. Easy to implement and adaptive systems that are also easily scalable and integrated. Low initial costs and low cost of operations. Minimal maintenance requirements and remote monitoring capabilities. Resilient to harsh environmental conditions and natural disasters.
Preferences	Prioritize sustainability and environmental responsibility in energy solutions. Value reliability and durability in power systems. Prefer modular and scalable solutions to accommodate changing energy needs. Interested in energy independence and self-sufficiency. Willing to invest in high-quality solutions for long-term benefits
Behavior	Actively seek out alternative energy solutions to reduce reliance on diesel generators Research and compare different power generation and energy storage technologies Join and participate in online groups and Social media to carry knowledge and experiences. Collaborate with local organizations and governments to advocate for renewable energy policies Prioritize energy efficiency and conservation in daily operations and activities High usage of customized power systems like solar home systems and diesel generators Interested in new energy technologies that can improve reliability and reduce costs Willing to adopt innovative solutions if they meet their needs and are affordable Rely on word-of-mouth and local networks for information on new products and services Prioritize durability and ease of use in power systems

(Please see the appendix for interviews)

2.4.2 Decision-making process for end users

2.4.2.1 Stakeholders in decision making:

For the "portable power users" as end users, the decision-making unit may include the following key individuals.

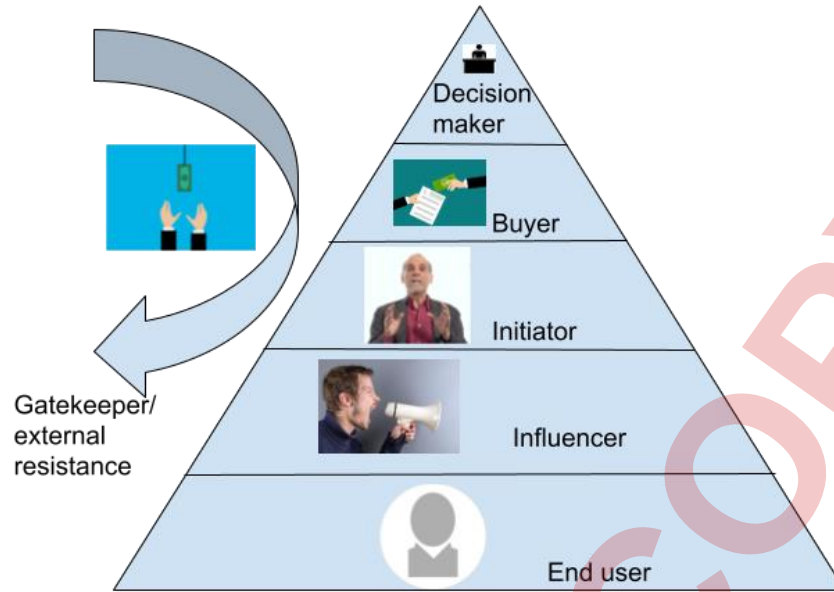


Figure 2.1: Five elements in a decision making unit

Table 2.3 Decision Making hierarchy for end users

Role	Who
Influencer	1. Friends/ neighbors /Sales executives/Marketing agencies/You-tubers
Initiator	2. Property owner and/or
Decision maker	3. Educated person in the family, property owner
Buyer	4. Head of the family, house/property owner,
Gatekeeper	5. Safety officers/Grid employees/Government regulating team
End-user	6. off-grid communities, remote industrial sites, and outdoor recreation enthusiasts

(Please see the appendix for interviews)

Based on this categorization in the decision-making hierarchy, the end user approaches to the firms or individual companies to install the energy storage as per the need and pain points. Hence such companies/firms involved in providing such solutions are the true customers for the businesses. These companies evaluate the strengths, weaknesses, opportunities and threats based on the customer profile (portable power users) and provide customized solutions that best match the energy solution at their end.

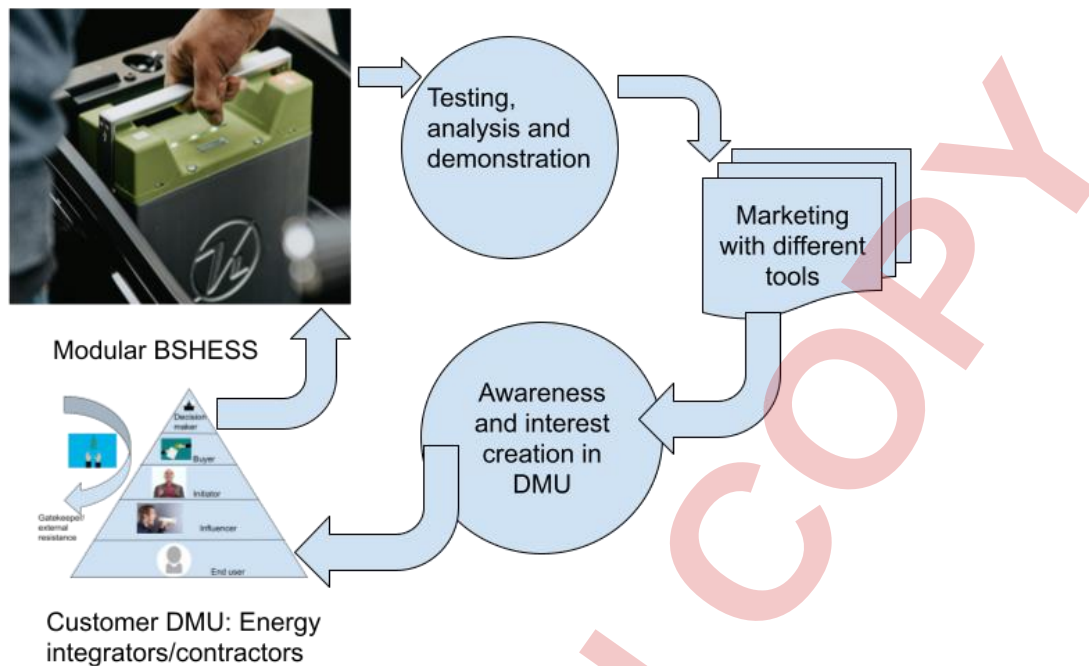


Figure 2.2: Plan for getting the first customer

2.4.2.2 Need and pain analysis for end-user profile:

Table 2.4 Need and pain analysis for end users

User types	Pains	Needs	Challenges
off-grid communities, remote industrial sites, and outdoor recreation enthusiasts	Frequent battery replacements Poor warranty compliances Complexity for portability decentralized power and renewable energy integration	Need for reliable and sustainable power solutions Cost reduction	Existing products have inverter systems. limited financial resources for the users Making aware the user of the offerings Geographic dispersion of portable power users alternative technologies, such as diesel generators or standalone renewable energy systems

(Please see the appendix for interviews)

In section 2.3 the opportunities, needs and pain points are explained in detail.

Furthermore, portable power users may lack the technical expertise to install, operate,

and maintain complex energy storage systems, requiring additional support and training. The fact that portable power users are also geographically dispersed also poses a challenge since it becomes hard to be actually provide support and services like maintenance. However, the market also faces several challenges. Portable power users may consider alternative technologies, such as diesel generators or standalone renewable energy systems, as competitors to customized hybrid energy storage systems. Demographics are often more vulnerable to natural disasters and harsh environmental conditions, which can damage energy storage systems and disrupt power supply (Guo et al., 2024). The two factors that could affect the adoption of customised hybrid energy storage systems include changes in regulation or policies on energy storage and renewable energy resources.

2.4.2.3 Regulatory and support environment:

To minimize carbon emission, the governments of the world are today encouraging the use of non-renewable energy sources. They also support energy storage technology in order to attain the energy security. They provide grants, subsidies, and tax incentives to help with this. The creation and use of hybrid energy storage systems have benefited from these grants. Connected grid rules are different depending on the area; some areas are connected easily while some areas are complex in term of infrastructure and legal connection rules. As is the case with most innovations, safety, reliability, and interoperability of products are best achieved by the government and businesses through adoption of standards and certification. Energy storage technology is one area in which government incentives may be encouraged. Grants, tax credits, or rebates are some examples of these incentives. These steps aid in lowering the initial expenses of

hybrid energy storage devices. Research and development is something that governments may fund. The efficiency, dependability, and affordability of hybrid energy storage systems will all increase as a result (Forum of Regulators, 2022; Ministry of Power, 2023). These expenditures can be applied for creating innovative technologies in the context of competition.

2.4.2.4 Competitive dynamics:

The market for hybrid energy storage is made up of both new and existing companies. Well-established businesses are well-represented in the market. Innovative business models and technology are introduced by newcomers. Businesses in this industry compete fiercely for clients based on pricing, dependability, and performance. Product differentiation is one of the key success factors for corporations. Through a variety of platforms, such as trade exhibitions, online marketing, system integrator and installer collaborations, and demonstrations, customers are outreached. Companies prioritize keeping clients by providing constant assistance and high system reliability (Aulet, 2024). Knowing about competitors and the environment for regulation aids in the creation of effective strategies for portable consumers in a market with a broad addressability.

2.5 CALCULATION OF TOTAL ADDRESSABLE MARKET (TAM) FOR THE BEACHHEAD MARKET

To calculate the TAM for the beachhead market, estimate the total revenue achievable. To calculate the TAM size for customized BSHESS, we target portable power users. Key demographics include rural communities and island communities. Remote

industrial sites, were also considered. Additionally, disaster relief and emergency response are important factors. Camping and outdoor recreation users are included in our analysis as well. After estimating the population, the average system size and cost for each segment is considered. This calculation is based on the data available in report of "State of housing in India 2013" stated in Table 2.5.

Table 2.5 Adoption and retention of renewable sources in rural India (Harrington et al., 2020)

	Adoption factors in %					Retention factors in %			
	Access	Awareness	Capacity	Price	Quality	Access	Capacity	Price	Quality
Kerosene lamp						41	77		36
Solar lantern	18	35	41	88	6	24	59		24
SHS	25	60	70	75	20	40	90	30	55
Microgrid		78	67						
Grid	100		46						

Table 2.6 Calculation of total addressable market

Total population: 1.4 billion	Rural houses 72 %	Urban 28 %	Off-grid
Total house: 331 million	239 million	92 million	Included in rural
Access to power backup	47.8 million (59.75 million (25 % of total houses) 80% are in rural)	61.64 million (67 %)	11.95 million (20 % of 59.75 million)
Capacity size kWh	0.5 kWh to 2 kWh	0.8 to 2.4 kWh	2 kWh to 5 kWh

Retention rate	40%	50%	60%
Total energy storage capacity with retention	9560 kWh to 38240 kWh	24656 kWh to 73968 kWh	13428 kWh to 33570 kWh
Total capacity (all)	47644 to 145778 kWh	*50000 per kWh	₹2.382 to 7.288 billion
Without retention	23900 kWh to 95600 kWh	49312 kWh to 147936 kWh	22380 kWh to 55960 kWh
Total market capacity without retention factor	95592 to 309496 kWh	*50000 per kwh	₹ 4.78to 15.475 billion

This approximation offers a starting point to estimate the TAM of customised hybrid energy storage systems for portative power users. The actual market size may vary depending on factors such as market penetration, competition, and regulatory environment. The rural communities segment represents the largest potential market due to the large population living in rural areas globally. Island communities and remote industrial sites also present significant opportunities for customised hybrid energy storage systems. Disaster relief and camping/outdoor recreation segments are smaller but still represent addressable markets. The total TAM is estimated at ₹2.382 to 7.288 billion, indicating a massive potential for growth in this market.

Market characteristics including laws, market changes, trends, perceived value, pricing sensitivity, the competitive environment, and legal and regulatory frameworks must all be taken into account in order to compute TAM precisely. For a precise TAM assessment, evaluate how market limitations, market share objectives, and substitute possibilities affect market potential. Additionally, assessing the target customers' willingness to pay for the product or service is crucial in determining the maximum revenue potential. Analyzing the persona profile for targeted customer effectively helps

to move the solution from actual customer to end use in the particular beachhead market (Aulet, 2024).

2.6 PERSONA PROFILE FOR THE BHM

A persona profile for the beachhead market, including technical, financial, and strategic decision-makers, helps companies understand their goals, pain points, and preferred communication channels, enabling them to tailor marketing strategies and content to meet their needs. The Table 2.7 qualifies different parameters that are involved in the process of decision-making for the BSHESS purchase. The Table 2.7 explain roles and thinking process which is linked to the actual interview conducted for this segment.

Table 2.7 Persona profile for the BHM

	Technical decision-makers	Financial decision-makers	Strategic decision-makers	Example situation (Renewable energy company Solutions Inc. (RES Inc.))
Job Title	Engineer, Technical manager, or similar	Financial manager, Accountant, or similar	Executive, CEO, or similar	Engineers, Project managers, Renewable energy specialists
Industry	Renewable energy, Power generation, solar farms or related fields	Same	Same	Renewable energy systems integrator
Education	Bachelor's or Master's degree in engineering, Physics, or related fields	Bachelor's or Master's degree in Finance, Accounting, or related fields	Bachelor's or Master's degree in Business Administration, Management, or related fields	Corporate trading for the domain
Experience	5+ years in the renewable energy or power generation industry	5+ years in the renewable energy or power generation industry	10+ years in the renewable energy or power sector	

Responsibilities	Design, installation, and maintenance of power systems	Budgeting, cost analysis, and financial planning	Long-term planning, strategy development, and decision-making	To integrate supercapacitors into grid-scale solar and wind power projects to improve grid stability, reduce energy costs, and maximize the value.
Decision-making criteria	Technical specifications, performance, reliability, and scalability	Cost, ROI, and financial feasibility	Long-term strategic implications, market trends, and competitive advantage	
Goals	Ensure efficient and reliable power supply, minimize downtime, and optimize system performance	Minimize costs, maximize ROI, and ensure financial sustainability	Ensure long-term sustainability, maintain market competitiveness, and drive growth	To deploy reliable and cost-effective energy storage solutions that enhance the performance and reliability of renewable energy systems.
Pain points	Limited access to reliable power sources, high maintenance costs, and	High upfront costs, limited budget, and difficulty in justifying investments	Limited resources, market uncertainty, and difficulty in	Grid Instability: frequency fluctuations, and voltage variations,

	difficulty in scaling up systems		balancing short-term and long-term goals	Reliability, Cost-effectiveness, competitive
Preferred communication channels	Technical forums, industry conferences, and online discussions	Financial reports, budget meetings, and online financial resources	Executive meetings, strategic planning sessions, and industry reports	Industry Conferences and Events, trade shows, and networking (industry-specific platforms, forums, and social media channels)
Preferred content	Technical specifications, case studies, and whitepapers	Financial reports, budget templates, and cost-benefit analyses	Strategic plans, market analysis, and industry trends	

2.7 IDENTIFYING THE NEXT 10 CUSTOMERS

Primarily, refining the persona's key characteristics, needs, and behaviors in the beachhead market can help find similar potential customers. Further enhancement in market size will increase by identifying similar market segments and sub-segments. For example, targeting to medium-sized renewable energy companies, can have a strong focus on decentralized energy management directly linked to the value propositions. Additionally, reaching out to existing industry contacts, such as suppliers, partners, or industry associations, to get introductions to potential customers will also help. Building a network with sales executives and finding new potential clients can also be facilitated by attending trade fairs, conferences, and industry events. In addition to introducing the value propositions, direct outreach—such as emails, phone calls, or LinkedIn messages—will also determine their level of interest in the company (IRENA, 2020; Tantau et al., 2021).

The outreach will highlight how tailored BSHESS solutions address their specific challenges. These "warm leads" are more likely to be receptive to the product and can provide valuable feedback. The detailed conversations with such next 10 potential customers will validate or invalidate the assumptions made in previous steps. In addition to discussing the goods to be supplied, their cost, and the value they will add, the actual demand for energy contractors will be identified. The responses gathered will also be useful in profiling more specialized offers and provide data on technological trends and advancements.

Thus, it positions the enterprise as a creative research and development partner. This will help in getting valuable input in achieving the objectives of the strategy. Additionally, it will improve the battery-supercapacitor hybrid energy storage system's entire marketing approach. We need to repeat the outreach and engagement process in order to reach the next ten potential beachhead market clients. This calls for a clear methodology. Additionally, we must develop a technical product by understanding a detailed methodology that aligns with our value propositions (Aulet, 2024).

CHAPTER 3.

METHODOLOGY

Business model development for a BSHESS entails indicating how the company is going to capitalize on value in the energy storage business. This chapter outlines the methodology involved in developing the revenue model covering the customer discovery process and value propositions. These were employed to achieve the research objectives and further help to set a business model. BSHESS hinges on several key components, including value propositions, and target customers. These combinations affect the total revenues that are possible for the BSHESS in several applications. 8 essential elements of the revenue model have been explained in the next section.

A business model is a thorough framework that includes several components and describes how an organization generates, delivers, and captures value. These include value propositions, channels, customer relationships, key activities, key resources, cost structure, and operations on customer groups. A revenue model, on the other hand, is a subset of the business model that focuses exclusively on how a firm makes money, including information on pricing policies, revenue streams, and sales channels. By establishing these components, companies can develop strategies that improve financial success and operational efficiency for specific revenue models and business models (International Solar Alliance (ISA), 2023; Remeňová et al., 2020).

3.1 ESTABLISHING THE REVENUE MODEL

The pricing structure, income streams, client categories, value proposition, and sales channels are important components of a revenue model, as shown in the figure. Frameworks such as the business model canvas, lean canvas, customer development, price strategies, and revenue model innovation can be used to build these areas efficiently. Businesses may develop a workable and lucrative revenue plan that supports their objectives by comprehending the significance of a revenue model and its constituent parts. The value proposition is directly related to this strategy (Osterwalder & Pigneur, 2013).

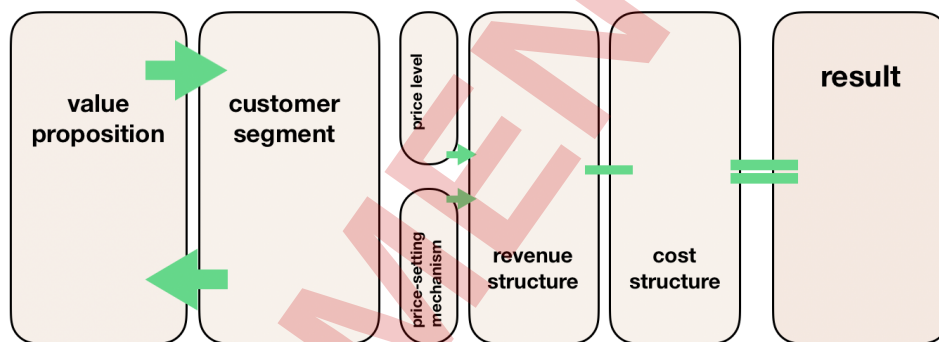


Figure 3.1: Revenue model framework. Image taken from (Osterwalder et al., 2015)

3.1.1 Value proposition: Finding excellence score and major value elements

The value proposition development process involved identifying the key benefits and value that customized hybrid energy storage systems could provide to portable power users. A value proposition acts as the link between customer requirements, preferences and the aspects of the customers. A value proposition is a concise statement that explains why a customer should choose an offering (a product or service etc.) over alternatives. Hence, the value proposition development process also involves evaluating

the competitive landscape and identifying the key differentiators. There are multiple methods available to develop a value proposition. Further 3 major methods are introduced to set an effective value proposition (Cote, 2020).

3.1.1.1 Methods for finding the value proposition:

Multiple methods for value propositions have been explored, including Peter Thomson's value proposition canvas, the direct method focusing on needs, and Steve Blank's intuitive template. Peter Thomson's value proposition canvas examines the various components of a company that contribute to a strong value proposition synchronizing business strategy and brand strategy. There are seven areas to explore, each occupying a section in the map. Figure 3.1.a contains all the elements involved in developing a value proposition by experimentation (Osterwalder et al., 2015).

Another approach that focuses on the end user's needs to find differentiation is the direct technique, which is depicted in Figure 3.1.b. Another way to define value propositions is to use Steve Blank's intuitive template. Blank realized that a straightforward formula was required to turn an idea into a succinct sentence. "We help (X) do (Y) by doing (Z)" is his approach. The value proposition statement for the business may be, using this template: "We serve portable power seekers by providing customized hybrid energy storage systems for power enhancements and long-term cost savings." More in-depth findings can be condensed into a clear and appealing value proposition that appeals to the target market using this straightforward but powerful technique. It is found that most of the methods line with each other. All of these methods require the selection of

optimum value to propose first by analyzing a set of possible values (Osterwalder et al., 2015).

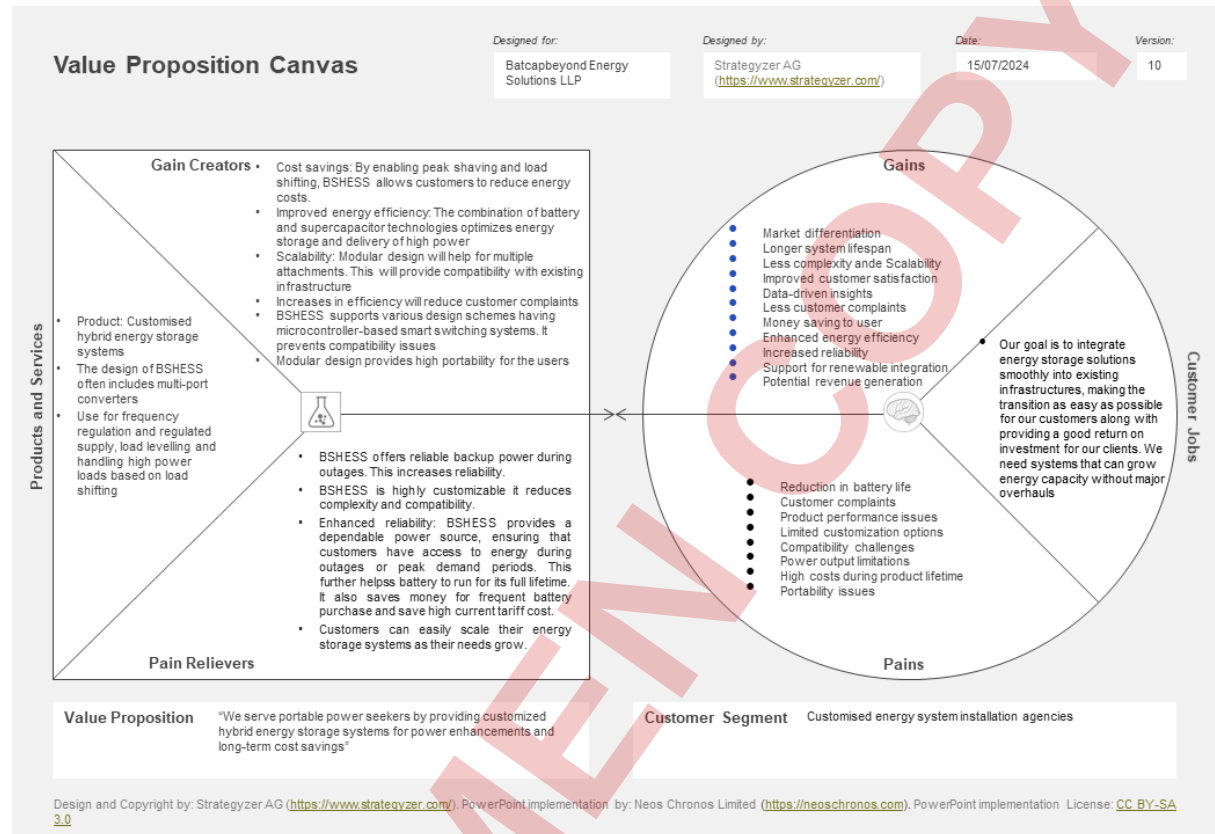


Figure 3.2 Value proposition canvas. This template is designed by www.strategyzer.com.

3.1.1.2 Analysis of values

According to Harvard-business-review, there are 40 total elements of value categorized into 5 categories table stakes, functional, emotional, individual, and inspirational. The most objective kinds of value are found at the base into a five-level pyramid. The higher the level is, the more subjective and personal the types of value it contains. Excellence in these elements will be defined as receiving a score of 65% or above on a scale of 0 to 10. Strong performance will be rated that well on six or more elements by at least

65% of decision-makers. It will help to benchmark the business model against competitors by surveying the customers on how proposed products and services may perform relative to rival offerings. The overall purpose of doing so is to examine what matters most to buyers and what are their wants, needs and fears (Almquist et al., 2016).

A thorough value matrix is created based on these values in order to identify the needs, desires, and anxieties of the intended client. The appendix contains the real client data. The characteristics, advantages, and experience of the suggested product are then compared or adjusted based on the analysis of the insights. With the help of analyzing data sources and conducting interviews with several participants, we can define the points of sustainable alignment of buyers' goals with key stakeholders' objectives. In addition to showing where attention is warranted, this method shall demonstrate the shortcomings of the items for the competitors. It may be noted that more advanced data analysis techniques such as the regression method can be used for further analysis. All the stated methods aim to identify common needs that can be transformed into a value proposition through the product. They then establish the customer segment and target market for the differentiated product.

Table 3.1 Analysis of values for the potential customers

Inspirational value: Vision, social responsibility, self-awareness
Individual value: Better productivity, better network, motivation, independence, time-saving
Emotional value: Reduced efforts, informative, transparency, wellness, reduce anxiety, reputation or standing sense of security and reliability, better control, aesthetics and design, feel empowered and capable, vendor commitment, availability, variety, configurability, stability, direct connection to solution provider, fast response, commitment, risk reduction, reach, flexibility, quality

Functional value: Cost-reduction, saves money, product-quality, scalability, easy to integrate, innovation, saves time, simplification, reduces risk, information indication
Customer mapping for fears (undesired outcomes): Customer complaints, reduced cycle life of battery, and long charging time. High operating expenses. portability issues, compatibility issues, customer complaints, power output limitations Customer mapping for needs (rational motivators): Scalable, cost-effective and modular systems that can handle pulse power requirements effectively. Less complexity and increased scalability, data-driven insights, Fewer customer complaints, money saving to user, enhanced energy efficiency, Increased reliability, Support for renewable sources integration with high power density storage, Customer mapping for wants (Wants (emotional drivers)): Increased customer satisfaction, market differentiation, affordable upfront costs and low operating expenses, minimal maintenance requirements and remote monitoring capabilities, Resilient to harsh environmental conditions and natural disasters. Longer system lifespan potential revenue generation
Product mapping (benefits, experience, features): Simplicity, cost-saving, portability, and scalable. Available in a variety of sizes, scalable, efficient Features and competitors' analysis: Application-specific customized, scalable and cost-effective, low upfront cost, reduced complexity, portable. (Fewer market threats due to customized and improved products) Product experience and competitors' analysis: Portable, easy to carry, low running cost.
Overall goal and differentiation: Portability, reduced complexity Increase efficiency, Reduction in import cost, (Fewer market threats due to customized and improved products)
Table stacks: Compliances and minimum viable product details derived from product mapping: Customized hybrid energy storage systems. The design of BSHESS often includes multi-port converters. Use for frequency regulation and regulated supply, load levelling and handling high power loads based on load shifting

3.1.2 Customer segment and target market

After finding the value proposition, this work focuses on the customer segment. The customer segmentation is done in section 2.3 based on primary points such as product usage, engagement levels to technology, demographics, purchase history, or customer feedback. The customer discovery phase was to carry out interviews and questionnaires with potential users of portable power products to establish their needs, preferences and issues about portable power solutions. This included identifying the key end-user segments, such as rural communities, island communities, remote industrial sites, disaster relief and emergency response, and camping and outdoor recreation. The customer discovery process also involves analyzing the competitive landscape and identifying the key players in the market (Aulet, 2024; Overall & Wise, 2015). In general, various people approached and based on the factors segmented into the group. After this, the people only focused on meeting the hypothesis. The customer segmentation data is attached in the appendix. There are many tools available for running customer feedback surveys and collecting data. Different tools were followed for customer segmentation and data gather platforms which are explained in the next section.

After speaking with actual clients, more data has been gathered via Google Forms and surveys. This is regarded as the study's main source. This work also have utilized secondary data sources such as transactional data, website analytics, social media insights, and other pertinent sources. Key segmentation variables are identified using this data. The most important factors for client base segmentation were identified through analysis of these data sources. Age, geography, frequency of purchases, past

purchases, interests, and any other characteristics that set separate client groups apart could be examples of these variables, as covered in chapter 2. Furthermore, the research methods and materials applied in this process are reasoned with the help of the client segmentation framework.

3.1.2.1 Framework for customer segmentation:

The development of customized pricing models to satisfy a range of needs is made possible by this comprehension of consumer segmentation. Offering different pricing tiers for firms of different sizes or giving price-sensitive clients discounts are examples of how this reflects different segments' differing willingness to pay. Segmentation also aids people's perception of value helping firms to fix prices that will suit the values of these segments say exclusive for luxury segments (Banduni, n.d.; Namvar et al., 2017). These are enhanced by dynamic pricing methods that change with the behavior of the user of the product. Concurrent ideas of price strategies and client segmentation have a big impact on consumer satisfaction, market dynamics, and a company's earnings (Gomes et al., 2023).

Table 3.2 Customer segmentation process

Defining the business objectives:	Forge strategic partnerships with at least three key players in the energy sector (e.g., solar panel manufacturers, technology providers) within the first two years to enhance product offerings and market reach. Continuously innovate and enhance BSHESS offerings, introducing at least two new product variations or improvements annually based on customer feedback and market trends. Attain a customer satisfaction score (CSAT) of 90% or higher by providing high-quality products and exceptional customer service. Incumbent market share rate should be at least 5% by first three years by marketing to energy integrators and backup power installation service providers.
Identifying segmentation criteria:	Demographic: Business size and service dealing type (backup power installation, vendors, renewable energy etc.) Geographic: Areas with high renewable energy adoption or frequent power outages. Off-grid consumers in rural and urban areas Psychographic: Needs-based, environmental concerns, early adopters Behavioural factors: Purchase history and engagement level such as attending webinars or downloading product information Need basis: Those who experience frequent power interruptions and require reliable backup solutions. Cost sensitivity Technical requirements: Understand the existing systems they provide. Estimating the needed energy capacity according to the characteristics of predetermined uses Partnership Potential: The second type of relationship is more about targeting companies that may be interested in building strategic partnerships, be it installation makers or energy providers.

Collecting data	Primary source: Face-to-face talk, calls, surveys/polls, Secondary sources: Online reviews, social sites, digital profiles, LinkedIn groups, Tracery data: Online reports, G2, TrustPilot, Gartner, YouTube, Getapp, Forrester, UserPilot, Google Analytics, Qualtrics, Hubspot
Data analysis for key segmentation variables	Business size and revenue, industry type, location, use patterns, purchase pattern and history, power reliability needs, cost sensitivity, system compatibility, capacity needs, and decision-making process
Creating customer segments based on similarities	Area-specific energy installation services, common vendors, distributors, customer services, company size
Developing segment profiles:	Preferences: Simplified setup, need of portable systems, compatibility, customer satisfaction Need: High power usage, cost-effectiveness, ability to grow energy capacity with minimum hardware Pain point: complexity and customer complaints
Evaluating segment potential with their priority	Top segment (hierarchy): Renewable energy, backup power installation, consumer electronics Market size: ₹2.382 to 7.288 billion, for the beachhead market and ₹1736 billion for follow-on markets. Profitability: Developing a manufacturing setup will give a minimum of 35% return on investment and 176% revenue growth. Growth potential: The renewed energy segment achieved a CAGR of 19.68% in this domain of business Strategic importance: Material development capacity, The decreasing import rate, Electronic board production capacity.
Defining engagement strategies	Communication channels: Gathering data: name of the company, location, size, phone numbers Direct calls for appointments

	Demonstration using ads, YouTube videos, on-site, trade fairs
Implementing, monitoring and measuring the effectiveness of each strategy	Measuring factors such as customer satisfaction, their loyalty, retention level, and increased revenue.
Refining and testing the segmentation framework	Within the first two years, establish strategic alliances with a minimum of three major energy-related companies (such as producers of solar panels or suppliers of technology) in order to expand your product line and market penetration. For enhanced product differentiation and quality services, it is important that a systematic feedback system is developed to capture client feedback frequently. Within six months, try to act on at least 75% of actionable input.

3.1.3 Revenue, cost-structure, price-setting mechanism and price level

The price mechanism has a direct impact on the revenue structure, which includes a variety of income sources. Prices are set by supply and demand in the price mechanism process. Prices usually rise in response to increased demand, which may result in higher income per unit sold and change the revenue structure. Pricing strategy adjustments may add or eliminate levels of prices that affect the nature of the revenues as well. Thus, a feedback loop is produced because the feedback strengthens the practice of the behavior. Thus, they might change prices and output to the existing market variations (International Solar Alliance (ISA), 2023). Thus, if strategic and tactic managers want to improve sales and stabilize or upgrade product pricing, those relationships must be understood.

3.1.3.1 The price setting strategy and price level

The BSHESS's pricing plan must consider system expenses. This covers the storage allocation between battery energy storage systems (BESS) and supercapacitor energy storage systems (SC-ESS), as well as the ramp rate needs. The price should also take into consideration the probability of power and energy compensation at particular confidence levels to ensure the solution's financial viability. One choice is to use a cost-plus pricing technique, where the price is set by marking up the whole cost of the system. This means a company which uses the cost-plus pricing technique calculates the selling price of a product as the total cost of production plus a markup percentage. The markup represents the targeted profit margin. Instead of concentrating on external factors like rival prices or consumer demand, this approach considers internal aspects like production costs (Luther, 2024).

The formula for cost-plus pricing is:

$$\text{Selling price} = \text{Total cost} + (\text{Total cost} \times \text{Mark-up percentage}) \quad (3.1)$$

Where, total cost includes direct materials, direct labour, and overhead expenses. Markup percentage is the desired profit margin expressed as a decimal. If the company wants to achieve a minimum of 20% profit margin. The markup percentage would be 0.20. The basis for the pricing is per kilo-watt-hour energy system. The prices will be variable based on parameters such as cost analysis and market research for competitors' prices for similar products. Additional factors include customization costs, scale of energy storage, and maximum current rating. In the BSHESS, the electronic control board and supercapacitors will be the extra units to integrate with the existing inverters system in the house. This system will be able to integrate with generally available power backup systems. Hence the major revenue generation components will be the control board and supercapacitors involved in the system. Based on the scale, discounts can be added to the cost structure. In the case of a supercapacitor and battery storage, the price level comes into view because it denotes the average price of a basket of goods and services that dominates market pricing for the revenue model.

3.1.3.2 Cost structure of BSHESS

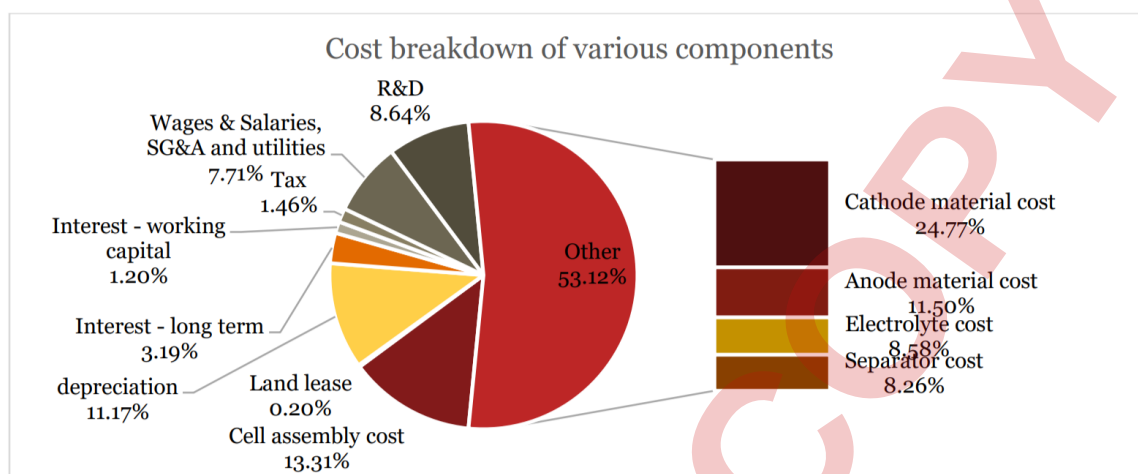


Figure 3.3 Cost structure to develop a supercapacitor/battery (Aayog, 2020)

Supercapacitors and batteries follow a cost structure of 54 % as material cost, 46% as indirect and overhead cost in mentioned in Figure 3.3. Experts says that the actual production cost is much lower than the available data. Such parameters are the capacity of the plant, its brand value and the skills of the technician, voltage of the cell (Dura et al., 2013). It is clearly shown that making cells in bulk reduces the production cost by a minimum of 3 times. Selling individual cells is much more profitable in the consumer electronics segment. The detailed data in the appendix supports this assumption which pridicts the profit as 5 to 6 times. The higher voltage of the pack reduces production costs (Bush, 2006).



Figure 3.4 Costs/kj have fallen quickly, with cost/kj dropping faster than cost/farad as the voltage is increased (Dura et al., 2013).

3.1.3.3 Revenue model based on pricing strategy and price level

In this context, market comparison, inflation, consumer demand, cost adjustments, economic indicators, and the consumer price index (CPI) are important components of price level. For example, it might set a premium price if comparable energy storage devices have high average prices. To stay competitive, a low market price could require changes. Inflation and other price level fluctuations can also affect customer behaviour and purchasing power, which in turn can affect sales. CPI, which helps determine the best pricing strategies by reflecting the state of the economy as a whole. The revenue model that aligns with the value proposition while considering other crucial factors can be established using the framework provided by the tables below (Clements & Nguyen, 1981).

Table 3.3 Framework for the revenue model

Production model	Manufacturing and assembly, lean manufacturing
Revenue stream	Product sales, direct sales (B2B)
Sales channels	Direct sales, online sales, distributors, resellers, etc.
What is paid	Money in return of per kilo watt energy system
Who pays	Client B2B, third-party manufacturers
For what is paid	Product (per kWh energy system), per-unit pricing
How it is paid	Credit, technology transfer fee, royalties on unit sales
How much is paid	Dynamic pricing
Price strategy	Cost-plus pricing
Customer lifetime value (CLV): ₹35000, CAC: ₹3011.2 per customer, Manufacturing cost: ₹13350, energy scale kWh: 1, The market price of looks-alike products ₹50000	
Application	Backup power, renewable energy, pulse power
The value proposition of competitors	Backup power application using an inverter and battery system. Long warranty life and harmonic-free supply.
The value proposition of the company	“We serve portable power seekers by providing customized hybrid energy storage systems for power enhancements and long-term cost savings”

Direct sales of the HESS units and supercapacitor modules to original equipment manufacturers (OEMs) in the backup power sector are one of the main sources of income. Negotiating per-unit pricing based on long-term supply agreements and

volume commitments would be part of the revenue model. Providing the HESS and supercapacitor technologies for various firms is a further way to make money. It also helps to penetrate the market with more offers without investing in the production plants' creation at the same time. A technology transfer fee at a regular schedule would also be necessary and other yearly fees would be calculated based on the licensee's income or number of units sold. Patents and license agreements liable to protect the intellectual property are to be taken with deeper consideration. Defining the detailed composition of these revenues will be made by referring to the behaviour of consumers and competitors. It also forecast the amount of profit and cost that will be associated with this business model for its execution. The detailed composition of these revenues will be adjusted concerning the reactions of consumers and competitors. It further predicts the profit and the expenses incurred by the business model for its operations.

3.1.3.4 Prediction, expenses and profit based on mini production plant of supercapacitor coin cell

The detailed calculation can be found in Appendix A. For a lean step to start the business, calculations for a small-scale supercapacitor plant were done. If the initial infrastructure is to develop quickly, It can be produced the coin cells for consumer electronics and sensor applications. The plan calculates the ROI as 114% and net profit ratio as 70%. Since, supercapacitor and battery production follow the same cost structure, these value are matching with several GOI reports (Aayog, 2020; Ministry of SSI, 2003; Pramanik, n.d.). The reports include the business plan of Lead-acid batteries and Lithium-ion batteries.

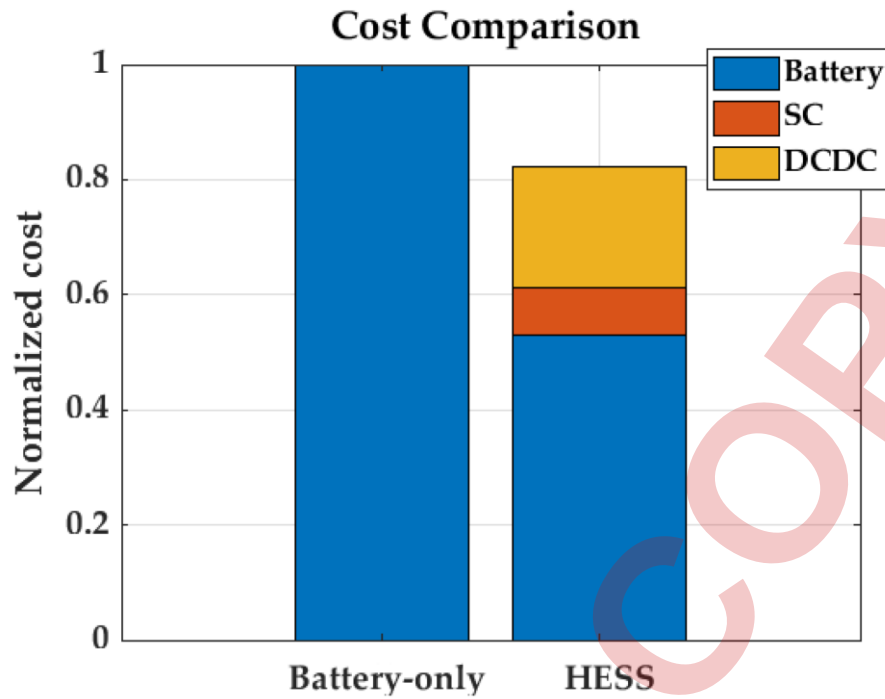


Figure 3.5 Cost-comparison of battery and BSHESS (Shan et al., 2024).

The production difference comes due to the difference in the specific energy of the battery and supercapacitor. An approximation is done based on the material required per kWh energy generation for the battery and supercapacitor. The factor for energy difference kWh per kg is about 10 (*Supercapacitors... What's Super about Them?* | Darcy Partners, n.d.). A few parameters also change this parameter, such as the maximum voltage of the cell, maximum current rating, and density of the active material. Hence the cost for a supercapacitor, based on the current path is expected to come down in the near future. Moreover, using BSHESS, the cost can be 60% lesser than using Li-ion battery on applications basis (P. Kumar et al., 2021b).

A hybrid bat-cap increases the life of the battery from a minimum of 25% based on the power level of the application. Overall, using these devices saves running costs 3 times for a bat-cap hybrid system and if the application is much more favourable to use

supercapacitor only (which is quite favourable in many consumer electronics appliances and micro-energy harvesting (MEH) systems), the user can save the money up to 1000 times due to 100 times more cycle life. In the worst case scenario, although, a supercapacitor initially costs 5-6 times more than a battery, it saves 1000 times later per kWh with clean and maintenance-free reliabilities. Further calculations indicate the lifetime value (LTV) of an acquired customer.

3.2 LIFETIME VALUE OF AN ACQUIRED CUSTOMER

The references come from the renewable energy market, where consumers use similar goods as home backup power systems and/or photovoltaic energy storage. The total revenue a company anticipates making from a customer throughout their relationship is known as the lifetime value (LTV) of an acquired customer. High LTV means that consumers are satisfied with a given product or service, they will continue using it time after time. It assists in figuring out the highest price at which a consumer may be acquired. It is an essential indicator for figuring out how profitable client acquisition initiatives will be in the long run (Jain & Singh, 2002). The following formula has been used to determine an acquired customer's LTV:

$$LTV = (ARPU \times \text{gross margin}) / (\text{Churn Rate}) \quad (3.2)$$

Where,

$$ARPU \text{ (in ₹ yearly)} = \frac{\text{Total Revenue}}{\text{Total Number of Users}} \quad (3.2a)$$

Average revenue per unit, or ARPU, is the total amount of money that a customer spends over a given period, such as a year. The percentage of revenue left over after deducting the cost of goods sold from total revenue is known as the gross margin. The percentage of customers who discontinue using a specific product or service within a given time frame is known as the churn rate.

Table 3.4 Monetary Values Projected to Consumer

Parameter	Value	Unit
Time window	1	Year
Marketing cost per kWh (amazon) (7% for item price > ₹ 15000)	700	₹
Shipping charge (heavy & Bulky Items)	370	₹
Closing fee	50	₹
Extra cost (travel regarding each 1kWh system in one visit)	300	₹
Total new customers acquired (in the first year)	100	₹
Total visits made (1 to 10 ratio of the successful deal)	1000	₹
Total extra cost	300000	₹
Total sales and marketing costs	301120	₹ .
Cost of customer acquisition to 1kWh system (CAC)	3011.2	₹
Product price per kWh (average)	13000	₹
Total costs to reach out to customers (this is 43% net profit which is near to the 35% gross profit of the production plant.)	15120	₹
Total number of users based on per kWh (minimum to maximum) (including retention rate)	47644 to 145778	kWh
(excluding retention rate)	95592 to 309496	kWh

Total revenue (based on TAM for the beachhead market) (minimum to maximum)	2.382 to 7.288	₹ (billion)
	4.78to 15.475	₹ (billion)
ARPU (approx.) (matching with market trends for per kWh sales)	50000	₹
Gross margin (based on the calculation of production plant for supercapacitors)	3500	%
Gross margin (based on the calculation of product, marketing and commissions)	4300	%
Churn Rate (average yearly)	5000	%
LTV (per kWh)	35000	₹
The average customer lifespan	2	Years

It is important to note here that developing a marketing or outreach network is much more economical than using other selling/marketing platforms. The LTV of an acquired customer is ₹35000. This means that, on average, each customer is expected to generate ₹35000 per kWh in revenue over their relationship with the company. If the LTV is lower than the cost of acquisition, the business may not be profitable.

3.3 COST OF CUSTOMER ACQUISITION

The entire amount of money a company spends in order to bring on a new client is known as the cost of customer acquisition (CAC). It includes a number of components, including advertising, sales, and marketing expenses. Additionally, it aids companies in determining the ROI (return on investment) of their marketing initiatives. Unsustainable company practices may be indicated by a high CAC in relation to the customer lifetime value (LTV) (Ali & Shabn, 2024). In order to ensure that they are successfully drawing clients while preserving profitability in the synthesis, characterization, and development of hybrid energy storage systems, organizations

must constantly analyze and optimize their CAC. The following is the formula to determine CAC:

$$CAC = \frac{\text{Total Sales and Marketing Costs}}{\text{Total New Customers Acquired}} \quad (3.3)$$

The customer acquisition cost (CAC) would be approximately ₹ 3011.2 per customer. This means that your company spends about ₹3011.2 to acquire each new customer for the BSHESS products. As we can see, CAS is less than LTV, hence it indicates the potential of the business.

3.4 COST CALCULATION OF SYSTEM PRODUCTION

Table 3.5 Cost calculation for BSHESS per 10 kWh system

Component	Range 1: Estimated Price ₹	Range 2: Estimated Price ₹	Description of 10 kWh system
Battery Pack	40,000	60,000	Cost of lithium-ion or other battery technologies for energy storage.
Supercapacitors	20,000	30,000	The cost associated with supercapacitor components for rapid energy discharge.
Inverter	15,000	25,000	Device that converts DC to AC power for usage in home appliances.
Control System	10,000	15,000	Includes management and monitoring systems for optimal performance.

Mounting Structure & Accessories	5,000	10,000	Structural components and electrical accessories needed for installation.
Installation costs	10,000	20,000	Labor and additional costs associated with the installation of the system.
Maintenance costs (Annual)	2,000	5,000	Estimated yearly maintenance expenses for upkeep and performance checks.
Total cost per 10 kWh	1,02,000	1,65,000	-
Total cost per 1 kWh	10200	16500	Average taken as ₹13350 per kWh module.

CHAPTER 4.

SYNTHESIS AND CHARACTERIZATION FOR ACTIVATED CARBON AND DEVELOPMENT OF HYBRID ENERGY STORAGE SYSTEM

This chapter covers the methodology for material synthesis and device development. A detailed overview of the research approach and methods has been given to investigate the development of portable BSHESS systems for portable power users. Technical prototype is a crucial form of creating a product prototype during the product development process. It entails making a model of the product to be launched on the market and this model can be altered to fit market standards. The goal of technical prototype development is to create a minimum viable product that allows testing and refinement. When developing a technical prototype, there are several critical factors to consider. Firstly, the prototype must be fully functional and able to perform the desired tasks effectively. Another issue arising with the prototype is scalability, or lack of, that is whether the selected design is suitable for mass production. In this process, the first about selecting and synthesizing the active energy material is done to proceed.

4.1 MATERIAL SYNTHESIS FOR DEVICE DEVELOPMENT

Nanotechnology has revolutionized supercapacitor technology, providing a new avenue for energy storage and improved performance. Nanofibers and nanoporous structures for supercapacitor electrodes have been developed to achieve this. The charge storage capacity is increased by these designs by giving electrolyte ions a broad surface area to

adsorb (Sarno, 2020). There are three types of supercapacitors: electric double-layer capacitors (EDLCs), hybrid supercapacitors, and pseudocapacitors. EDLCs work based on Helmholtz double-layer formation on the electrodes. Hybrid supercapacitors combine the capabilities of EDLCs and pseudocapacitors, whereas pseudocapacitors depend on redox reactions that take place on the electrode material's surface (Yaseen et al., 2021). Due to their outstanding properties, carbon materials particularly activated carbon employed actively in the electrode of the supercapacitor. These include little chemical activity, big surface area, and sound electrical conductivity. In addition, the highly developed porosity prepares good adsorption for electrolyte ions, which could improve the energy storage capacity of AC. Usually, organic precursors like coal, wood, or coconut shells are pyrolyzed at high temperatures without oxygen to create activated carbon. It is followed by activation using physical or chemical methods to create a highly porous structure. The AC comes from biomass, which is a plentiful and renewable source of carbon for the production of activated carbon in energy storage applications (Aparna et al., 2022).

4.1.1 Materials required for synthesis for activated carbon:

Bitter gourd pericarp (BGP) is the biomass source used in the manufacture of activated carbon. The activation agent is potassium hydroxide (KOH), and the activated carbon is cleaned with 0.1 M hydrochloric acid (HCl). The pericarp is peeled, chopped into tiny pieces, and repeatedly rinsed with deionized water in order to create activated carbon from BGP. After that, it is dried in a hot air oven set to 60°C for an entire day. The dried BGP is then carbonized at a rate of 4°C per minute for an hour at 400°C in an inert atmosphere to produce carbon powders. The carbon powder is activated by

swirling it with a 10 weight percent KOH solution for a whole day while keeping the BGP carbon powder to KOH ratio at 1:4. After six hours of heat treatment at 220°C, the resultant mixture is dried for twelve hours at 80°C. The dry slurry is pyrolyzed at 700°C for 1.5 hours at a rate of 5°C per minute in a tubular furnace using nitrogen gas as the last activation step. After that, 0.1 M HCl is used to wash the product till the pH reaches 7. The final activated carbon product, known as AC 700 and further characterized, is then produced by drying it once more for 12 hours at 60°C (Aparna et al., 2022).



Figure 4.1 Carbonization of BGP

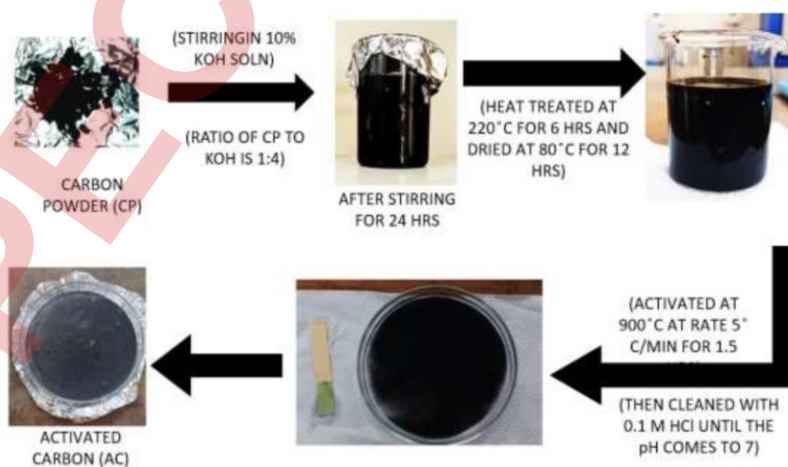


Figure 4.2 Activation of carbon powder

4.2 CHARACTERIZATION OF ACTIVATED CARBON

The chemical, morphological, and structural characteristics of activated carbon are evaluated using a range of methods. These are necessary to comprehend how effective it is in processes like energy storage and adsorption. X-ray diffraction which is often abbreviated as XRD is an analytic technique ordinarily employed to identify phase presence and the degree of crystallinity in a given material. Fourier transform infrared spectroscopy, or FTIR, is required to identify the functional groups available on the carbon surface. use scanning electron microscopy (SEM) to examine the porosity and surface composition. Utilizing Brunauer–Emmett–Teller (BET) analysis, adsorption capability is assessed by estimating surface area and pore size distribution. Raman spectroscopy also shows that the carbon material has flaws and a graphitic structure (Aparna et al., 2022).

4.2.1 X-ray diffraction (XRD)

A non-destructive approach for examining a material's crystal structure and crystallographic properties is X-ray diffraction (XRD). Constructive interference of the diffracted rays occurs when X-rays, which have wavelengths similar to the inter-planar distances of crystal lattices, are diffracted over high-density planes. This aids in figuring out the diffraction's Miller indices (hkl). XRD uses Bragg's Law, which is

$$n\lambda = 2d\sin(\theta) \quad (4.1)$$

The incident X-rays' wavelength is denoted by λ , the interplanar distance by d , and the angle of diffraction, or Bragg's angle, by θ . The diffracted photons from two planes

experience a route modification of $2d \cdot \sin(\theta)$ when X-rays of a particular wavelength collide with a lattice. Constructive interference results when this path-change distance equals the entering wavelength. High-intensity electrons are used to bombard a target material, usually molybdenum (Mo) or copper (Cu), to produce X-rays. XRD techniques come in two varieties: single crystal XRD and powder XRD. Single Crystal XRD rotates single crystal materials to determine Bragg's angles, while Powder XRD analyzes powdered samples without sample rotation. Using copper as the target material and a $K\alpha$ wavelength of $1.54 \text{ \AA}$, powder XRD was used in this investigation. The Bragg angle ranged from 10° to 90° .

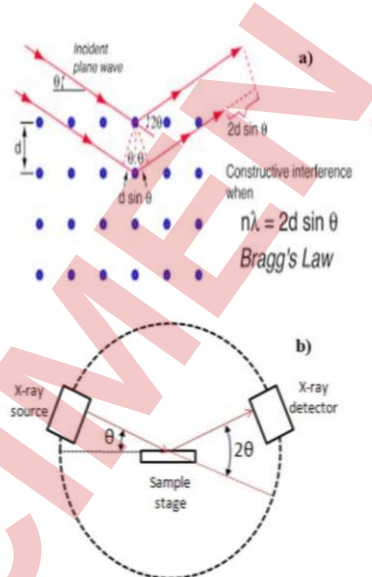


Figure 4.3 (a) Bragg's law, (b) Working of Powder XRD

4.2.2 Scanning electron microscopy (SEM)

The sample's morphology, topology, and compositional contrast are examined using the SEM technique. High-intensity electrons' interaction with the atoms on the sample's surface is the foundation of SEM. When an atom is struck by a very powerful electron beam, the incoming electrons destroy the atom's own electrons. The detector picks up

these knocked-out electrons, which are known as secondary electrons (SE). These provide the sample's morphology and topography. Secondary electrons reveal details about the sample's morphology since they are released by atoms that are present on its surface. The nucleus deflects some incident electrons as they go through the atoms. Backscattered electrons (BSE) are the name given to these electrons. The basis for backscattering is the atomic number-based effective repulsion of other electrons within an atom. Therefore, the compositional contrast of close atomic number elements is studied using backscatter electrons. Particles, rods, tubes, and other morphological structures can all be effectively analyzed with SEM. Studying particle aggregation and dispersity is beneficial. The morphology of the samples is examined in this study using SE electrons. Images with a resolution of up to $2\ \mu\text{m}$ and a magnification of up to 100kx were obtained using an input voltage of 30 kV.

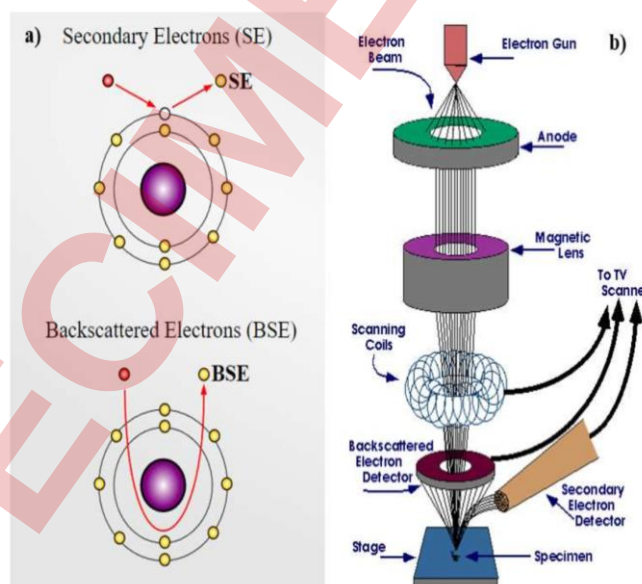


Figure 4.4 Principle of SE and BSE, (b) Working of SEM

4.2.3 Raman spectroscopy

A common analytical method in physics, chemistry, and materials science for examining rotational, vibrational, and other low-frequency modes in a system is Raman spectroscopy. Its foundation is the Raman Effect, which deals with molecules' ability to scatter light. The majority of light that interacts with a sample is scattered (Rayleigh scattering) without undergoing any energy change. The Raman Effect is caused by interactions with molecular vibrations, which alter the energy of a tiny portion of the scattered light. To determine the sample's molecular structure and chemical makeup, this energy change in the scattered light is quantified. When light interacts with materials, it scatters inelastically, which is known as the Raman Effect. The dispersed light and incident light are separated and analyzed using a spectrometer. Peaks at specific frequencies (Raman shifts) in the resulting Raman spectrum represent the different vibrational modes of the molecules in the sample. The Raman spectrum of a material is used to identify and characterize its molecular structure, crystallography, and chemical composition.

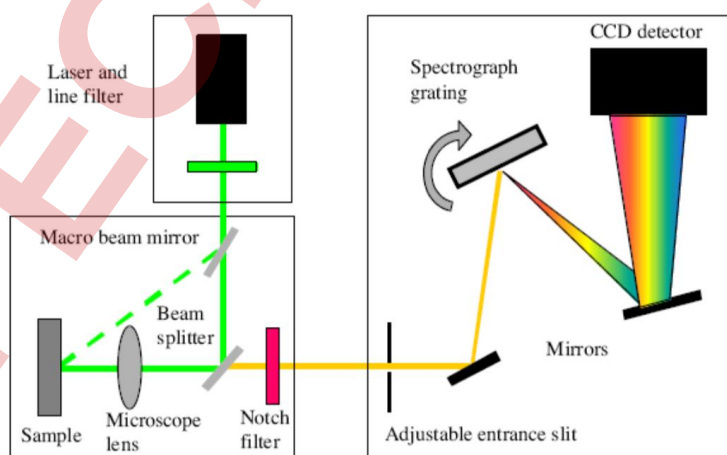


Figure 4.5 Working of RAMAN spectroscopy

4.2.4 Electrochemical studies

Electrochemical analysis focuses on the interactions that happen between substances with the exchange of electrons. To comprehend and describe the behaviour of electrochemical systems, these investigations are carried out utilizing electrochemical techniques and procedures. Applying a potential that changes linearly with time and then reverses direction is known as cyclic voltammetry or CV. Redox reactions, electron transfer pathways, and electrode surface stability are all covered by this method. During the investigation, the redox potential peaks and the current response to changing voltage are examined. The time it takes for a material to charge and discharge to attain a particular potential is studied using chronopotentiometry (CP) or galvanostatic charge-discharge (GCD). Electrochemical impedance spectroscopy (EIS) measures how an electrochemical cell reacts to small amplitude alternating current signals throughout a frequency range. A three-electrode setup including a working electrode, counter electrode, and reference electrode submerged in an aqueous electrolyte solution is used for electrochemical investigations. Storage materials are characterized through electrochemical investigations. We may be able to examine and classify the kind of energy storage mechanism used in the active material with the aid of the test profile.

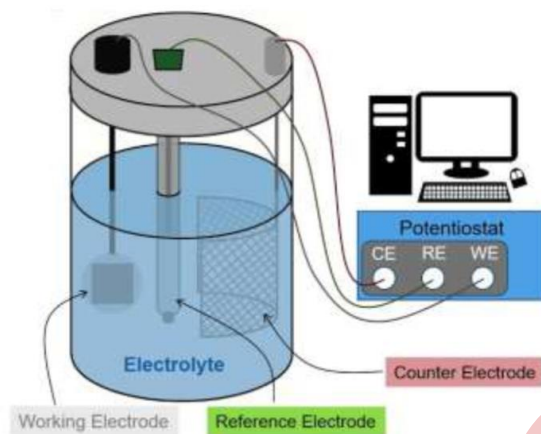


Figure 4.6 3-electrode system for electrochemical studies

4.2.5 Impedance analyser

An impedance analyzer, sometimes called an impedance spectrometer or impedance meter, is a specialist instrument. It is used to ascertain the electrical impedance of a system at various frequencies. Electrical impedance, which is made up of X and R , is the opposition given to the AC in any circuit. When a small alternating voltage signal is applied to the system under test, the impedance analyzer measures the current response that results. The impedance analyzer can produce a figure called an impedance spectrum by altering the applied signal's frequency and measuring the accompanying current. The complex impedance is a function of frequency as:

$$Z = R + jX \quad (4.2)$$

It is displayed in this spectrum. The impedance spectra obtained from an impedance analyzer provide crucial information on the electrical properties and behavior of the system under study. Once the required material has been successfully validated by these

characterizations, the properties of the materials help in the development or modeling of the required prototype device.

4.3 PROCESS FOR DESIGNING AND BUILDING THE PROPOSED DEVICE

4.3.1 Identify key assumptions for the product

Critically identifying the customer's problem and confirming important assumptions that a problem exists are crucial steps in the product design and development process. A precise problem definition that expresses the needs of the client comes first. This study makes the assumption that sellers of customized energy systems require a simple method for integrating HESS with current energy storage technologies. By gathering information about customer behavior and pain areas through surveys, interviews, and observation, a firm can better understand its customers' experiences. Key assumptions regarding the problem's severity and target consumer segment are established following data analysis. After that, these assumptions are ranked according to their significance and level of risk, with an emphasis on verifying the most important ones. A potential solution is established through further research, prototyping, and testing (Aulet, 2024). The Table 4.1 below summarizes the objective for this and is being used as a framework to test these key assumptions further.

Table 4.1 Key assumptions driven by field study

Identify the target audience	Potential customers as customized energy system installation agencies and end users as off-grid and backup power system users.
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Identification of problem statement/hypothesis (customer pain point or need)	Customized energy solution providers must enhance their services wider market reach. They must be able to integrate renewable sources and meet high power demands overcoming compatibility issues. This will help clients navigate modern energy markets, achieve energy capacity expansion, and attain more customers for easy portability needs.
Proposed solution (key features and benefits)	Customized energy solution providers should utilize BSHESS to create scalable, resilient energy solutions. This integration will optimize energy usage, reduce costs, and support clients in achieving sustainability goals.
Assumptions and their priority based on customer survey (rank of the assumptions)	Diverse client needs Demand for cost-effective and sustainable solutions Client focus on sustainability Potential of BSHESS Importance of system resilience Increasing complexity in energy markets Evolving energy technologies Lack of comprehensive advisory services and software tools
Identify the riskiest assumptions (These are the assumptions that should be tested first)	Diverse client needs Demand for cost-effective and sustainable solutions Increasing complexity in energy markets

Feasibility of implementation	The appendix contains the viability of these eight assumptions. Surveying all of the aforementioned assumptions is possible.
Expected performance of product	It is anticipated that the energy solution based on BSHESS will offer an energy management system that is resilient, affordable, and sustainable. It can deal with difficult power loads, can easily integrate with renewable power systems and, energy use to address various needs of the customers.
Willingness to pay for product	More than 60% of clients' agreeing to the value proposition is considered as willingness to pay

4.3.2 Testing key assumptions for the product

The process typically starts with listing all assumptions related to the product which have been shown in the previous section. This work conducted surveys/interviews with potential customers to gauge their interest in such a system compared to traditional battery-only solutions. Additionally, this work conducted testing on a prototype to check where it the expected performance is achievable in contrast with battery-alone systems. By systematically testing these assumptions, whether through customer feedback, prototype testing, or performance evaluations, the business can identify which beliefs are valid and which need to be adjusted or discarded. This process ultimately leads to a more successful launch and adoption of the hybrid energy storage system (Osterwalder & Pigneur, 2013). The table below summarizes different methods,

which help to define the minimum viable business product and detailed data of testing can be found in the appendix.

Table 4.2 Assumption testing method from asking (left side) towards building (right side)

Qualitative research	Quantitative research	Demand testing	Prototype testing	Product analysis
User interviews (based on problem statement)	Scenario testing	Ad testing	Impersonator	A/B testing
User Feedback(based on value proposition)	Paraphrase testing	Dry wallet	Wizard or Oz	Feature adoption
Ethnography (based on target customer)	Quant survey	Fake door	Wireframe testing	“Ship it and see”

In essence, there are vast methods that can be applied to effectively validate the assumptions of the BSHESS. Through encouraging the understanding of different wants and desires of the clients, methods such as user interviews, surveys and ethnography help in the identification of the motivation and pain point. Methods such as feature adoption and scenario testing guarantee that the system meets user needs and successfully resolves their energy management issues. Customized energy solution providers can verify the hypotheses as BSHESS offering to successfully satisfy market demands by utilizing these various techniques (Busetto et al., 2020). The post-development of a minimal viable product is applied to several of the techniques listed in the above Table 4.2. These have not been conducted for this work. Based on the feedback from the interview, the features of the minimum viable product have been finalised.

4.3.3 Define the minimum viable business product (MVBP)

Determining the key characteristics and capabilities of the hybrid battery and supercapacitor storage system is necessary to define the minimum viable business product (MVBP). For early adopters to be satisfied and the company concept to be validated, these must be part of the initial product offering. Delivering core value while reducing development expenses and time to market should be the MVBP's primary goal. Identifying key features, concentrating on the user experience, verifying assumptions, creating a prototype, and getting feedback are all important phases in defining the MVBP. It should be noted that MVBP may encompass a variety of energy storage capacities (Rocha et al., 2022). Iterative adjustments must be made in response to input on the prototype's concept, and the prototype should show that the system can match the value proposition. Below is a Table 4.3 that maps the interrelated key points for defining minimum viable product. For the hybrid battery and supercapacitor storage system, the MVBP might include a functional prototype that combines a lithium battery and a supercapacitor bank. This framework helps to acquire a paying customer or a letters of intent (LOIs).

Table 4.3 Framework for acquiring a paying customer

Key steps	Description	
Identify core features/functions for early adopters	Determine essential functionalities needed to meet customer needs.	Energy storage capacity of 100 watt-hours. Quick charge and discharge capabilities. Basic connectivity to an ESS for monitoring. Less cost and time
Focus on user experience	Ensure the user interface is intuitive and easy to operate.	User-friendly operation, displaying real-time data on battery status, energy consumption, and alerts
Validate assumptions	Test key assumptions about customer	Surveys or focus groups to confirm priorities like fast charging

	preferences for such a product	capabilities, cost-effectiveness and reliability
Prototype development	Create a working prototype that incorporates identified core features.	Functional prototype combining a lithium battery and supercapacitor to demonstrate energy storage and release
Gather feedback	Create a working prototype/model that incorporates identified core features and gather feedback on that.	Using various surveys and tests to collect data on the experience of key consumers, in order to see how well a product is performing and what could be done to make it easier for consumers to understand.

4.3.4 Acquire paying customer of a letters of intent (LOI)

Getting letters of intent or paying customers will ensure market demand before full-scale production begins. During this stage, prospective customers are asked to express interest in purchasing the goods, usually by submitting a letter of intent (LOI) or written commitment. With LOI, it is possible to regulate the further development of new goods and receive help in obtaining capital. Identification of the target client and provision of plenty of consideration to him or her needs and wishes are the vital aspects of this plan (Aulet, 2024). Therefore, a structured and appealing value offer is produced. There are several examples of such projects that may be helpful for the promotion of the product and getting useful feedback: direct marketing, topical conferences, and events, and Business & Industrial networking events. Through field demonstration, this action has secured a letter of intent from the market segment. The Table 4.4 below lays out the interrelated crucial aspects for acquiring letters of intent (LOIs) or paying clients for BSHESS.

Table 4.4 Process for acquiring paying customers/LOI

Key steps	Description	Details
Identify target customers	Determine potential customers who would benefit from the hybrid energy storage system.	Residential homeowners, Commercial businesses, Industrial applications
Conduct market research	Engage in thorough research to understand customer needs, preferences, and pain points.	Surveys, Interviews, and Focus groups on LinkedIn
Develop a value proposition	Reach out to identify target customers through various channels to present the product and gather feedback.	Improved energy efficiency, cost savings, and enhanced performance compared to traditional systems
Create prototypes for demonstration	Reach out to identify target customers through various channels to present the product and gather feedback.	Functional prototypes to allow potential customers to experience the product first-hand experience.
Engage with potential customers	Reach out to identify target customers through various channels to present the product and gather feedback.	Direct outreach, networking events, Industry conferences, LinkedIn conversations
Offer incentives for early commitment	Encourage potential customers to commit by providing incentives.	Discounts for early orders, exclusive access to new features, extended warranties
Secure letters of intent (LOIs)	Work towards obtaining formal LOIs from interested customers to validate demand.	Formal agreements indicating intent to purchase once available

The business has drawn a model of the hybrid battery and supercapacitor system and presented a business proposal of how to produce such an item as well. At the event, the business presents several interested businesses and shows how system can make certain savings in energy cost and ensure reliability. To incentivize early commitments, the business plan offered a detailed market survey for potential industries/partners. After presentations and discussions, three parties expressed strong interest and the business idea successfully received a funding of 60L. Following this, the business will immediately embark on marketing; at the very least as a minimum viable product, the business will ensure that the hybrid system is sold once it is marketed. Besides providing evidence of the market's need for such a product in funding this startup, it also provides evidence that there's a genuine interest for such technology in the market.

For launching of the hybrid energy storage system, the immediate necessity is to develop and simulate a variety of hybrid systems. Also, there is an immediate need to develop large-scale cylindrical supercapacitors and research for active material. After this, the end product as a whole integration of battery, supercapacitor, load and supporting electronics must be developed.

4.3.5 Methodology for making cylindrical cells

The components of cylindrical cells include active material, coated on electrodes (cathode and anode), a separator, an aqueous electrolyte, a current collector, and sealing components. The active material for this experiment is the synthesized activated carbon AC-700. The electrode slurry is prepared using active material, binding agent, conducting agent, and solvent. In this study, the binding agent is polyvinylidene fluoride (PVDF), the conducting medium is carbon black (CB), and the solvent is *n*-methyl pyrrolidine (NMP). The slurry was prepared by mixing 90% active material (AC 700), 5% binding agent (PVDF), and 5% conducting medium (CB). After this, solvent (NMP) is added to mix the slurry uniformly. The electrode slurry is brush coated on to current collectors and the coated current collectors are dried at 60 °C for 12 hours. Once the electrodes are dried up, a single-cell of cylindrical cell is assembled by placing a separator. The separator is a filter paper between two coated sides of electrodes such that the active material coated faces each. The assembled cell is wrapped with a sheet of aluminium foil and placed inside a plastic cylinder of similar size and sealed. The aqueous electrolyte for this study is 1M KOH. The preparation steps involved etching the foil, making the slurry, cutting the electrode, coating the electrode, and drying the electrode. The cell is intended to use in a BSHESS. For validating the basic idea a

simulation model of a hybrid system containing a supercapacitor is simulated and studied in detail. This set a motivation to use actual supercapacitors in such systems (Lokhande et al., 2019).

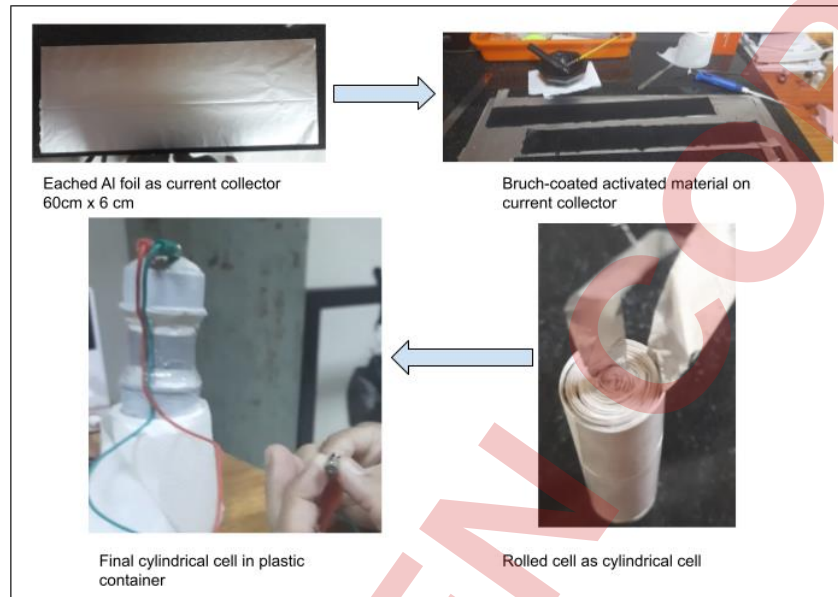


Figure 4.7 Cylindrical-cell-making process for supercapacitor

4.3.6 Hybrid energy storage system

In order to satisfy the demands of portable power users, customized hybrid energy storage systems were designed and modelled during the device development process. This necessitated looking at several device architectures and assessing how well they performed in terms of energy density, power density, and cycle life, including batteries and supercapacitors. To guarantee dependability and efficacy, simulation and testing were carried out under a variety of operating scenarios, such as load-based switching and a broad range of loads. To guarantee safe operation, one of the main factors for the hybrid energy storage system was to house the parts in a sturdy, well-ventilated

enclosure. To obtain the required voltage and capacity, the lithium-ion battery pack was initially put together in parallel using separate supercapacitor cells connected in series.

A specially created printed circuit board (PCB) that functioned as the primary control and interface board made it easier to integrate the battery and supercapacitor modules. Current and voltage sensors were included on this PCB to track the health and charge level of both parts. Through regulation of the energy storage system's charging and draining, the control electronics guaranteed optimal use in accordance with application standards. The control board's microcontroller unit (MCU) applied sophisticated energy management strategies by running firmware. Power will be split between the battery and supercapacitor based on the load profile, charge level, and degradation state (Gómez et al., 2023). The hybrid system was outfitted with a number of security elements to guarantee dependability and safety. Thermal management systems, overvoltage protection, and overcurrent protection are among the characteristics. The enclosure was built with sufficient cooling components, such as fans or heat sinks. Value creation for the user is the aim of the simulation and product; the results of these developments are discussed in the next chapter.

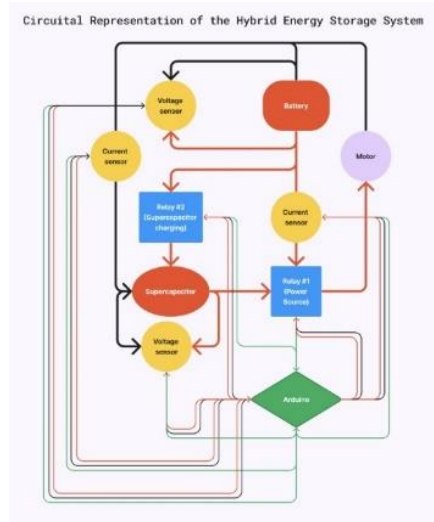


Figure 4.8 Block-diagram of BSHESS

The initial parameters for the system are as follows:

Total load consumption = Volt* Ampere*Hours = $V \times I \times t$ = $80 \times 10 = 800W$ or $0.8KWh$

Battery Rating = $12V/75Ah = 12V \times 75Ah = 900VAh$ or $0.9KWh$ (approx.)

Super Capacitor Rating:

We know that $Q = C \times V$, or $Q = \text{Ampere} \times \text{Time}$, $C = Q/NV$, $C = 175/12$ $C = 6.25$ Farad

Therefore supercapacitor rating require: $6F/12V$ (approx)

CHAPTER 5.

FINDINGS AND DISCUSSION

5.1 THE VALUE TO CUSTOMERS

The study conducted on potential customers demonstrates the value and services the goods aim to offer in fulfilling their energy storage needs. Delivering a dependable, high-quality solution that combines the advantages of a supercapacitor and a battery, such as faster charging and discharging and a better energy density, is part of this. Easy integration for current renewable energy, backup power, and pulse power appliance systems is another benefit. Furthermore, value-proposition provides thorough customer service at every stage of the buying process, including pre-sale consultations to assist clients in selecting the best system. This aids in deciding on their particular hardware demands, installation services, and maintenance and monitoring requirements in order to optimize system performance and profit from the purchase. To guarantee that everything is put up properly and securely, the company would offer expert installation after the sale. "We serve portable power seekers by providing customized hybrid energy storage systems for power enhancements and long-term cost savings," is the value proposition offered to customers.

5.2 FULL LIFE CYCLE USE CASES

This refers to the complete tracking of all of the interactions the client experiences with the product or service. They start this action as soon as they recognize a need and keep going till they are marketing the brand. Understanding the client journey, identifying

issues, and enhancing the entire experience all depend on this concept. In addition to improving customer happiness, this thorough mapping helps guide marketing plans, customer service programs, and product development—all of which contribute to the expansion of the business. Its eight essential phases can be used to quantify the value proposition (Patti et al., 2020).

Table 5.1 Process flow for the product in the context of business idea

1.	Recognizing a need	Decentralized power generation solutions that can integrate with renewable energy sources with high power density, long cycle life, and safe and fast charging capabilities. Scalable, cost-effective and modular systems that can handle pulse power requirements effectively.
2.	Discovering the product	The customer discovers the product online and/or through existing vendors/contractors and via personal outreach.
3.	Evaluating its suitability	The customer found the system is the best fit for their needs by seeing a demo.
4.	Purchasing	The product will be purchased through the vendor over email or personal call.
5.	Installing	The customer installs the product at the location of end user and provides manuals/video tutorials and other supporting material for the reference of the customer.
6.	Using	The end user starts using the system in their backup power setup, renewable energy system and pulse power applications.
7.	Assessing value	Over time, the customer evaluates the value gained, such as reduced energy bills of the user, increased reliability and better performance.
8.	Payment	The customer will pay the business on the basis on commission per system. The end user will pay for the product using a credit card/cash in return of a bill and warranty card to the customer.
9.	Receiving support	If the end-users will encounter any issue, they will reach out to business. The business will provide support via the vendor.
10.	Sharing feedback or referrals	Customers and end user will be approached for feedback on a regular basis.

5.3 QUANTIFICATION OF VALUE PROPOSITION

Quantifying value proposition requires key performance indicators (KPIs), which are essential for effectively measuring the success of a BSHESS business. KPIs should relate to the company's strategic objectives; It should target key areas including product and market development, customer service and organizational performance. The KPIs should adhere to the SMART criteria, where S stands for specific, M is measurable, A is achievable, R is relevant, and T is time-bound. The customer acquisition cost (CAC), which measures the total cost of acquiring a new customer is one of the important KPI. An optimal value proposition can be evaluated by focusing on these (van de Ven et al., 2023).

The quantization for this is as follows: (a) potential increase in energy efficiency by 15-30% compared to standard systems, (b) Supercapacitors can charge and discharge in seconds, providing power density that is 10-100 times greater than traditional batteries. For instance, if a battery system can deliver 100 kW, a hybrid system could potentially deliver up to 1,000 kW, allowing for quicker response times, (c) if a renewable energy project typically incurs \$100,000 annually in energy costs, a 20% improvement could save \$20,000 per year. This is because a current lithium-ion battery is expected to last for roughly about 5-10 years but with a hybrid system this could take even 10-15 years. If the replacement cost of a battery is \$500,000, extending the lifespan by 5 years could save \$100,000 in replacement costs over that period. Additionally, the lower maintenance frequency can lead to further savings, estimated at 10-15% of annual operational costs. These very specific quantization helps to define the core capability of the business.

5.4 DEFINING THE CORE CAPABILITY OF THE BUSINESS MODEL

Finding and expressing the essential features that give the product uniqueness and attract for buyers is known as "defining the core." It might contribute to the hybrid system's positioning as the industry's best option. This is the most strategic concept upon which all product specification, positioning and selling propositions are benchmarked to provide coherence and clarity about the proposition that the offering brings to the market. When asked to rate the relative importance of the different criteria, the poll participants selected cost reduction as the most important criterion. Respondents' top investment goals were, predictably, consistency, availability, less stress, lower risk, and lower expenses. A fundamental study of rivals and market segments is necessary to optimize these factors (Papula & Volna, 2013).

Table 5.2 Core capabilities of the business idea

Key capabilities: Service through the product	Customized hybrid energy storage system integrates high power density supercapacitor and high energy density lithium-ion battery.
Key capabilities: Service through the material	The business also aims to develop activated carbon for energy applications. Understanding the electrochemical properties and performance characteristics of carbon-based materials is critical for assessing their suitability for supercapacitor applications.
Key capabilities: Service through the consultancy	The business aims to help customers to find hardware selections for versatile load demands.
Key capabilities: Cost-effectiveness	The business aims to deliver products without import costs. Also, the product will take 20% of the energy cost of the load calculation for the customer's side.
Competitive advantages	With a lifespan of 30% longer than conventional battery systems and 20% higher efficiency, our product provides a cost-effective solution for residential, commercial, and industrial customers seeking reliable, high-power energy storage.
Target applications	This could include renewable energy integration, electric vehicle charging, or industrial applications requiring high-power, short-duration energy storage.

Minimum technology readiness	The business currently holds a TRL of 8-9.
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5.5 COMPETITORS AND MARKET ANALYSIS

The product is flexible and could be advertised as the latest innovative solution that provides high performance in comparison with the value proposition. “The business prospectus for the proposed business outlines our hybrid storage system that integrates the proficiency of supercapacitors and the energy density of a lithium battery,’ the pitching for the business entails. With 20% better efficiency per kilowatt and 30% lower lifetime costs than its nearest rivals, the system is the sensible option for backup power applications and renewable energy integration. There is a ten-year warranty on it. On its part, by pointing out the benefits and competitive position of the hybrid storage system to the current market, can get customer willing to part with their extra cash to make purchase with the company. In addition, this positioning also offers solid support to numerous marketing, sales and product specification (Aulet, 2024).

Table 5.3 Close competitors and their insights

Company Name	Address	Products Offered	Serving Market	Potential Gaps/Issues
Tata Power Solar	Mumbai, Maharashtra	Solar panels, EPC services, solar rooftops	Residential, commercial, industrial	High competition and price sensitivity
Adani Solar	Ahmedabad, Gujarat	Mono & bifacial solar modules	Utility-scale projects	Supply chain constraints
Servotekh Power Systems	New Delhi	Solar panels, inverters, batteries	Government and commercial sectors	Limited brand recognition outside major cities

Waaree Energies Ltd.	Mumbai, Maharashtra	Solar panels, EPC services	Residential and commercial	Dependence on government policies
Vikram Solar	Kolkata, West Bengal	Photovoltaic modules	Commercial and industrial	High operational costs
Goldi Solar	Surat, Gujarat	Solar panels	Residential and industrial	Market saturation in certain regions
Loom Solar Pvt. Ltd.	Faridabad, Haryana	Solar panels	Residential	Limited manufacturing capacity
Amplus Solar	Gurgaon, Haryana	Distributed solar assets	Industrial and commercial	Regulatory hurdles
Mahindra Susten	Mumbai, Maharashtra	EPC services for solar projects	Commercial and industrial	Project execution delays
Sunsure Energy	Bengaluru, Karnataka	Large scale rooftop solutions	Commercial	Competition from established players
Fourth Partner Energy	Hyderabad, Telangana	Rooftop solar installations	Commercial and industrial	Need for innovative financing solutions
HM Solar	Surat, Gujarat	Solar installation services	Residential	Customer acquisition in competitive markets
Rayzon Solar	Ahmedabad, Gujarat	PV modules	Residential and commercial	Scaling production to meet demand
Nujum Solar	Surat, Gujarat	Rooftop solar installations	Residential	Limited brand visibility
AEG Power Solutions :	Netherlands (http://www.aegps.com)	Covering peak demands, shifting peak currents, frequency control and load distribution	Hybrid energy storage system for space heating. Voltage regulation, full off-grid power generation.	Modular storage container through batteries only
Jakson Group	India (https://www.jakson.com/distributed-energy/)	solar rooftop systems and battery energy storage systems	Hybrid energy storage system for microgrid application, also used DG sets in bulk.	Modular storage container through batteries only

Hybrid Energy Storage Solutions Ltd.	Spain (http://www.hesstec.net)	Focused on the calculation of the optimal sizing and operation strategy for different scenarios, at the same time that generate the optimal business model.	Hybrid Energy Storage Solutions through batteries, supercapacitors and flywheel	High maintenance due to company specific components
VoltaGrid	(http://www.voltagrid.com)	Off grid and on grid	Gas generation	Modular storage through container through batteries only
MNA ENERGY SDN BH	Malaysia (https://www.mnabattery.com/about/)	Graphene, Ultracapacitor, and Lithium-Ion Battery to form the “GUC hybrid energy storage system	UPS and Back-up Applications	The product is used for energy module swapping, it does not come with BSHESS
SPEL	Pune	Advanced energy storage solutions, supercapacitors, lithium-ion batteries.	Energy storage applications, renewable energy sectors.	Competition from established battery manufacturers; reliance on R&D for innovation may delay product rollout.
ARCI	HYD	Research and development in advanced materials for solar applications	Research institutions, solar manufacturers.	Limited direct market presence; primarily focused on R&D rather than commercial solar installation.
Nano and Soft Matter Sciences (cens),	Bengaluru	Nanotechnology solutions for solar energy applications.	Academic and industrial research sectors.	Limited commercialization of research; potential difficulty in scaling to mass production.
Nano Science and Technology (INST)	Mohali	Research institutions, solar technology developers.	Consultancy in energy materials- non-conventional electrifications to houses	Limited commercialization of research; potential difficulty in scaling to mass production.
Golden Arrow Wireless Pvt. Ltd.	Uttar Pradesh	Wireless solutions integrated with solar technology.	Telecommunications, renewable energy sectors.	Niche market focus may limit growth; competition from larger firms with

				broader product ranges.
Avghni Solar	Hyderabad	Solar rooftop and ground mount systems for residential and commercial use	Residential communities and businesses in Andhra Pradesh and Telangana	Regional focus may limit expansion opportunities; competition from larger national firms.
Amplus Solar		On-site solar projects, battery storage solutions, energy management services	Commercial and industrial customers across various sectors	High initial investment costs for customers; regulatory challenges in different states.
Su-Kam Power Systems Ltd.		UPS systems, batteries, solar solutions	Residential and commercial sectors globally	High competition in the inverter market; need for continuous innovation to retain market leadership.
Microtek		UPS systems, inverter batteries, home power backup solutions	Residential and industrial sectors	Market saturation with many competitors; potential quality control issues affecting brand reputation.
SolarSquare	Maharashtra	Rooftop solar panel systems, maintenance services	Residential homes and housing societies	Customer concerns regarding installation quality; reliance on subsidies for affordability.
Vortex Solar	Uttar Pradesh	On-grid, off-grid solar systems, solar plant AMC services	Residential customers	Dependence on government subsidies and policies; competition from other solar providers may affect market share.
Gegadyne Energy	Mumbai	Solar power solutions and energy management systems.	Residential and commercial sectors.	Emerging company status may affect customer trust; need for stronger marketing to compete with established brands.
Batcap beyond energy solutions LLP	12. This work	Hybrid energy storage systems	consumer electronics and renewable energy	Limited reach and depending on research and development

Due to pricing battles and difficulties obtaining project permissions and funding, the Indian solar business is getting oversaturated. High operating expenses and difficulties acquiring raw materials are problems for businesses like Adani Solar. Smaller businesses can set themselves apart, though, by providing specialized hybrid energy storage systems, serving diverse market niches, and providing other services like frequency regulation, load leveling, and backup power alternatives. A solid basis for product specifications, sales, and marketing is provided by this positioning (Aman Mohd, 2024).

5.6 SPECIFICATIONS AND CHARACTERIZATION RESULTS FOR THE MATERIAL AND DEVICE

Activated carbon (AC) made from the biomass of bitter gourd pericarps and activated with KOH and 700°C is one of the product specifications. Pore creation is confirmed by SEM images, AC generation is confirmed by RAMAN spectroscopy, and EDLC behavior is demonstrated by electrochemical investigations. AC is used as the electrode, while aluminum sheet and carbon cloth are used as the current collectors to create a cylindrical cell. A 10-watt motor is used to test the performance of the capacitor. The amorphous form of AC is confirmed by XRD.

5.6.1 X-RAY diffraction (XRD):

The figure displays the XRD patterns of activated carbon, known as AC 700, and carbon powder (CP) prior to activation. A broad and intensified peak is also seen at position 25 and 43 (2-theta values), which represents (002) and (101) planes in both CP and AC

700 from XRD. This demonstrates that both samples contain carbon (Kalyani & Anitha, 2013). On the other hand, CP contains impurity peaks that AC 700 does not. When activated carbon is synthesized, heat treatment removes the carbon's crystalline form and creates an amorphous form with fewer impurities. Cleaning and drying procedures are used to remove these contaminants. It is for this reason that large peaks in the XRD patterns point towards the fact that the sample contains amorphous carbon (Xu et al., 2018).

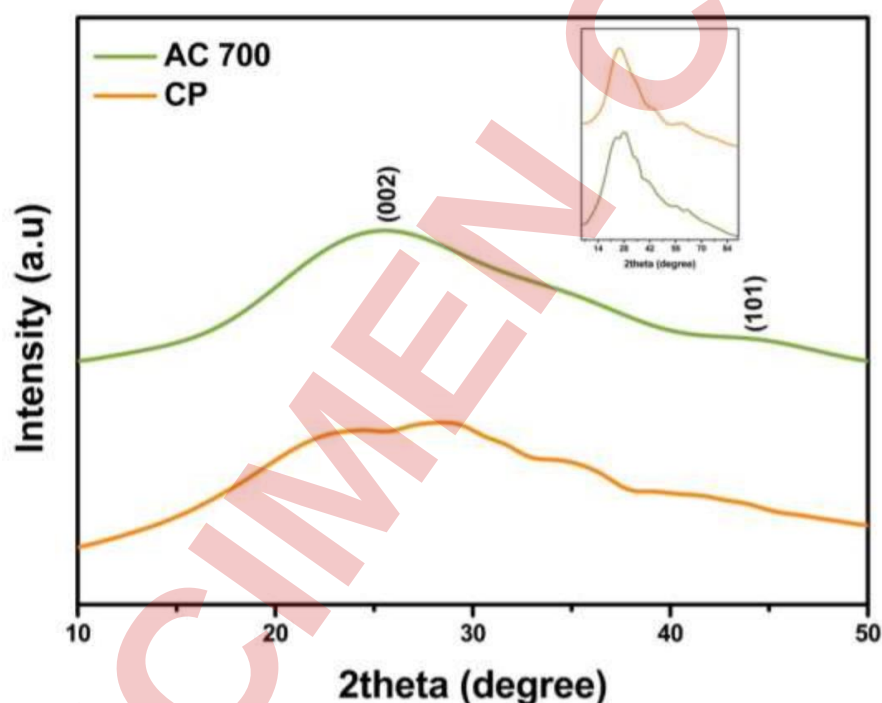


Figure 5.1 XRD graph of AC 700 and CP

5.6.2 Scanning electron microscopy (SEM):

The figure displays the SEM micrographs of the carbon powder and AC 700. Carbon is present in both CP and AC, according to the XRD study. SEM pictures, however, clearly show the impact of chemical activation (Dungani et al., 2022). The amorphous solid structures of carbon are visible in the SEM pictures of CP. The appearance of

nanometer-sized pores in the AC 700's SEM images, however, indicates that KOH has activated the carbon. Based on SEM scans, the typical pore diameter of AC 700 is around 2 μm . SEM analysis thus validates carbon activation.

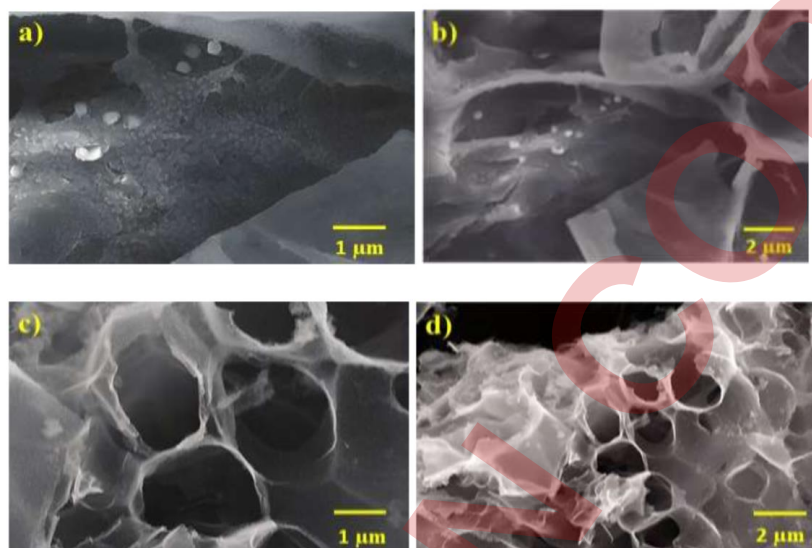


Figure 5.2 SEM images of (a & b) Carbon powder, (c & d) AC 700

5.6.3 RAMAN spectroscopy:

The figure displays the AC 700's Raman shift spectrum. A 532 nm excitation wavelength Nd-YAG laser is used for the Raman spectroscopy. The Raman shifts for the D and G bands were 1351 and 1568 cm^{-1} , respectively. This is in line with published research that supports the creation of activated carbon (Y. Liu, Liu, et al., 2017). The intensity of the sample's defective carbon is represented by the D band, while the graphitic carbon is represented by the G band. 1.75 is the I_D/I_G ratio, which is the ratio of the D to G band intensities. The high percentage of chemical activating agent (KOH) supplied during synthesis may be the cause of this high I_D/I_G result, which indicates an increase in faults.

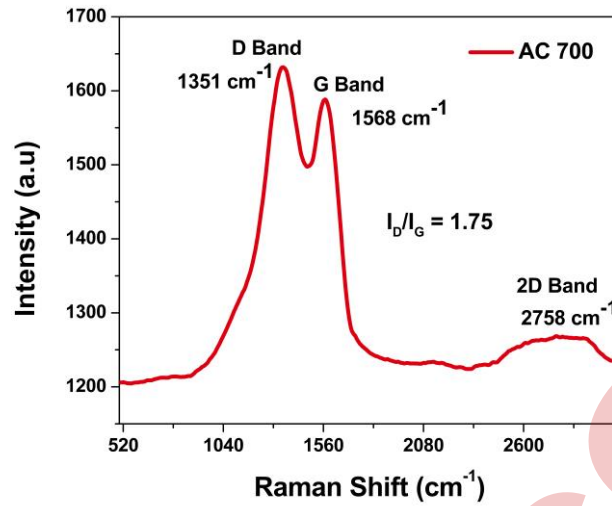


Figure 5.3 Raman spectra of AC 700

The following formula can be used to determine the separation between AC 700 flaws:

$$L_d = 2.4 \times 10^{-10} \times \lambda^4 \times (I_d/I_g)^{-1} (nm) \quad (5.1)$$

where L_d is the distance in nm between the said defects. One of the more popular measures of the laser's excitation wavelength or λ is in nanometers (nm). The signal intensity in the non-defective region is represented by I_G while that in the defect-related region is represented by I_D . The L_d for AC 700 is determined to be 10.98 nm. The density of faults is calculated using the formula below:

$$n_d = \frac{(1.8 \pm 0.8) \times 10^{22}}{\lambda^4} \times \frac{I_d}{I_g} cm^{-2} \quad (5.2)$$

The defect density per square centimeter (cm^{-2}) is denoted by n_d . λ is the laser's excitation wavelength, usually expressed in nanometers (nm). The AC 700's n_d measures $5.024 \times 10^{37} cm^{-2}$.

5.6.4 Electrochemical studies:

A three-electrode setup is used for the electrochemical investigations. Platinum plate served as the counter electrode, saturated calomel as the reference electrode, and active material as the working electrode. The figure displays the AC 700's cyclic voltammogram with two distinct electrolytes. The electrolytes used in the experiment were 1 M Na₂SO₄ and 6 M KOH. Different voltage scan rates, such as 10 mv/s, 20 mv/s, 30 mv/s, 40 mv/s, 50 mv/s, 75 mv/s, and 100 mv/s, were used. The CV profile produced with 6 M KOH is represented by AC 700 KOH, and the CV profile produced with 1 M Na₂SO₄ is represented by AC 700 Na₂SO₄. The voltage window falls between -0.9 and +0.2 V. It is also observed that the profile is similar to the electric double-layer capacitance (EDLC) profile (Du et al., 2015; Thongwichit et al., 2016). This confirms that the activated carbon in the active material has EDLC behaviour for its charge-discharge mechanism. It is observed that the profile is similar however the area under the curve for KOH electrolyte is higher than Na₂SO₄ electrolyte. This could be due to the concentration difference in the electrolytes used. The specific capacitance from the CV profile can be calculated using the formula:

$$c_p = \frac{I}{2 \times m \times \Delta v \times \text{scan rate}} f/g \quad (5.3)$$

Where c_p is the specific capacitance, Δv is the voltage difference or potential window, m is the active mass of the active material present in the working electrode. The figure and the Table 5.4 compare the specific capacitance obtained for each scan rate from the CV profile for both KOH and Na₂SO₄ electrolytes. It is observed that specific capacitance decreases with an increase in scan rate. It is also observed that the specific

capacitance obtained when KOH as the electrolyte is higher than the specific capacitance obtained using Na_2SO_4 as the electrolyte. This is because the current obtained for each scan rate is higher for KOH due to its higher concentration. The figure show the CV profiles.

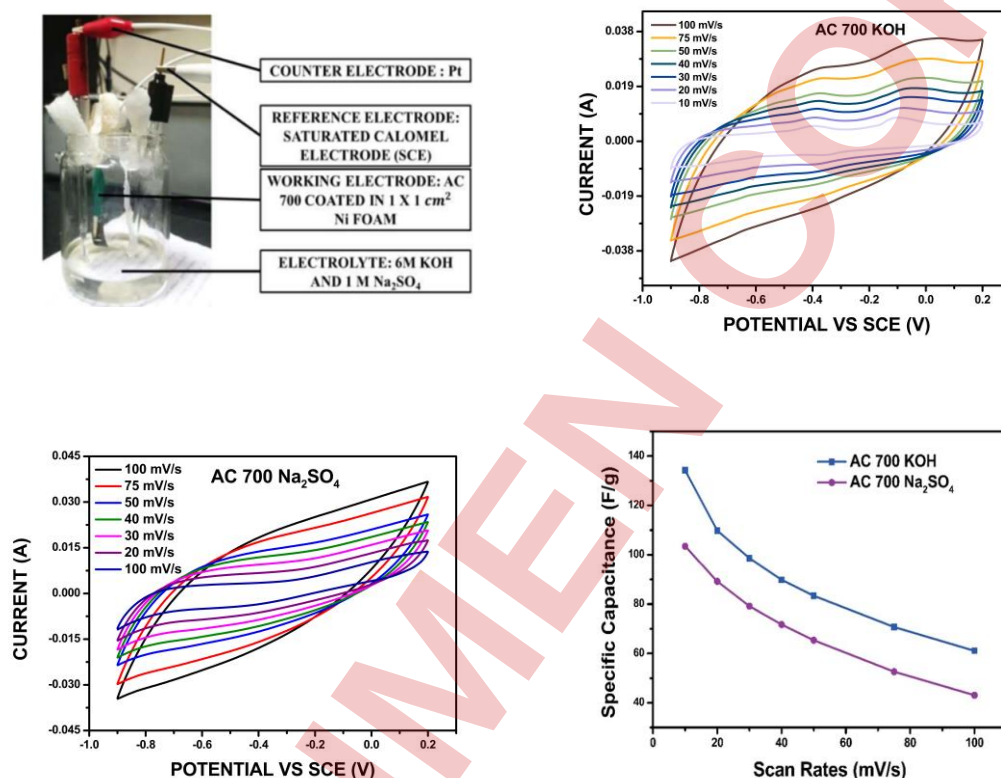


Figure 5.4: (a) 3-electrode system setup, (b) CV profile of AC 700 KOH for different scan rates, (c) CV profile of AC 700 Na₂SO₄ for different scan rates, (d) Specific capacitance value at different scan for AC 700 in KOH and Na₂SO₄

Table 5.4 The CV profile's specific capacitance for AC 700 in KOH and Na₂SO₄ electrolytes

Scan rates (mV/s)	The specific capacitance of AC 700 using KOH (F/g)	The specific capacitance of AC 700 using Na ₂ SO ₄ (F/g)
100	61.07	43.09
75	70.70	52.6
50	83.42	65.36
40	89.87	71.72

30	98.61	79.15
20	109.79	89.23
10	134.21	103.42

The three-electrode system is used to study the Chrono-potentiometry (CP) or galvanostatic charge-discharge (GCD) cycle at various current densities between 2 and 10 A/g. The charge-discharge curves for KOH and Na₂SO₄ at various current densities are displayed in figures (a) and (b), respectively. Both KOH and Na₂SO₄'s charge-discharge curves fit the EDLC capacitor's triangle profile, demonstrating the AC 700's EDLC behavior. The charge-discharge curve of AC 700 in KOH and Na₂SO₄ at a current density of 3 A/g is displayed in Figure (c). It has been noted that AC 700 takes longer to charge and discharge in Na₂SO₄ than it does in KOH. This observance might be due to a lower concentration of Na₂SO₄ which requires a longer time to discharge the charges that are stored. Figure (d) shows that the specific capacitance of AC 700 in Na₂SO₄ is higher than that of KOH. The specific capacitance from GCD is calculated using the formula:

$$c_p = \frac{I}{m \times \Delta v / \Delta t} \text{ f/g} \quad (5.4)$$

Where C_p is the specific capacitance, Δt is the discharge time and ΔV is the potential difference. The potential difference is 1.1 and 1.0 V for KOH and Na₂SO₄ respectively. The Table 5.5: shows the discharge time and specific capacitance of AC 700 in KOH and Na₂SO₄.

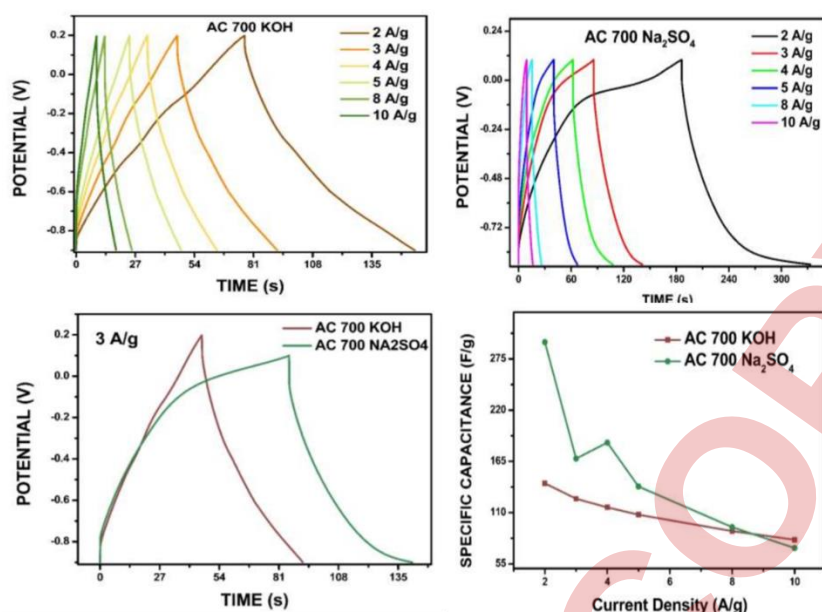


Figure 5.5 Charge-discharge curve for various current densities for AC 700 using (a) KOH, (b) Na₂SO₄, (c) KOH and Na₂SO₄ at 3 A/g, (d) specific capacitance of AC 700 using both the electrolytes

Table 5.5 The GCD curves' discharge duration and specific capacitance for AC 700 KOH and Na₂SO₄

Current density (A/g)	The discharge time of AC 700 KOH (s)	The specific capacitance of AC 700 KOH (F/g)	The discharge time of AC 700 Na ₂ SO ₄ (s)	The specific capacitance of AC 700 Na ₂ SO ₄ (F/g)
2	77.8	141.15	146.5	293
3	45.8	124.9	56	168
4	31.8	115.63	46.3	185.2
5	23.7	107.72	27.6	138
8	12.4	90.18	11.8	94.4
10	8.9	80.91	7.2	72

5.6.5 Electrochemical impedance spectroscopy (EIS)

The three-electrode system is used to study electrochemical impedance spectroscopy (EIS). The nyquist curve in figure plots the real and imaginary impedance of the system using AC 700 for KOH and Na₂SO₄. The nyquist plot is fitted to plot the analogous circuit. The nyquist plot is tabulated in the Table 5.6, and the ESR, R_s, Charge Transfer

Resistance (R_{ct}), and Warburg Resistance are computed from the equivalent circuit. Compared to KOH, the ESR for AC 700 in Na_2SO_4 is lower. Due to a higher electrolyte concentration, AC 700 in KOH has a higher charge transfer resistance than Na_2SO_4 . (Peng et al., 2014).

Table 5.6 Ohmic values obtained from EIS nyquist plot

	AC 700 with KOH	AC 700 with Na_2SO_4
R_s (Ohm)	0.769	3.7
R_{ct} (Ohm)	3.9	3.3
W (Ohm $s^{0.5}$)	3.5	3.5
ESR (Ohm)	5.46	4.37

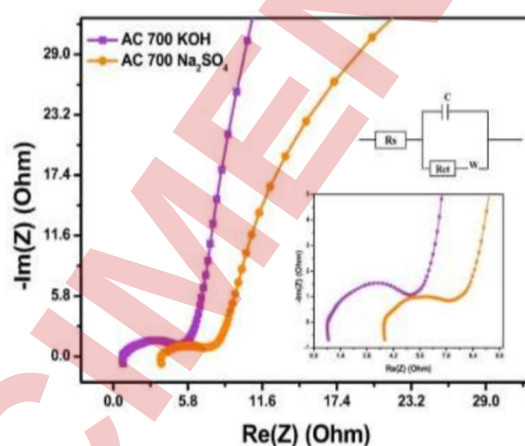


Figure 5.6 Nyquist plot of AC 700 in KOH and Na_2SO_4

5.6.6 Impedance analysis on LCR meter:

The cylindrical and pouch cell are tested using an impedance analyzer to measure series resistance (R_s) and specific capacitance (C_s) based on input parameters. The frequency range is 10 to 100 kHz. The impedance value increases with frequency, with a sudden drop at 95 kHz indicating system operation up to 95 kHz. The series resistance and

specific capacitance are linear only at high-frequency range, indicating non-linearity in single stack cells. Higher rigidity provides stability and ions separation without leakage.

The theoretical power of the cell is calculated using the formula:

$$P = V \times I \text{ (watts)} \quad (5.5)$$

The value of power obtained during charging is following theoretical calculation,

$$V \times I \text{ (watts)} = 1.13 \text{ V} * 0.029 \text{ A} = 0.032 \text{ Watts} \quad (5.5a)$$

This is the obtained power. The capacitance (C_s) of cell is 42 mf. The total energy of the system is calculated using the formula:

$$E = 0.5 \times C \times V^2 \quad (5.6)$$

Where E is in Joules (J), C is in farad and V is in volts.

The power density of the system can be calculated from the formula

$$\text{Power density (P}_g\text{)} = \text{Energy density} / \text{Discharge time.} \quad (5.7)$$

Capacitance retention is maintained by the system even at very high frequencies. A 10W motor is attached to the SS cell, which charges for 10 seconds using a DC source. It is estimated that the power handling is 27 mW. While high series resistance results

from low cell stiffness, capacitance values rise with frequency. More cells should be stacked, and a robust, stiff sealing should be made. A BSHESS system is used to assess a supercapacitor stack's performance for load-based switching.

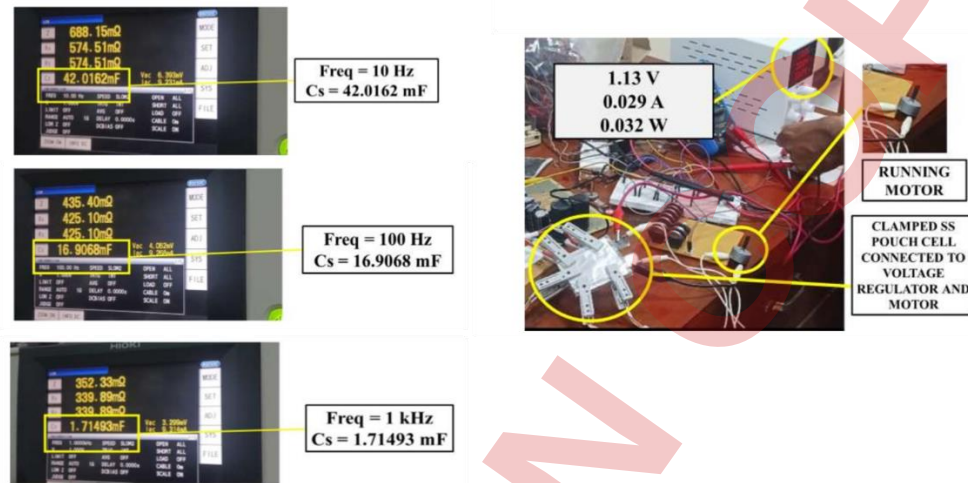


Figure 5.7 Performance of CC cell on load

5.6.7 Battery and supercapacitor hybrid energy storage system (BSHESS)

In a BSHESS system, power bursts are shared dynamically between a battery and a supercapacitor based on the loading pattern. This strategy takes benefits from both energy storage technologies, batteries can provide a longer energy supply and supercapacitors can tackle high-frequency power variation. The load condition real-time monitoring is also done by the system where supercapacitors are favoured when load is high while batteries are favored if the load becomes stable or low. Real-time load condition monitoring is done by the system, which gives supercapacitors priority when power demand is high and switches to batteries when load stabilizes or drops. This technique proves beneficial in avoiding higher levels of stress during a high load and, in turn, optimizes performance and battery access. Therefore, there is a possibility

that BSHESS will attract new clients through elasticity of operating expenses and energy efficiency (Chen & Zhang, 2021).



Figure 5.8 Load based switching between battery and supercapacitor in BSHESS

5.7 CUSTOMER ACQUIRING PROCESS

Designing a customer acquisition plan for the BSHESS at TRLs 10 and 12 is a strategic business issue. This is in order to stabilize the structure of the product and to prepare it for the market place. The first task here must be to ensure that there are major suppliers for high quality parts which are crucial to the company's production. These include lithium-ion batteries, electronic components, and supercapacitors, in addition to building connections that guarantee reasonable prices and dependable delivery dates. Establishing target market segments, such as residential clients or corporate enterprises,

is necessary to have a solid clientele. The next process is the advertising campaign as an attempt to popularize the new hybrid energy storage technology. One can form effective partnerships with companies in related industries like the above suppliers of renewable energy products hence getting exposure and accessing their customers. To a chosen group of clients, it is possible to arrange pilot programmes that let practice and receive valuable feedback. This can also be utilized to improve the product and produce endorsements for advertising campaigns. Potential clients will be drawn in throughout these procedures by filling in the gaps and emphasizing special advantages like increased productivity and cost savings (Aulet, 2024).

5.7.1 The steps for addressing the gaps during the customer-acquiring process

This work establishes the various gaps in the market after a careful market gap analysis was conducted in a bid to establish the unmet customer needs as well as the nature of market competition. It was then the need to analyze the offerings of competitors to identify markets where the product offered the potential for more added value. Many of these concepts relate to value propositions like efficiency, charging rates, ease of use and even product lifespan. The business will be able to learn about the concerns and expectations of potential customers through surveys and interviews. This will improve the system's current operational and physical characteristics. Based on the findings the business will address the existing technological gap by thoroughly testing the prototype. In addition to ensuring that the company markets the appropriate hybrid energy storage system in the market, such efforts will assist the company in bridging some of these technology gaps (Blum Andrew F. and Long, 2016). This offers a good foundation on which one can establish the customer decision-making unit or DMU.

5.7.2 Determine the customer's decision-making unit DMU

A collection of people inside an organization is called a Decision Making Unit (DMU). Although it can also apply to large B2C purchases, this group as a whole has an impact on the purchasing process, especially in B2B transactions. Each role at DMU contributes to the decision-making process according to their respective responsibilities and viewpoints. End users, who use the product or service and offer crucial feedback; influencers, who shape the views of initiators and decision-makers; initiators, who recognize issues that need to be addressed; economic buyers, who handle financial restraints and bargain for prices; decision-makers, who have the power to complete purchases; and gatekeepers, who manage access to important stakeholders are all important roles within the DMU (Charnes et al., 1978). Gathering comprehensive data on the many characteristics of each member is essential for the first stage of mapping the Decision-Making Unit (DMU). Table 5.6 indicates these characteristics.

5.7.3 Map the process to acquire a paying customer

Having a working product for a demo is essential before acquiring a paying customer. This implies that the technology must first be validated in a laboratory. The next step is to showcase the technology in a relevant environment so that its performance may be assessed under real-world working conditions. Based on this strategy, it is important to complete a few steps before involving the clients in actual sales decisions, in order to accomplish this demonstration as presented in Table 5.7 below. These procedures proved valuable as they establish that the prototype works well within conditions that closely resemble with its application (Vashishtha & Sharma, 2016).

5.7.4 Map the sales process to acquire a customer

The sales process of BSHESS is similar to that of a battery manufacturer or a solar panel manufacturer. The sales process of a BSHESS to acquire new customers typically involves several key stages. These are explained in Table 5.8. Once the market and customer research is done satisfactorily, a potential lead can be generated via direct outreach, trade exhibitions, online marketing campaigns, and demos. This establishes the initial contact with potential customers (Filho et al., 2023). This contact can be converted to an actual client using a personalized proposal that includes product specifications, prices, and delivery dates according to his/her needs. After discussions to settle the conditions, a formal contract is signed. After the product is delivered on time, the company needs to provide the consumer with thorough assistance. This covers customer service and technical support to ensure satisfied customers. This uses the feedback to provide useful data. The company may find areas for development, improve client interactions, and eventually raise conversion rates for the BSHESS by outlining this sales process. The following chapter concludes these entire processes and lessons learned.

Table 5.7 Customer's decision-making unit DMU

Role	Who	Demographic	Geographic	Behavior	Firmographic	Technographic	Psychographic
End-user	6. Operator/Member of family/Employee	Age: 25-50, Income: Varies, Education: Varies	Urban and rural areas, off-grid locations	Frequent usage during outages or travel Purchase driven by need for reliability and convenience	N/A (individual consumers)	Familiar with portable power technologies Preference for user-friendly and economical devices	Values sustainability and reliability Interests (application specific) and self-sufficiency
Influencer	1. Friends/Formal contacts/Sales executives/Marketing agencies/You-tubers	Age: 20-45 Income: Varies Education: Higher education preferred	Online/Urban areas with high connectivity	Influence purchasing decisions through recommendations or reviews	N/A (individuals or small firms)	Knowledgeable about energy solutions and trends	Values innovation and social proof
Initiator	2. The business owner/CTO of business/Research engineer	Age: 30-55 Income: Higher income due to position Education: Technical or managerial background	Typically urban or suburban areas	Identifies problems requiring solutions; motivated by efficiency improvements and cost cutting	Small to medium enterprises	Familiar with energy management systems and technologies	Values efficiency, cost-effectiveness, and innovation
Buyer	4. Procurement officer/Finance manager	Age: 35-60 Income: Higher income due to position Education: Business or finance background	Typically urban or suburban areas	Evaluates options based on cost, efficiency, and supplier reliability	Small to Large enterprises	Proficient in procurement technology and systems	Values cost savings, reliability, and vendor relationships
Decision maker	3. Senior management/Executive team member	Age: 40-65 Income: High income due to seniority Education: Advanced degrees preferred	Urban centers	Makes final purchasing decisions based on strategic goals	Large enterprises	Familiar with strategic planning tools and analytics	Values long-term benefits, strategic alignment, and organizational impact
Gatekeeper	5. Safety officers/Grid employees/Government regulating team	Age: 25-50 Income: Moderate income Education: Varies	Urban and rural areas	Controls access to decision-makers; filters information	N/A (individuals within organizations)	Controls availability of energy with the existing resources	Values efficiency in communication and organizational protocol

Table 5.8 The key steps involved in acquiring a paying customer

Step	Key Points	Details
Identify a relevant test environment and plan for testing	Design of experiment: Relevant test conditions and other parameters must be identified in advance for Customized BSHESS	Electrical simulation and physical experiments are selected as modes of testing the BSHESS for value propositions.
Identification of objective during testing	A matrix for key questions must be set to match value propositions	Question 1: What is the cost-saving if the BSHESS is used in place of BESS per KWh energy setup Question2: How much is the battery life improvement Question 3: What are improvements in battery charging time? Question 4: What is power enhancement per kWh battery system if BSHESS is implemented? Question 5: What is qualitative analysis for performance enhancement and efficiency?
Prepare the test system	1. Modify the simulation model or laboratory prototype or both to meet test environment requirements 2. Scale up the system and integrate it with other components	While experimenting with the model, it is necessary to follow safety and regulatory standards. Also, the system must be scalable to integrate with other renewable energy systems.
Conduct the demonstration	Implement the test plan and collect performance data. Monitor system behavior under various operating conditions	Record data for questions 1 to 5 mentioned in the above row.
Analyse the results	Evaluate system performance against established metrics. Identify areas for improvement and potential risks	Use collected data to assess the effectiveness and limitations of the hybrid storage system
Document the findings	Create a comprehensive report including test plan, data, analysis, and conclusions	Utilize the report to support further development and funding applications

Engage with stakeholders	Share demonstration results with relevant stakeholders	Engage potential customers, investors, and industry partners for insights and future development
Spread awareness and generate interest in potential paying customer	Develop informative blog posts, whitepapers, and videos that educate customers about the benefits. Engage with potential customers on platforms like LinkedIn, Twitter, and industry forums by sharing valuable content and responding to queries. Collaborate with renewable energy providers, electric vehicle manufacturers, or system integrators to reach their customer base and co-promote the hybrid storage system	Content marketing, social media, webinar hosting and possible collaborations will be conducted after the development of the prototype. The benefits will highlight how the system outperforms competitors in terms of performance, cost, and value, using objective data and third-party certifications
Purchase	Offer an easy-to-use online ordering platform or dedicated sales team to guide interested customers through the purchasing process.	A billing and supply chain software will help to Ensure that the system is delivered to the customer within the promised timeframe, minimizing delays and disruptions to their project timeline
Post-purchase support and feedback collection	Offer guidance and resources to help customers properly install and configure the system, ensuring optimal performance. Provide a comprehensive warranty and maintenance plan that covers any issues or malfunctions, demonstrating the commitment to customer satisfaction. Encourage satisfied customers to share their experiences with the system, which can be used to attract new customers and demonstrate the value of the product	Development of manuals/supporting literature and warranty/maintenance cards will be developed for this. Customer success stories will be collected for further outreach to new and existing customers.

Table 5.9 Sales process of the product

Sales process	Key points	Activities
---------------	------------	------------

Awareness	Creating awareness through targeted marketing campaigns, social-media interactions, participating in industry events	Publishing educational content about hybrid systems and running targeted ads on social media platforms
Interest	Generating interest with detailed product information, offering webinars and demonstrations, showcasing case studies	Hosting webinars, demonstrating system integration with BSHESS
Evaluation	Facilitating product evaluation against competitors, providing technical specifications and offering cost-benefit analyses and testimonials	Creating a comparison chart highlighting the advantages of the hybrid system over traditional solutions
Purchase	Making sure that the purchasing procedure is simple	Using a dedicated sales crew to answer questions and an online purchase platform
Post-purchase Support	Providing excellent customer support, offering installation assistance and maintenance services and following up for feedback	Conducting follow-up calls to gather customer feedback and address any issues post-installation

CHAPTER 6.

CONCLUSION AND FUTURE WORK

6.1 SUMMARY OF THE THESIS

This thesis examines the business prospects of energy solutions based on supercapacitors in hybrid energy systems. This thesis proposed a business plan for customized BSHESS. This thesis aims to identify business opportunities, develop a hypothesis, validate theory, and validate the theory through field research. Chapter 1 explains the financial and technical background emphasizing the potential of growth with such innovative solutions across various fields. Supercapacitors offer unique qualities such as high power output and fast charge/discharge capabilities. These can transform energy storage solutions in sectors like consumer electronics, backup power, renewable energy, and the IoT. Hybrid solutions are developed to cover technical gaps and adapt to different industries. Research is ongoing to develop innovative, economic and environment friendly materials for supercapacitors and their hybrid integration. Industry 4.0 is transforming energy module and supercapacitor manufacturing, with India's energy transition focuses on sustainable energy management, with a national framework for energy storage systems (ESS). The global energy market is expected to experience a 30% annual growth rate between 2021 and 2030, reaching a total value of \$15 billion.

Chapter 2 highlights customer discovery for a product through the development of a hypothesis and use of hypothesis testing and validation. It takes a systematic method, which is the market segmentation, the needs and pains analysis and the end user profile.

The research is chosen for the supercapacitor technology-based market, focusing on industrial and commercial facilities that can benefit from supercapacitor technology. Energy integrators and contractors can use supercapacitors to improve energy storage systems, enhance charging rates, and integrate with renewable energy systems. Due to the optimal integration of renewable power solutions and enhanced technologies, the demand in the market for customized BSHESS is predicted to soar significantly to USD 20.12 billion between 2023 and 2030. The discussed government regulations and incentives affect the market share by encouraging the capital flow to sustainable energy alternatives. The end user prioritizes sustainability, reliability, and self-sufficiency, and seek alternative energy solutions. The Total Addressable Market (TAM) for the beachhead market is estimated to be between ₹2.382 and 7.288 billion, with rural communities being the largest potential market.

Chapter 3 of this work provides a breakdown of the process of business model design for a BSHESS with the energy storage value optimisation in mind. The revenue model includes pricing strategies, revenue streams, and sales channels. The value proposition, which emphasizes the benefits and value of specially created hybrid energy storage systems for portable power users, is an essential component. The process used includes their identification of the competitors' strengths and weaknesses, identification of key differentiation factors, and the formulation of a value proposition statement. The direct technique and Steve Blank's instinctive value proposition canvas are employed to develop value propositions, to assess which are offered in this article, using a range of possible values. The study uses Google Forms and LinkedIn surveys to segment its customers in addition to secondary sources including transactional data, website analytics, and social media insights. Establishing strategic alliances, consistently

innovating, reaching a 90% or greater customer satisfaction rating, and preserving a 5% market share rate are among the company's goals.

Customer segments are created based on similarity of responses in interviews. New partnership with key energy sector firms and implementing a scientific feedback mechanism for better product differentiation and quality service delivery shall be embarked on by following this work. The cost-plus pricing technique is used, and the company's revenue generation components include electronic control boards and supercapacitors. The market size ranges from ₹2.382 to 7.288 billion for the beachhead market and ₹17.36 billion for follow-on markets. Each customer is expected to generate ₹35000 per kWh in revenue over their relationship with the company. The CAC for BSHESS products is approximately ₹3011.2 per customer. The total cost per 10 kWh is approximately ₹102000 to 165000 per kWh module. Moreover, this chapter explains the cost calculation for a small-scale supercapacitor manufacturing plant and the return on investment is calculated to be 35% after 18 months.

Chapter 4 discusses the methodology for material synthesis and device development for portable BSHESS systems. Activated carbon, with large surface area, is used in supercapacitors due to its potential for improved energy storage capacity. Activated carbon is prepared with biomass and analysed by XRD pattern, FTIR, SEM, BET surface area analysis, Raman analysis and electrochemical analysis. X-ray diffraction (XRD) is used to identify phase presence and crystallinity, while SEM examines morphology, topology, and compositional contrast. Raman spectroscopy examines rotational, vibrational, and low-frequency modes in a system. Electrochemical studies focus on interactions between substances with the exchange of electrons using

techniques like cyclic voltammetry, chrono-potentiometry, and electrochemical impedance spectroscopy.

An impedance analyzer measures electrical impedance to allow the description of a system's properties or behavior. Assumptions made while designing the product involve the user group, the problem to be solved, the solution to the problem, the riskiest assumptions and feasibility of implementation. The minimum viable business product (MVBP) for a hybrid battery and supercapacitor storage system involves defining assumptions, conducting surveys, and testing a prototype. Various methods can be used to validate assumptions and ensure the system delivers core value while reducing development expenses and time to market. Formal LOIs are secured to validate demand and confirm intent to purchase. The business has developed parameters for a hybrid battery and supercapacitor system, receiving funding of 60L after a detailed market survey. The system will be designed for portable power users and features a well-ventilated enclosure, printed circuit board, and security features. The targeted energy rating of this will be 1 kWh.

Chapter 5 discusses the near matching competitors which combines supercapacitors and batteries and finally forms an idea of the business model. The business model includes customized solutions for improved performance and efficiency in renewable energy applications and backup power systems. The goal is to increase energy efficiency by 15-30%, provide faster response times, and save 20% per year in energy tariffs overall. The company also aims to develop activated carbon for energy applications and help customers find hardware selections for versatile load demands. The core business includes, offering specialized hybrid energy storage systems, serving diverse market

niches, and providing. The prototype currently holds a Technology Readiness Level (TRL) of 6-8.

A plan which was developed to obtain new clients included key suppliers, target market segments, and efficient partnerships. In order to identify a market gap that has not been able to be filled by current competition, a market gap analysis was conducted. In particular, business-to-business (B2B) buying/ purchase decision involves a group or 'Decision-Making Unit' (DMU) within a purchasing organization. This company's primary goals will be to raise awareness through focused marketing campaigns, spark interest through webinars and product demos, and make it easier for customers to evaluate products, streamline the purchasing process with a dedicated sales force, and offer post-purchase assistance. A three-electrode configuration was also used for electrochemical investigations of AC. Because KOH has a higher concentration than other electrolytes, it was shown to have a higher specific capacitance. The system's impedance spectroscopy (EIS) plotted the real and imaginary impedance of AC 700 for KOH and Na₂SO₄, with higher ESR in Na₂SO₄ due to higher electrolyte concentration.

6.2 PLAN FOR SCALING OF BUSINESS

Long-term success in energy storage solutions is driven by having a plan for business scaling. Expanding manufacturing capacities through the purchase of new production facilities, equipment upgrades, and ongoing product quality improvement is one of the main scaling strategies. Rapid expansion without a large initial investment can also be facilitated by forming strategic alliances with contract manufacturers and component suppliers. Diversifying product offerings can also help meet the needs of end users.

Creating many scaled copies for various purposes is part of this. Additionally, the company will be able to serve a larger customer base by combining with home solar-plus-storage systems. Another essential requirement for growth is working with regional distributors and setting up sales and service offices in strategic areas. Table 6.1 below contains the key points for scaling the hybrid battery and supercapacitor storage business:

Table 6.1 Process for scaling the business

Scaling strategy	Key points
Expanding manufacturing capabilities	Possess more manufacturing space. Upgrade equipment to boost output Streamline processes to maintain quality Establish strategic partnerships with suppliers and contract manufacturers
Diversifying product offerings	Develop hybrid systems tailored for specific applications Offer a wider range of power ratings, energy capacities, and form factors Address needs of various market segments (backup power, peak power, commercial/industrial, residential) Reduce reliance on standalone batteries
Penetrating new geographic markets	Establish sales and service partners in key regions Partner with local distributors and system integrators Adapt products to meet regional standards and preferences Leverage digital marketing and e-commerce platforms
Driving innovation and cost reduction	Invest in R&D to increase cycle life, power capability, and energy density. Continuous innovation can lower production costs. Work together with suppliers, industry partners, and research institutions. In a market that is maturing, stay ahead of the curve.

6.2.1 Calculation of TAM for follow-on markets

Understanding the potential income of an item or service inside a particular market requires knowledge of the total addressable market (TAM). The overall market demand

for a product, assuming a company can attain a 100% market share, is represented by the TAM. This calculation can guide strategic decisions, such as product development, marketing investments, potential market entry and assess market opportunities. There are several methods to calculate TAM, including the top-down and bottom-up approaches. The top-down approach involves the use of industry statistics and reported figures to provide an overall market size While the bottom-up approach involves the use of existing sales data and generally accepted prices to arrived at the overall market size. This work follows the bottom-up approach for the calculation of TAM. It also provides the clear vision of the future number of revenues for the company and could be helpful in establishing the different strategic directions. Because it helps in proving that it is possible to enter the market and allocate resources the computation makes it easier to set realistic sales goals.

The total energy storage capacity being used in India, especially for backup power is expected to increase in future financial years. According to the National Electricity Plan (NEP) 2023 by the Central Electricity Authority (CEA), the energy storage capacity requirement is expected to reach approximately 82.37 GWh by 2026-27, which includes 34.72 GWh from Battery Energy Storage Systems (BESS) and 47.65 GWh from Pumped Storage Plants (PSP).As per projections, the probable load requirement for the same year would be 411.4 GWh of which 236.22 GWH would be provided by BESS and 175.18 GWH by PSP. Moreover, the total requirement of energy storage for various sectors is expected to be 178.5 GWh by 2022, which reveals projected value of total opportunity of more than 2700 GWh by 2032. Utilization rate of renewables like solar and wind energy is rising which makes it imperative to have energy storage systems, thereby making the role of backup systems very important in Indian energy system.

Table 6.2 TAM for follow on markets

Capacity	Company working in the matching domain	The average cost (₹)
10 kWh	Loom solar	500000
5 kWh	Loom solar	250000
84 Wh	BPE	5000
192 Wh	Energy intelligence pvt. Ltd.	12000
1 kWh	Business idea	50000
Average selling price (ASP) ₹ per kWh	50000	
Number of potential customers (based on kilo-watt-hour demand)	34720000	
Total Addressable Market (TAM) (100% market share) in ₹	50000*34720000= ₹1736000 million or ₹1736 billion	

6.2.2 Developing a product plan: Technology development roadmap

This roadmap outlines a step-by-step plan for the development process of the product right through to its introduction to the market. It includes areas of production, marketing, sale strategies, and even technology innovation. This is why this product plan's first stage involves creating a comprehensive map of the proposed technical evolution. This indicates that the product plan is very vital toward proposing the best outcome. This plan should address technology development, manufacturing, and market entry. It must also include financial planning and key milestones. These elements increase the likelihood of successfully bringing the BSHESS to market. This product plan further helps to design and improve an optimum technology development plan on a predetermined time scale. The steps required to advance BSHESS are outlined in this technology development roadmap. Moving from Technology Readiness Level

(TRL) 4 to TRL 8 is its goal. Full commercial deployment is denoted by TRL 8. Important benchmarks like prototype creation, thorough testing in pertinent settings, and verification of performance and safety standards are all included in the plan. For this reason, each stage must be properly defined in terms of resources, time frame and roles of responsibility. Periodicity of updates and improvements to the roadmap would show the potential problem with project flow and fix it in order to retain the schedule.

Milestone	Key activities	First segment		Second segment		Final segment	
		0-5 months	6-10 months	11-14 months	15-18 months	19-21 months	22-24 months
Ideation: Developing hypothesis as many as possible		This work has covered this.					
(b) Estimation: Validating each hypothesis to find customer segments and end users	Key customer segments: residential users, commercial enterprises, or industrial applications						
(b) Prioritization: Selecting the highest matching hypothesis along with the business objective	Residential users, remote sites and consumer electronics for portability needs						
(d) Product roadmap Setting up the minimum viable product development plan	A 1kWh system: (a) Potential increase in energy efficiency by 15-30% compared to standard systems, (b) power density that is 10-100 times greater than traditional batteries, (c) battery lifetime use up to 90 %, (d) annual cost saving a minimum of						

	15% based on base-load tariffs.						
(i) Manufacturing and supply chain strategy: quality control process, production capacity, lead times, and cost optimization	A detailed cost and performance comparison with outsourcing of electronic components and PCB development						
(ii) Testing in relevant environments	Testing the device for objectives in section (d)						
(iii) Validation of performance and safety requirements	Analysis for RoHS and safety parameters as per government of India guidelines						
(iv) Market entry and commercialization strategy: Pricing strategies, regulatory requirements and distribution channels	B2B selling for 50000 per kWh scale. Primarily amazon.in would be used to understand this						
(v) Financial planning and funding: venture capital, grants, or strategic partnerships	Identifying potential sources of funding, estimate sales volumes, and forecast potential revenues						

6.2.3 Business model

The business plan is influenced from Engineering Procurement and Construction (EPC) model (International Solar Alliance (ISA), 2023). The structure of creating, delivering and capturing value within manufacturing firms is explained by a paradigm known as the manufacturer business model. Usually this model includes raw material identification. It includes producing finished goods and then marketing the goods to the

consumers themselves or through agents. Single operations can lead to existence of increased profit margins. Manufacturers use a variety of sales strategies. They have the option of using conventional distribution techniques or Business-to-consumer (B2C) sales. Every strategy has an impact on total profitability, customer relations, and operational performance. Many businesses are investigating new business models as the industrial sector expands. They are using e-commerce and digital technology. Hence, their coverage and the ability to supply market needs are improved.

A direct-to-consumer strategy gives producers more authority over their brand. Pricing and customer relations are improved by this tactic. Additionally, it enables faster reactions to shifts in the market and customer preferences. High capital investment is still a crucial factor, though, because setting up production facilities and buying equipment cost a lot of money. This industry is risky due to its capital-intensive nature. However, long-term cost savings can be achieved by deliberate investments in digital technologies and automation. Productivity can also be increased by these investments. They stimulate innovation and competitive advantage when properly managed.

The business model could focus on manufacturing and selling hybrid energy storage systems to automotive manufacturers, renewable energy developers, and commercial enterprises looking to improve energy efficiency and reduce costs. In this business model, the revenue streams could include direct sales to other service providers in this domain. The company needs to establish key partnerships with suppliers of battery and inverter systems. These partnerships will help integrate their systems into existing infrastructures. The model will emphasize innovation in product development. It will focus on enhancing battery lifespan, superior performance and efficiency and

maximizing power delivery of the system. This differentiation can strengthen the product's position in a competitive market. Additionally, it focuses on cost-effectiveness for backup power and renewable energy storage applications. The business model for a hybrid battery and supercapacitor energy storage system is presented using the Business Model Canvas framework. For a detailed model kindly refer the appendix.

Table 6.3 Business plan of the BSHESS product

Key partners	Key activities	Value propositions	Customer relationships	Customer segments	Key resources	Channels	Cost structure	Revenue streams
1. Battery and supercapacitor manufacturers 2. Inverter suppliers 3. Solar OEMs 4. Renewable energy developers 5. System integrators	1. R&D for hybrid storage system design 2. Manufacturing and assembly 3. Software development for energy management 4. Marketing via demonstrations 5. Development of variety in capacity.	1. Improved performance and efficiency for solar and renewable energy applications 2. Faster charging and higher power density compared to batteries alone 3. Longer lifespan and lower maintenance costs, 4. Customizable and scalable systems to meet specific energy needs	1. Direct sales to solar OEMs and renewable energy developers. 2. Partnerships with system integrators and installers 3. Technical support and after-sales service	1. Energy backup services (shops and distributors) 2. Renewable energy developers for utility-scale storage 3. Commercial and industrial customers for behind-the-meter applications	1. Intellectual property for hybrid storage system design. 2. Manufacturing facilities and equipment 3. Technical expertise in energy storage and power electronics 4. Brand reputation and customer relationships	1. Direct sales team for solar and renewable energy installation firms. 2. Partnerships with system integrators and installers for commercial/industrial customers 3. Online marketing and e-commerce for smaller-scale applications	1. R&D and engineering costs. 2. Manufacturing and assembly costs 3. Sales and marketing expenses 4. Installation and maintenance costs	1. Product sales to solar and renewable energy developers, commercial/industrial customers. 2. Licensing fees for hybrid storage distributors. 3. Direct payment from EPC firms.

6.3 TECHNOLOGY ASSESSMENT

6.3.1 Current use cases

Supercapacitors were used in 2021 to provide a hydropower-based distribution utility with black-start support when in a temporary microgrid configuration as part of a DOE-funded field demonstration. Supercapacitors also have been deployed in combination with solar photovoltaic generation to power the West Thumb Ranger Station in Yellowstone National Park. Supercapacitors are used in a flexible alternating current transmission system (FACTS) and in high-voltage direct current transmission to alter the impedance of the line to regulate power factor and transmission capabilities by injecting or absorbing reactive power. They are used in renewable systems integration to improve the power quality of fluctuating renewable generation (Durvasulu et al., 2023).

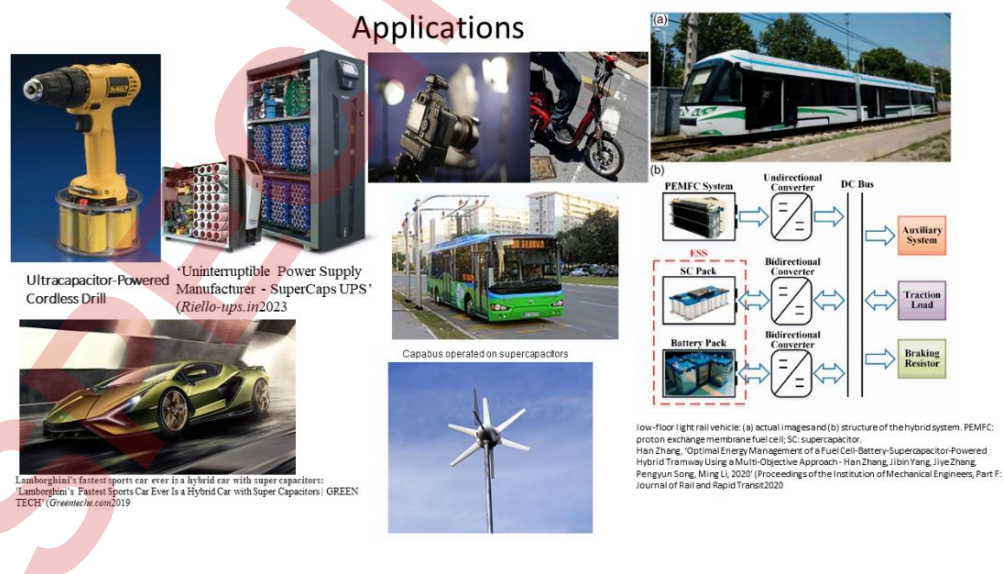


Figure 6.1 Application of supercapacitor

6.4 KEY LEARNINGS

The main requirements for hybrid energy storage are lower cost and simplicity, scalability for high-power applications. Potential risks include high power from batteries that may cause battery failure. People are aware of the benefits of supercapacitors but need demonstrations. End users have mixed feedback, and consultants/integrators are responsible for changing the trend. High power demand using hybrid energy storage is a need when supercap cost is comparable. Most integrators agree to use battery+supercap combining but not solely supercapacitors alone. Early adopters include installation firms working in decentralized energy storage. Over the course of this entrepreneurial journey, significant progress has been made in three critical steps: materials discovery, development, and technology demonstration. From the standpoint of materials discovery, the company focused on identifying the best materials. By creating methods to build cylindrical cells using activated carbon, our study successfully solved the primary concerns of stiffness and capacitance during the development phase. During the technology demonstration, the created systems' capabilities were displayed, their performance in practical applications was verified, and insightful input for upcoming revisions was gathered.

6.5 FUTURE WORK:

Looking ahead, the next steps will focus on refining the technology as well as expanding the outreach efforts for attracting potential funding for setting up a plant. Constant feedback will be the primary activity for the alpha version of the prototype. We will also expand market outreach. For reliability and consistent performance of the cylindrical cells, activated carbon generation should be optimized. The toughness of the

cell and the ability of the cell to maintain reliable, efficient capacitance storage are two aspects which need further research. Effective sealing materials which prevent leakage and degeneration of cell structures that are of longer service can be employed for this purpose. In addition, the performance of the capacitive properties can be improved from the design and structure features of cylindrical cells. One of these techniques is rolling a larger cell area linearly. This technique raises the total capacitance of the cell substantially. There are ways to further improve the amount of capacitance per given volume, and thus construct more efficient, powerful and energy-storing cells by tweaking the winding pattern or the shape of the cells. These improvements in cell design, sealing and manufacturing will improve cylindrical activated carbon cells performance.

APPENDIX A.

LEAN MANUFACTURING CALCULATIONS

A.	Fixed capital	Value	Unit	B.5	Total working (per month) (B.1+B.2.2+B.3+B.4.10)	105900	₹
A.1	Machinery and equipment			B.5.1	Total working (for 3 months)(3*B.5)	317700	₹
A.1.1	Cost of production unit	600000	₹	B.5.2	Total working capital (for 1 year)(12*B.5)	1270800	₹
A.1.2	Testing equipment	650000	₹	C.	Total Investment required (3 months)(A.3+B.5.1)	1742700	₹
A.1.3	Total (A.1.1+A.1.b)	1250000	₹		Total Investment required (1 year) (A.3+B.5.2)	2695800	₹
A.2	Electrification and other			D.	Financial analysis	Value	
A.2.1	Electrification (10% of machinery)	125000	₹	D.1	Total fixed capital as A.3)	1425000	₹

A.2.2	Cost of tools, die etc	20000	₹	D.1.1	Total working capital (for 1 year)(same as B.5.2)	1270800	₹
A.2.3	Cost of office equipment	25000	₹	E.	Depreciation		
A.2.4	Pre-operative expenses	5000	₹	E.1	Depreciation on plant and machinery per year @ 10% of A.1.3	125000	₹
A.2.5	Total Electrification and other costs(A.2.1TOA.2.4)	175000	₹	E.2	Depreciation on testing tools @20% of A.2.2	4000	₹
A.3	Total fixed capital (A.1.3+A.2.5)	1425000	₹	E.3	Depreciation on office equipment @20% of A.2.3	5000	₹
B.	Working capital			E.4	Total depreciation (E.1 to E.3)	134000	₹
B.1	Raw materials (per month)	50000	₹	F.	Total cost of production (per year)(D.1.1+E.4)	1404800	₹
B.2	Salaries per month	40000	₹	G	Turnover per year		

B.2.1	Salaries overhead per month (15%)	6000	₹	G.1	working day in a year 22 days per month	264	days
B.2.2	Total salaries (B.2.1+B.2.2)	46000	₹	G.2	Total units of capacitors produced per year (110 units per day)	29040	units
B.3	Utilities (per month)	3100	₹	G.3	Average cost per capacitor	150	₹
B.4	Other Contingent Expenses (per month)				Total revenue	4356000	₹
B.4.1	Rent	3000	₹		Total profit per year(first year	3085200	₹
B.4.2	Postage and stationery	800	₹		Net Profit Ratio =(Net Profit × 100)//Total Turnover in %	70.8264463	
B.4.3	Telephone	1000	₹		Return on Investment in %=(Net Profit × 100)/Total Investment of 3 months	177.035634	
B.4.4	Advertisement	500	₹		Return on Investment in %=(Net Profit × 100)/Total Investment of 1 year	114.444692	

B.4.5	Repair and maintenance	500	₹		Interest and depreciation cost		
B.4.6	Transportation expenses	1000	₹		22% average interest on total capital investment(yearly)	279576	₹
B.4.7	Insurance	500	₹		Total depreciation (E.1 to E.3)	134000	₹
B.4.8	Consumables	200	₹		Total Interest and depreciation cost	413576	₹
B.4.9	Other misc. expenses	500					
B.4.10	Total contingent expenses (per month) (B.4.1 to B.4.9)	6800	₹				

SPECIMEN COPY

APPENDIX B.

B.1 ENGINEERING PROCUREMENT CONSTRUCTION (EPC) MODEL (INTERNATIONAL SOLAR ALLIANCE (ISA), 2023)

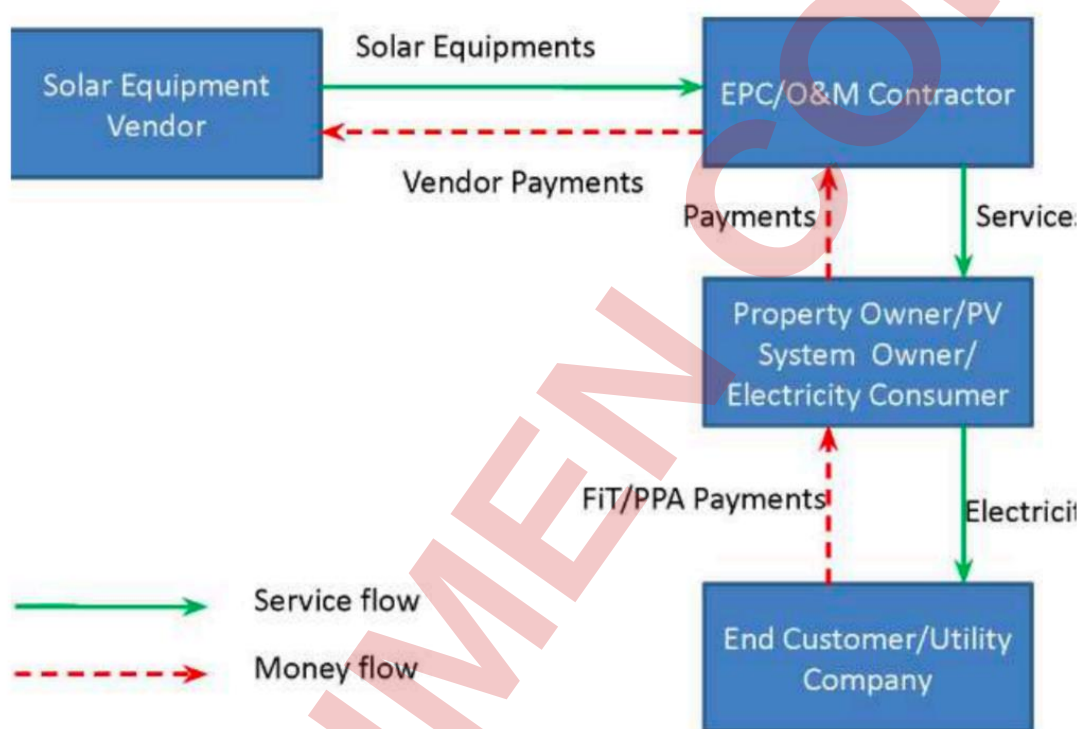


Figure 6.2 Engineering Procurement Construction (EPC) model

B.2 COST TABLE: AVERAGE PRICE ₹50000 PER KWH

Example energy product	kWh rating	Price	Per kilo watts hour		Link to the product
Hybrid Battery Energy Storage System, For Commercial, Capacity		50000	50000	Li-Battery with solar	https://www.aajjo.com/solar-renewable-energy-products/hybrid-battery-energy-storage-system-for-commercial-capacity-1-kw-in-ahmedabad-surya-solar-and-waters/product#MoreDetails
MuscleGrid Super 6.2 KW	75kw h	365000	50000	2 Li-Battery	https://musclegrid.in/products/musclegrid-super-combo-6-2-kw-hybrid-solar-

Hybrid Combo with Lithium Battery and Bi_Facial Mono Perc Solar Panels (2 Battery 8 Panels (4600w))				8 Panels (4600w)	system-with-lithium-battery-and-bi-facial-mono-perc-solar-panels-2-battery-12-panels-6900w
White And Green 80kwh 100kwh 120kwh 150kwh Lifepo4 Lithium Battery With Ac Charger Integrated Energy Storage System	80		51000		https://www.tradeindia.com/products/80kwh-100kwh-120kwh-150kwh-lifepo4-lithium-battery-with-ac-charger-integrated-energy-storage-system-c8278914.html
Loom solar	10		500000	50000	
BPE	0.84		5000		
ENERGY INTELLIGENCE PLTD	0.192		12000		

B.3 INTERVIEWS CONDUCTED

Hypothesis:

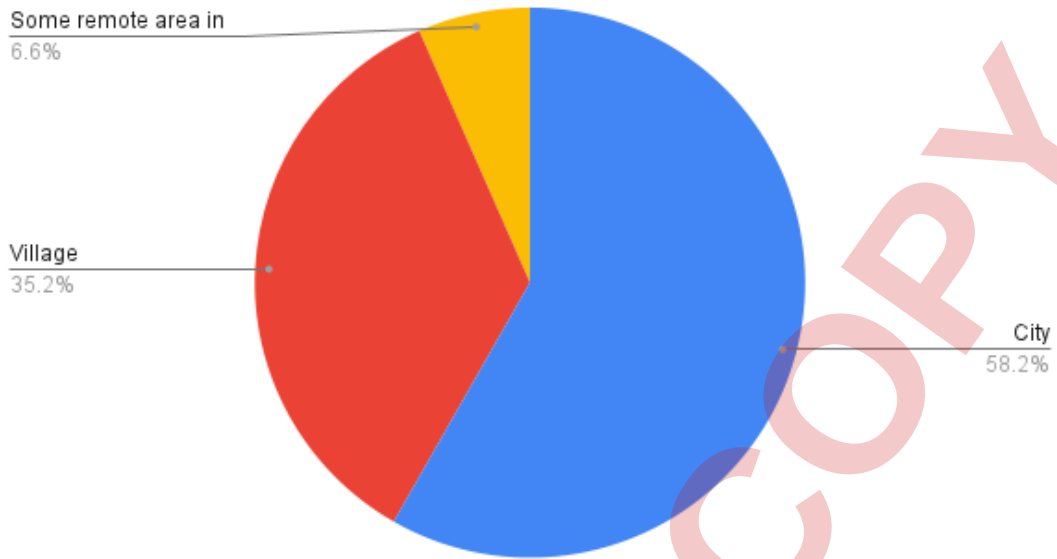
"Customised energy solution providers must enhance their services with customer satisfaction. They must be able to integrate renewable sources and meet high power demands overcoming compatibility issues. This will help clients navigate modern energy markets, expand energy capacity, and attain more customers for easy portability needs."

Interview details

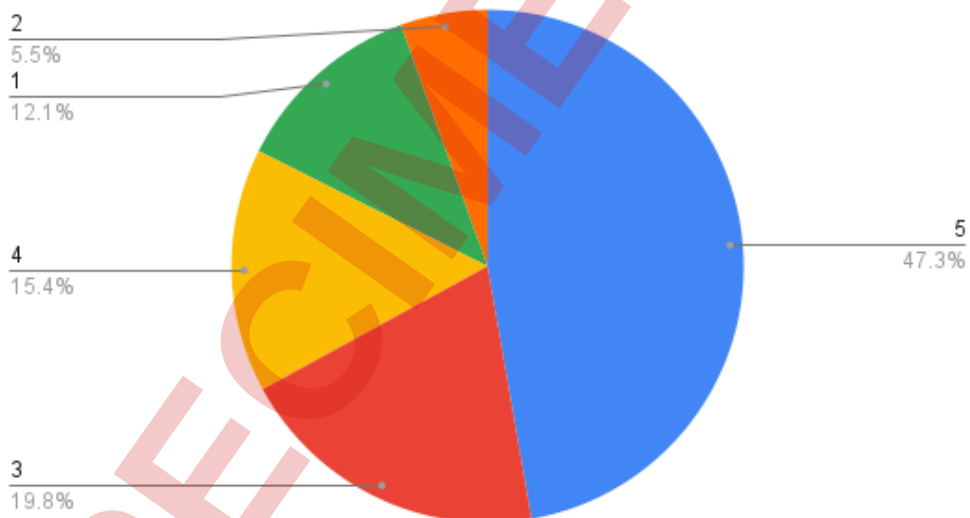
	Type	Regarding customer/end user	Description
18	Field visit	customer	Local shops/electric hardware stores
20	Field visit	customer	Local contractors/service providers in solar energy domain
11	Field visit	End-user	Houses retaining power backup
13	Field visit	End-user	Local shops for battery business
80	Telephonic	customer	
5	Video meet	customer	
92	Questionnaire survey	End-user	Mix categories from village, city and remote locations
29	Face to face	mix	

Summary of the questions asked as per the hypothesis from the customer segment:

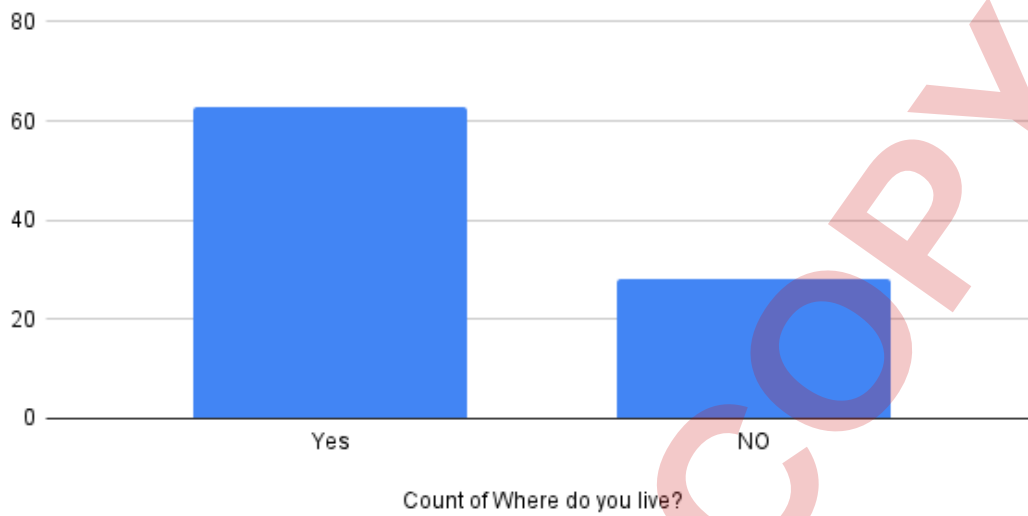
Count of Where do you live?



Do you purchase a power backup system based on the recommendation of a local store who deals with selling and



Do you have a power backup system or UPS at your home/workplace that contains a battery?



COUNTA of Features required for the firm	Feature required for the firm					
expectation from improved tech	cost	effectiveness	Fast Charging and Discharging	High Energy and Power	Long life	Grand Total
efficiency		10.56%	2.82%	2.82%		16.20%
simplification	16.20%		8.45%	5.63%	2.82%	33.10%
wider operations	16.90%	14.08%	5.63%	11.27%	2.82%	50.70%
Grand Total	33.10%	24.65%	16.90%	19.72%	5.63%	100.00%

COUNTA of Feature required for the firm	Feature required for the firm					
Requirement for portability from the customers	cost	effectiveness	Fast Charging and Discharging	High Energy and Power	Long life	Grand Total
2	2.82%	2.82%				5.63%
3	16.90%	14.08%		3.52%	2.82%	37.32%
4	13.38%	5.63%	14.08%	4.93%	2.82%	40.85%

5		2.11%	2.82%	11.27%	16.20%	
Grand Total	33.10%	24.65%	16.90%	19.72%	5.63%	100.00%

Portability is linked with cost an effectiveness. Effectiveness is linked to compatibility, smooth operation, prevention of overloading etc,

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SPECIMEN COPY

CURRICULUM VITAE

1. **NAME** : Sandeep Kumar

2. **DATE OF BIRTH** : 15 July 1988

3. **EDUCATIONAL QUALIFICATIONS**

Bachelor of Technology 2007 – 2011

Institution : Gautam Buddha Technical University, Lucknow

Specialization : Electronics and Communication Engineering

Master of Science 2022 – 2024

Institution : Indian Institution of Technology, Madras

Specialization : Energy storage devices

Registration Date : 20 Jan 2022

DOCTORAL COMMITTEE

CHAIRPERSON:

Dr. Bhattacharya S S
Professor
Department of Mechanical Engineering

GUIDE(S):

Dr, Tiju Thomas
Professor
Department of Mechanical Engineering

MEMBERS:

Dr. Hemaprabha E
Assistant Professor
Department of Mechanical Engineering

Dr. Satyanarayanan Seshadri
Associate Professor
Department of Applied Mechanics and
Biomedical Engineering